13. Courses of Study and Scheme of Assessment

BE CSE (AI&ML) **(2023 REGULATIONS)**

(Minimum No. of credits to be earned: 166)

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER I										
THEORY										
1	23N101	Calculus and its Applications	3	1	0	4	40	60	100	BS
2	23N102	Basics of Electrical and Electronic Systems	3	0	0	3	40	60	100	ES
3	23N103	Applied Chemistry	3	0	0	3	40	60	100	BS
4	23N104	Computational Thinking	3	1	0	4	40	60	100	ES
5	23G105	English Language Proficiency	3	1	0	4	40	60	100	HS
PRACTICALS										
6	23N110	Python Programming Laboratory	0	0	4	2	60	40	100	ES
7	23N111	Basic Science Laboratory	0	0	4	2	60	40	100	BS
8	23N112	Engineering Practices	0	0	4	2	60	40	100	EEC
MANDATORY COURSES										
9	23IP15	Induction Programme	-	-	-	Grade	-	-		MC
		Total: 30 hrs	**15**	**3**	**12**	**24**	**380**	**420**	**800**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER II										
THEORY										
1	23N201	Transforms and its Applications	3	1	0	4	40	60	100	BS
2	23N202	Discrete Mathematics	3	1	0	4	40	60	100	BS
3	23N203	Computer Organization and Architecture	3	0	0	3	40	60	100	ES
4	23N204	C Programming	3	0	0	3	40	60	100	ES
PRACTICALS										
5	23G___	Language Elective	0	0	4	2	60	40	100	HS
6	23N211	Hardware Programming Laboratory	0	0	4	2	60	40	100	ES
7	23N212	C Programming Laboratory	0	0	4	2	60	40	100	ES
8	23N214	Engineering Graphics	0	0	2	1	60	40	100	ES
MANDATORY COURSES										
8	23N215	Activity Point Programme	-	-	-	Grade	-	-		MC
9	23Q213	Foundations for Problem Solving	0	0	2	0	60	40	100	MC
		Total 29 hrs	**12**	**2**	**16**	**21**	**440**	**460**	**900**	

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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER III										
THEORY										
1	23N301	Linear Algebra	3	1	0	4	40	60	100	BS
2	23N302	Probability, Stochastic Processes and Statistics	3	1	0	4	40	60	100	BS
3	23N303	Data Structures	3	0	0	3	40	60	100	PC
4	23N304	Software Engineering	3	0	0	3	40	60	100	PC
5	23O305	Engineering Economics	3	1	0	4	40	60	100	HS
PRACTICALS										
6	23N310	Object Oriented Programming Laboratory	0	0	4	2	60	40	100	ES
7	23N311	Data Structures Laboratory	0	0	4	2	60	40	100	PC
8	23Q313	Building Communication Skills	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23K312	Environmental Science	2	0	0	0	-	-	-	MC
10	23N315	Activity Point Programme	-	-	-	Grade	-	-	-	MC
11	23TC01	Heritage of Tamils / **தமிழர் மரபு**	1	0	0	1	100	0	100	HS
		Total 31 hrs	**18**	**3**	**10**	**24**	**480**	**420**	**900**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER IV										
THEORY										
1	23N401	Foundations of Optimization Techniques	3	1	0	4	40	60	100	BS
2	23N402	Database Management Systems	3	0	0	3	40	60	100	PC
3	23N403	Design and Analysis of Algorithms	3	1	0	4	40	60	100	PC
4	23N404	Operating Systems	3	1	0	4	40	60	100	PC
5	23N405	Machine Learning -1	3	0	0	3	40	60	100	PC
PRACTICALS										
6	23N410	Database Management Systems Laboratory	0	0	4	2	60	40	100	PC
7	23N411	Machine Learning Laboratory	0	0	4	2	60	40	100	PC
8	23Q413	Problem Solving	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23O414	Indian Constitution	2	0	0	0	-	-	-	MC
10	23N415	Activity Point Programme	-	-	-	Grade				MC
11	23TC02	Tamils and Technology / தமிழரும் தொழில் நுட்பமும்	1	0	0	1	100	0	100	HS
		Total: 31 hrs	**18**	**3**	**10**	**24**	**480**	**420**	**900**	

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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER V										
THEORY										
1	23N501	Machine Learning -2	3	1	0	4	40	60	100	PC
2	23N502	Artificial Intelligence	3	1	0	4	40	60	100	PC
3	23N503	Embedded Systems	2	2	0	4	40	60	100	ES
4	23N504	Computer Networks	3	0	0	3	40	60	100	PC
5	23N ___	Professional Elective - 1	3	0	0	3	40	60	100	PE
PRACTICALS										
6	23N510	Computer Networks Laboratory	0	0	4	2	60	40	100	PC
7	23N511	Application Development Laboratory	0	0	4	2	60	40	100	EEC
8	23Q513	Aptitude Skills	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23N515	Activity Point Programme	-	-	-	Grade	-	-	-	MC
		Total: 28 hrs	**14**	**4**	**10**	**23**	**380**	**420**	**800**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VI										
THEORY										
1	23N601	Data Privacy and Security	3	1	0	4	40	60	100	PC
2	23N602	Parallel and Distributed Systems	3	1	0	4	40	60	100	PC
3	23N603	Deep Learning	3	0	0	3	40	60	100	PC
4	23N604	Natural Language Processing	3	0	0	3	40	60	100	PC
5	23N ___	Professional Elective - 2	3	0	0	3	40	60	100	PE
PRACTICALS										
6	23N610	Deep Learning Laboratory	0	0	4	2	60	40	100	PC
7	23N611	Innovation Practices	0	0	2	1	60	40	100	EEC
8	23Q613	Enhancing Arithmetic Problem Solving	0	0	2	1	60	40	100	EEC
MANDATORY COURSES										
9	23N615	Activity Point Programme	-	-	-	Grade	-	-	-	MC
		Total : 25 hrs	**15**	**2**	**8**	**21**	**380**	**420**	**800**	

At the end of 6th semester, the students are required to earn the minimum number of activity points from the AICTE mandated ACTIVITY POINT PROGRAMME to qualify for the award of BE/BTech degree (Refer section 4 (vii) (c) of 2023 Regulations)

* As per AICTE norms; Total 60 hrs; Grade: Completed/Not Completed; Not counted for CGPA

Grade Competed/Not Completed; Not counted for CGPA

CA -Continuous Assessment FE-Final Examination

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S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VII										
THEORY										
1	23N701	Big Data and Advanced Database Systems	3	0	0	3	40	60	100	PC
2	23N ___	Professional Elective – 3	3	0	0	3	40	60	100	PE
3	23N ___	Professional Elective – 4	3	0	0	3	40	60	100	PE
4	23N ___	Professional Elective – 5	3	0	0	3	40	60	100	PE
5	23NO ___	Open Elective – 1	3	0	0	3	40	60	100	OE
PRACTICALS										
6	23N710	Big Data and Advanced Database Systems Laboratory	0	0	4	2	60	40	100	PC
7	23N720	Project Work - I	0	0	4	2	60	40	100	EEC
		Total: 23 hrs	**15**	**0**	**8**	**19**	**320**	**380**	**700**	

S.No	Course Code	Course Title	Hours / Week			Credits	Maximum Marks			CAT
			Lecture	Tutorial	Practical		CA	FE	Total	
SEMESTER VIII										
THEORY										
1	23N ___	Professional Elective - 6	3	0	0	3	40	60	100	PE
2	23NO ___	Open Elective - 2	3	0	0	3	40	60	100	OE
3	23N820	Project Work - II	0	0	8	4	60	40	100	EEC
		Total: 14 hrs	**6**	**0**	**8**	**10**	**140**	**160**	**300**	

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LANGUAGE ELECTIVES

23G001	Communication Skills for Engineers
23G002	Basic German
23G003	Basic French
23G004	Basic Japanese

PROFESSIONAL ELECTIVES FOR BE DEGREE PROGRAMME

Course Code	**Course Title**
23N001	Software Testing and Quality Assurance
23N002	Software Project Management
23N003	Recommender Systems
23N004	Agile Software Development
23N005	Open Source Technology
23N006	Human Computer Interaction
23N007	Information Retrieval
23N008	Full Stack Development
23N009	DevOps
23N010	Advanced Data Structures
23N011	Ethical Hacking
23N012	Compiler Design
23N013	Blockchain Technology
23N014	Cloud Computing
23N015	Data Exploration and Visualization
23N016	Internet of Things
23N017	Computer Vision
23N018	Evolutionary Computing Algorithms
23N019	Neuro Fuzzy System
23N020	Generative AI

Professional Electives for BE Honours / BE Honours with specialization in same discipline and BE Minor degree programmes

VERTICAL I: Cyber-Physical Systems

23N031	Virtualization
23N032	Edge Computing
23N033	Network Design
23N034	Software Defined Networks
23N035	Semantic Web Technology
23N036	Speech Processing
23N037	Cognitive Analytics
23N038	Augmented Reality
23N039	Robotics
23N040	Network Optimization
23N046	Network Management & Security

VERTICAL II: Security and Privacy

23N041	Information Security
23N042	Cyber Security and Digital Forensics
23N043	Web Services and Security
23N044	Privacy and Security in Online Social Networks
23N045	Malware Analysis
23N047	Quantum-Safe Cryptography

OPEN ELECTIVES

23NO01	Bioinformatics
23NO02	Design Thinking
23NO03	Ethics of Artificial Intelligent
23NO04	Graph Theory
23NO05	Intellectual Property Right
23NO06	Social and Economic Network Analysis

SELF DIRECTED LEARNING COURSES

23IC01	Experiencing Entrepreneurship
23IC02	Venture Development

Summary of Credit Distribution

BE COMPUTER SCIENCE AND ENGINEERING (AI&ML)										
S. No	**Course Category**	**Credits Per Semester**								**Total Credits**
		I	**II**	**III**	**IV**	**V**	**VI**	**VII**	**VIII**	
1	HS	4	2	5	1	0	0	0	0	12
2	BS	9	8	8	4	0	0	0	0	29
3	ES	9	11	2	0	4	0	0	0	26
4	PC	0	0	8	18	13	16	5	0	60
5	PE	0	0	0	0	3	3	9	3	18
6	OE	0	0	0	0	0	0	3	3	6
7	EEC	2	0	1	1	3	2	2	4	15
8	MC	0	0	0	0	0	0	0	0	0
	TOTAL	24	21	24	24	23	21	19	10	**166**

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SEMESTER – I

23N101 CALCULUS AND ITS APPLICATIONS

3 1 0 4

DIFFERENTIAL CALCULUS: Functions of two variables, limits and continuity, partial derivatives, chain rule, extreme values and saddle points, Lagrange multipliers, Taylor's formula for two variables. (9+3)

INTEGRAL CALCULUS: Double and iterated integrals over rectangles, double integrals over general regions, Fubini's theorem, area and volume by double integration, reversing the order of integration, double integrals in polar form. (9+3)

FIRST ORDER ORDINARY DIFFERENTIAL EQUATIONS: Basic concepts, separable differential equations, exact differential equations, integrating factors, linear differential equations, modeling - mixing problems, Newton's law of cooling, decay and growth problems. (9+3)

SECOND ORDER LINEAR DIFFERENTIAL EQUATIONS: Homogeneous linear equations of second order, homogeneous linear ODEs with constant coefficients, Euler–Cauchy equations, solution by variation of parameters, free oscillations mass spring systems, electric circuits. (9+3)

VECTOR CALCULUS: Gradient and directional derivative of a scalar field, divergence and curl of a vector field. Integration in vector field – line integrals, path independence of line integrals, Green's theorem in the plane, divergence theorem of Gauss and Stokes theorem. (9+3)

Total L: 45 +T: 15 = 60

TEXT BOOKS

1. J. Hass, C. Heil, Maurice D.W "'Thomas' Calculus", Pearson Education, New Delhi, 2018.
2. Erwin Kreyszig, "Advanced Engineering Mathematics", Wiley India, New Delhi, 2018.

REFERENCES

1. H. Anton, I. Bivens, S. Davis, "Calculus", John Wiley and Sons, USA, 2016.
2. Wylie C. R and Barrett L. C, "Advanced Engineering Mathematics", Tata McGraw-Hill, New Delhi, 2019.
3. Michael D. G, "Foundations of Applied Mathematics", Dover Publications, INC, New York, 2013.
4. Gilbert Strang, "Calculus", Wellesley Cambridge Press, USA, 2017.

23N102 BASICS OF ELECTRICAL AND ELECTRONIC SYSTEMS

3 0 0 3

DC CIRCUIT: Current -voltage -power- energy, electrical circuit elements: resistors - inductor- capacitor, source of electrical energy. Ohm's law - Kirchhoff's laws, series and parallel circuits, Maxwell's loop current method, Network theorems: superposition theorem - thevenin's theorem - Norton's theorem - maximum power transfer theorem. (9)

AC CIRCUITS: Single phase AC circuits: Average and RMS values of sinusoidal wave form - RLC Circuit - Phasor representation - active, reactive apparent power – power factor, analysis of RLC Circuit, three phase circuit: star and delta connection - phase and line quantities - balance and unbalance systems. (9)

ELECTROMAGNETISM AND MAGNETIC CIRCUITS: Electromagnetic induction; induced currents, Faraday's law, induction and energy, motional emf and Lenz's law. Magnetic field - magnetic circuit - inductance and mutual inductance - magnetic materials – ideal transformers and real transformers. (8)

SEMICONDUCTOR DEVICES: Basic diode concepts - diode circuit: half wave rectifier - full wave rectifier - bridge rectifier-special purpose diodes - zener diode – transistor fundamentals – transistor biasing - bipolar junction transistors - basis amplifier concept - loading effect - power supplies and efficiency. (10)

OPERATIONAL AMPLIFIERS: Op-Amp Basics, Ideal characteristics of Op-Amp, practical Op-Amp circuits, differential and common mode operation, Inverting and non-inverting amplifiers, Op-Amp as Adder, Subtractor, integrator and differentiator. (9)

Total = L: 45

TEXT BOOKS

1. Edward Hughes, John Hiley and Keith Brown, "Electrical and Electronic Technology", Pearson education, 2020
2. Murugesh Kumar K, Basic Electrical Science and Technology", Vikas Publishing House, 2016
3. Theraja B L, "Basic Electronic Solid State", S. Chand & Company Ltd., New Delhi, 2010

REFERENCES

1. Leach DP, "Digital Principles & Applications", Tata McGraw Hill, 2021
2. Hambley A R., "Electrical Engineering Principles and Applications", Pearson education, 2018
3. Boylestad R. L., and Nashelsky L., "Electronic Devices and Circuit Theory", Pearson Education, Noida, 2013

23N103 APPLIED CHEMISTRY

3 0 0 3

ELECTRONIC MATERIALS: Inorganic semiconductors – Elemental – Si and Ge - band theory, doping, compound semiconductors – band gap engineering – applications. Organic semiconductors – conjugated polymers – mechanism of charge transport, doping, states of aggregation, material properties – thermal, mechanical, electrical, chemical, electrochemical.

Applications – OLED, OPV – working principle. Liquid crystalline materials – display application. (9)

PROCESSES IN ELECTRONICS MANUFACTURE: Microchip fabrication – overview, photo resists – chemistry, types. Fabrication facilities – clean rooms - maintenance, ultrapure water– specification, production processes – ion exchange, reverse osmosis, continuous electrode ionisation. PCB fabrication – electroless and electroplating of copper – principle, bath chemistries and process parameters, formation of copper track on plastic board. (9)

ELECTRONICS PACKAGING AND PROTECTION: Packaging materials-encapsulants and underfills - adhesives – chemical types, application methods, factors influencing adhesion, soldering alloys – phase diagrams, lead free alloys, phase change materials for cooling. Conducting inks for printed electronics - metal and carbon based – graphene, CNT– synthesis, structure, electrical properties. Corrosion in electronics – types, protection – vapour phase inhibitors. (9)

ELECTROCHEMICAL POWER SOURCES: Electrochemical cells – emf, electrode potential, dependence of emf on electrolyte concentration – Nernst equation. Batteries **–** performance characteristics. Materials, construction, reactions, characteristics of lechlanche cell, primary lithium batteries, lead - acid battery and lithium-ion batteries. Super capacitors – EDLC – fundamentals, electrode materials, electrolytes, pseudo capacitors– materials. (9)

CHEMICAL SENSORS: Sensors – basic components. Electrochemical sensors- potentiometric transducers – principle, ion-selective electrodes – configurations, response functions and selectivity, applications –potentiometric titrations, water quality monitoring - pH, Hardness, fluoride ion sensors Amperometric transducers – principle, application - glucose biosensors, conductivity sensors – principle – application in conductometric titrations. Colorimetric sensors - Beer-Lambert's law, components, application - determination of ferric ion in water sample. Chemiresistive sensors - principle, application – environmental monitoring – CO_2 sensor. Micro electrodes for sensors – fabrication. (9)

Total L : 45

TEXT BOOKS:

1. Shashi Chawla, "A Textbook of Engineering Chemistry", Dhanpat Rai and Co. , 2005.
2. Cowie J.M.G,ValeriaArrighi "Polymers: Chemistry and Physics of modern materials", CRC Press, 2007.

REFERENCES:

1. Bansi D. Malhotra "Handbook of Polymers in Electronics", Rapra Technology Ltd., 2002.
2. Peter Van Zant "Microchip Fabrication: A Practical Guide to Semiconductor Processing", Mc Graw Hill, 2014.
3. Derek Pletcher and Frank C. Walsh. "Industrial Electrochemistry",Chapman and Hall , 1993.
4. Florinel-Gabriel Banica, " Chemical Sensors and Biosensors – Fundamentals and Applications", John Wiley & Sons Ltd, 2012.

23N104 COMPUTATIONAL THINKING

3 1 0 4

INTRODUCTION: Computational thinking - Logical thinking - Flow charts - Algorithmic thinking - Characteristics of algorithms – Pseudocode - Example problems (9+3)

PROBLEM SOLVING AND DECOMPOSITION: Defining the problem - Devising a solution - Decomposition - Other effective strategies - Patterns - Example problems (9+3)

ABSTRACTION AND MODELING: Generalization - Abstraction - Modeling– Examples (8+3)

ITERATIVE LOGIC: Iterator - Variable - Filtering - Dynamic filtering - Example problems (9+3)

CASE STUDIES: Text processing - Pattern search - Linear search - Sorting (10+3)

Total L: 45 +T: 15 = 60

TEXT BOOKS:

1. Karl Beecher, "Computational Thinking: A Beginner's Guide to Problem Solving and programming", 1st Edition, BCS Learning & Development Limited, 2017.
2. G Venkatesh and Madhavan Mukund, "Computational Thinking: A Primer for Programmers and Data Scientists", 1st Edition, Notion Press, 2021.

REFERENCES:

1. R.G.Dromey, "How to Solve it by Computer", Pearson Education, Second Edition, 2008.
2. Peter j Denning , Matti Tedre, "Computational Thinking", The MIT Press, 2019
3. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", Third Edition, Pearson Education, 2017
4. Peter William Mcowan (Author), Paul Curzon (Author), "Power of Computational Thinking, The: Games, Magic And Puzzles To Help You Become A Computational Thinker ", World Scientific Europe Ltd, 2017

23G105 ENGLISH LANGUAGE PROFICIENCY

3 1 0 4

VOCABULARY: Etymology - Prefixes and suffixes – Synonyms – Antonyms – Guessing meanings from context – Word formation - Single-word substitutes - Different forms of a word – Phrasal verbs – Collocations. (9+3)

LISTENING AND SPEAKING: Understanding listening – Listening techniques - Introducing oneself and others –Seeking and sharing information– Description - Conversation skills – Extempore speaking – Speech practice in varied formal contexts.(9+3)

GRAMMAR: Wh-questions – Yes/no questions – Parts of speech – Articles – Prepositions – Gerunds – Conjunctions - Degrees of comparison – Tenses – Modal verbs – Adverbs - Direct and indirect questions. (9+3)

READING: Reading strategies: Skimming and scanning, Predicting – Reading comprehension: techniques – Practice reading. (9+3)

WRITING: Discourse markers – Dialogue writing - Completing sentences – Jumbled sentences – Paragraph writing – Writing compare & contrast paragraphs – Letter writing. (9+3)

Total: 45 +15 = 60

TEXTBOOKS:

1. Shoba K.N.and Lourdes Joavani Rayen, Communicative English, Cambridge University press, Cambridge, 2021.
2. Raymond Murphy, Intermediate English Grammar, Cambridge University Press, New Delhi, 2020.

REFERENCES:

1. Raymond Murphy, English Grammar in Use, Cambridge University Press, New Delhi, 2020.
2. Sudharshana N P and Savitha C, English for Engineers, Cambridge University Press, New York, 2018.
3. Anne Laws, Writing Skills, Orient Black Swan, Hyderabad, 2011.
4. Helen Naylor with Raymond Murphy, Essential English Grammar, Cambridge University Press, New Delhi, 2019.

23N110 PYTHON PROGRAMMING LABORATORY

0 0 4 2

1. Algorithm and Flowchart
2. Programs using Decision Making statements and Looping Statements
4. Applications using Set, Lists, Tuples, Dictionary
5. Applications using Functions
6. Iterative
7. Text processing
8. Sorting algorithms
9. Linear Searching
10. Application Debugging

Total P: 60

REFERENCES:

1. Romano, Fabrizio, “Learn Python Programming: A Beginner's Guide to Learning the Fundamentals of Python Language to Write Efficient, High-Quality Code”, 2nd Edition. India, Packt Publishing, 2018.
2. R. Nageswara Rao, “Core Python Programming”, Second edition, Dream tech press, 2019
3. Vijay Kumar Sharma, Vimal Kumar, Swati Sharma, Shashwat Pathak, “Python Programming: A Practical Approach”, United States, CRC Press, 2021.
4. Meenu Kohli, “Basic Core Python Programming: A Complete Reference Book to Master Python with Practical Applications”,First edition, BPB Publications, 2021.
5. Cogliati, Josh, “Non-Programmers Tutorial For Python 3”,Platypus Global Media, 2019.

23N111 BASIC SCIENCE LABORATORY

0 0 4 2

Physics (Any eight experiments)

1. Determination of Hysteresis loss of a ferromagnetic material
2. Determination of resistivity of metal and alloy using Carey Foster bridge
3. Determination of capacitance using LCR bridge
4. Study of reverse bias characteristics of Germanium diode and determination of its band gap
5. Study of I-V characteristics of solar cell and determination of its efficiency
6. Thermistor: Measurement of temperature and band gap
7. Study of characteristics of Photo Diode
8. Operational Amp. (741) – Inverting and non inverting modes
9. Determine the Planck’s constant using Photo Electric method.
10. Determine the refractive index of given specimen using Brewster’s angle apparatus.

Demonstration:

1. Determine the excitation potential of Argon using the Franck Hertz method.
2. Determine the number of turns & radius of the coil and magnetic field of ‘Helmholtz’ coil using Biot-Savart’s Law – apparatus
3. Find the Hall voltage, Hall coefficient, mobility & charge density of the given ‘Ge’ crystal using Hall apparatus.
4. Determine e/m (charge to mass ratio) of the electron by Thomson’s method.
5. Determine the charge of an electron by Millikan’s oil drop method.
6. Determine the wavelength of the laser source by Michelson Interferometer.
7. Determine the thickness of the film by Mach Zehnder Interferometer.
8. Find the finesse and free spectral region of the Etalon.
9. Analyze the mercury spectrum and find wavelength using polynomial equation.

Total P: 30

REFERENCES:

1. Department of Physics,” Physics laboratory observation”, 2022.
2. Jerry D Wilson; Cecilia A Hernandez Hall, “Physics laboratory experiments”, Boston, MA: Cengage Learning, 2016.

CHEMISTRY (Any eight experiments)

1. Determination of total hardness of water by EDTA method.
2. Demineralisation of water using ion-exchange resins and determination of water quality - pH, TDS and conductivity.

3. Electroplating of copper and Nickel and determination of cathode efficiency of electroplating.
4. Construction of phase diagram for a simple eutectic system.
5. Demonstration of different forms of metallic corrosion using ferroxyl indicator test and determination of rate of galvanic corrosion using zero-resistance ammeter.
6. Designing a battery and determination of its characteristics.
7. Potentiometric estimation of ferrous iron.
8. Preparation of a chloride ion sensor and ascertaining Nernstian response by construction of calibration curve.
9. Conductometric estimation of a mixture of strong and weak acids.
10. Photocolorimetric estimation of Ferric ion in a water sample.

Total P: 30

REFERENCE:

1. Laboratory Manual Prepared by the Department.

23N112 ENGINEERING PRACTICES

0 0 4 2

1. Study of PC Components
2. Assembling and disassembling of PC.
3. Study of passive and active components (resistors, capacitors, inductors, diodes and transistor).
4. Generation of Signals (DSO, Function generator).
5. Construction of series and parallel circuits using resistors.
6. Study of Digital IC Technologies.
7. Verification of Logic Gates and Implementation of Boolean equations
8. Design and Implementation of Arithmetic Circuits.
9. Design and Implementation of Registers.
10. Design and Implementation of Counters

Total P: 60

REFERENCES

1. Hambley A.R., “Electrical Engineering Principles and Applications”, PHI Learning Pvt. Ltd., New Delhi, 2017.
2. Thomas L. Floyd, “Digital Fundamentals”, Pearson Education, USA, 2015.
3. Wikibooks Contributors,”How to assemble your Desktop PC”, Platypus Global Media, 2011.
4. Govindarajalu B,”IBM PC and CLONES: Hardware, Troubleshooting and Maintenance”, TATA McGraw-Hill Education,2008.

23IP15 INDUCTION PROGRAMME

2 1 0 3

As per AICTE guidelines

Total L: 30

SEMESTER – II

23N201 TRANSFORMS AND ITS APPLICATIONS

3 1 0 4

LAPLACE TRANSFORMS: Laplace transform, inverse transform, linearity, s-shifting, transforms of derivatives and integrals, unit step function, t – shifting, Dirac’s delta function, periodic functions, differentiation and integration of transforms. (9+3)

APPLICATION OF LAPLACE TRANSFORMS: Convolution, solving differential equations with constant coefficients and variable coefficients, integral equations, systems of ODEs by using Laplace transform technique. (9+3)

Z TRANSFORM: Z-transform, relationship of Z-transforms to Laplace transforms, some useful properties, inverse Z-transforms, solution of difference equations. (6+2)

FOURIER SERIES: Fourier series – even and odd functions, half range expansion, convergence of Fourier series, basic concepts of PDE’s, wave equation, solution of one dimensional heat equation and steady state two dimensional heat equation. (12+4)

FOURIER TRANSFORMS: Fourier integral, Fourier cosine and sine integrals, Fourier transform, Discrete Fourier transform, Fast Fourier transform – DIT algorithm. (9+3)

Total L: 45 + T: 15 = 60

TEXT BOOKS

1. Erwin Kreyszig, “Advanced Engineering Mathematics”, Wiley India, New Delhi, 2018.
2. Dean G. D, “Advanced Engineering Mathematics with MATLAB”, CRC Press, USA, 2017.

REFERENCES

1. Peter V.O Neil, “Advanced Engineering Mathematics”, Cengage, New Delhi, 2018.
2. Wylie C. R and Barrett L. C, “Advanced Engineering Mathematics”, Tata McGraw-Hill, New Delhi, 2019.
3. Jain R. K and Iyenger S. R. K, “Advanced Engineering Mathematics”, Narosa Publishing House, New Delhi, 2018.
4. Alexander D Poularikas, “Transforms and applications primer for engineers with examples and MATLAB”, CRC press, USA, 2010.

23N202 DISCRETE MATHEMATICS

3 1 0 4

SETS, RELATIONS AND FUNCTIONS: Introduction to Sets - Set Operations – Computer representation of Sets - Sequences and Summations - Cardinality of Sets - Relations and Their Properties - Closures of Relations - Equivalence Relations - Partial Orderings – Functions: Injective, Surjective, Bijective, and Composition. (9+3)

MATHEMATICAL LOGIC AND PROOFS: Propositional Logic - Applications of Propositional Logic - Propositional Equivalences - Predicates and Quantifiers - Nested Quantifiers - Rules of Inference - Introduction to Proofs: Proof by contraposition, Proof by contradiction. (9+3)

COUNTING AND COMBINATORICS: The Basics of Counting - The Pigeonhole Principle - Permutations and Combinations - Generalized Permutations and Combinations - Generating Permutations and Combinations. (9+3)

RECURRENCE RELATIONS: Recursion – Recurrence relations: Linear First-order recurrence relations – Linear Second order recurrence relations – Divide and conquer recurrence relations. (9+3)

ALGEBRAIC STRUCTURES: Algebraic Systems: properties – Groups, Semi groups, Monoids: Homomorphism of Semi groups and monoids – Sub semi groups and sub monoids – Rings: Structure and Properties. (9+3)

Total L: 45 +T: 15 = 60

TEXT BOOKS:

1. Discrete Mathematics and its Applications - Kenneth H. Rosen 7th Edition -Tata McGraw Hill Publishers, 2007
2. Tremblay J. P. and Manohar R., “Discrete Mathematical structures with application to Computer Science”, Tata McGraw Hill, 2011.

REFERENCES:

1. Ralph. P. Grimaldi, “Discrete and Combinatorial Mathematics: An Applied Introduction”, Pearson Education, Fifth Edition, New Delhi, 2014.
2. Seymour Lipschutz and Mark Lipson,” Discrete Mathematics”, Schaum‘s Outlines, Tata McGraw Hill Pub. Co. Ltd., Third Edition, New Delhi, 2013.
3. Thomas Koshy,” Discrete Mathematics with Applications”, Elsevier Publications, Boston, 2004.

23N203 COMPUTER ORGANIZATION AND ARCHITECTURE

3 0 0 3

DIGITAL CIRCUITS: Binary Arithmetic - Boolean Algebra - Basic Theorems and Properties of Boolean Algebra - Simplification of Boolean Functions - Digital Logic Gates - Karnaugh Map Method - Design of Combinational Circuits - Flip-Flops - Design of Sequential Circuits. (10+3)

COMPUTER ORGANIZATION & PROCESSOR DESIGN: Stored program organization (Von Neumann architecture) - Computer Registers -Stack organization - Instruction Formats - Addressing modes - RISC Vs CISC - Quantitative Principles of computer design. (9+3)

MEMORY AND I/O SYSTEMS: Memory Hierarchy - Associative Memory - Cache Memory - Mapping policies – Cache optimization; I/O Systems: Introduction - Interrupts - Modes of Transfer - DMA. (9+3)

PARALLELISM: Pipelining - Pipelining Hazards - Overcoming Hazards - Instruction Level Parallelism – Dependencies (9+3)

MULTIPROCESSOR SYSTEMS: Symmetric and Distributed shared memory architectures - Challenges – Cache Coherence – Snooping protocol (8+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. M. Morris Mano, Michael D. Ciletti "Digital Design: With an Introduction to the Verilog HDL, VHDL and System Verilog”, Pearson Education., USA, 2018, Sixth Edition
2. John L. Hennessey, David A. Patterson, "Computer Architecture: A Quantitative Approach", Elsevier India Pvt. Ltd, New Delhi, 2015.

REFERENCES:

1. Morris Mano, "Computer System Architecture", Prentice Hall of India, Prentice Hall of India, 2007.
2. Hennessy, John L., and Patterson, David A. “Computer Organization and Design MIPS Edition: The Hardware/Software Interface. Netherlands, Elsevier Science, 2020”
3. Thomas L. Floyd "Digital Fundamentals", Pearson Education., USA, 2015
4. Carl Hamacher, "Computer Organization", Tata McGraw Hill Publishing, New Delhi, 2002.
5. William Stallings, "Computer Organization and Architecture", Pearson Education / Prentice Hall of India, New Delhi, 2006.

23N204 C PROGRAMMING

3 0 0 3

C PROGRAMMING BASICS: Introduction to C programming - C Program Structure - Program Compilation & Execution - Character Set - Keywords - Data Types - Library functions - Control Statements: Branching and Looping. (6)

ARRAYS AND STRINGS: Single Dimensional Array, Two Dimensional Arrays, Multi-dimensional Array: Initialization, Unsized Array Initialization. - Strings: Defining, initializing of strings, Processing strings, Array of Strings - (8)

FUNCTIONS: Definition of Function - Prototypes - Storage Classes - Scope Rules - Recursion - Command Line Argument. (9)

POINTERS: Pointer Declaration - Operations on Pointer - Passing Pointers to a Function - Calling Function: Call by Value, Call by Address - Return Statement - Passing Arrays to Function - Pointers and One Dimensional, two dimensional Array - Array of Pointers - Dynamic Memory Allocation. (11)

STRUCTURES, UNION AND FILES: Definition - Processing a Structure - Typedef - Array of Structure, and Pointer to Structure - Passing Structure to Functions. - Self-Referential Structures - Nested Structures - Union - Introduction to Files - File Access - File Organization – File Operations. (11)

Total L: 45

TEXT BOOKS:

1. Byron S. Gottfried, Jitendar Kumar Chhabra "Programming with C", Tata McGraw Hill Publishing Company., New Delhi, 2018., fourth edition
2. Kernighan B. W., Ritchie D. M. "C Programming Language (ANSI C)", Prentice Hall of India Private Limited., New Delhi, 2010, second edition

REFERENCES:

1. Herbert Schildt "C – The Complete Reference", Tata McGraw Hill Publishing Company., New Delhi, 2010, fourth edition
2. Pradip Dey and Manas Ghosh "Programming in C", Oxford University Press., New Delhi, 2018
3. Yashavant P. Kanetkar "Let Us C", BPB Publications., 2017, 16th edition
4. H. M. Deitel, P. J. Deitel "C How to Program", Pearson Education., New Delhi, 2013, seventh edition

23N211 HARDWARE PROGRAMING LABORATORY

0 0 4 2

1. Design of combinational circuits
2. Design of sequential logic circuits
3. Arithmetic Operations in 8086 microprocessor
4. Searching and Sorting in 8086 microprocessor
5. Recursion in 8086 microprocessor
6. String Operations in 8086 microprocessor
7. Study of ARM programming environment
8. Interfacing LED and Switch with ARM
9. Interfacing 4x4 Matrix Keypad with ARM
10. Interfacing the LCD with ARM

Total P: 60

TEXT BOOKS

1. Morris Mano M, Michael D Cilettim, “Digital Design”, Pearson Education, New Delhi, 2008
2. Walter A Triebel, Avtar Singh, "The 8088 and 8086 Microprocessors – Programming, Interfacing, software, Hardware and Applications", Pearson Education, New Delhi, 2009.

REFERENCES

1. M. Morris Mano, Michael D. Ciletti "Digital Design: With an Introduction to the Verilog HDL, VHDL and System Verilog”, Pearson Education., USA, 2018, Sixth Edition
2. Ray A K, Bhurchandi K M , "Advanced Microprocessors and Peripherals", Tata McGraw-Hill Publishing Company Limited,New Delhi, 2009.
3. Steve Furber , "ARM System-On-Chip Architecture", Pearson Education Limited, USA, 2010.
4. John F. Wakerly. Digital Design: Principles and Practices USA: Pearson Education, 2018.

23N212 C PROGRAMMING LABORATORY

0 0 4 2

1. Formatted I/O statements.
2. Decision Making statements: Simple If, If – else, Switch- case.
3. Looping Statements: For, While, Do – while.
4. Single dimensional arrays and multi-dimensional arrays.
5. Operations on Strings.
6. Pass by value and pass by address, Recursion using functions.
7. Structures and nested structures.
8. String handling operations using pointers.
9. Operations on arrays using pointers.
10. File operations using command line arguments

Total P: 60

REFERENCES:

1. Byron S. Gottfried, Jitendar Kumar Chhabra, "Programming with C", Tata McGraw Hill Publishing Company., New Delhi, 2018,Fourth edition
2. Herbert Schildt, "C – The Complete Reference", Tata McGraw Hill Publishing Company., New Delhi, 2010,Fourth edition
3. PradipDey and Manas Ghosh, "Programming in C", Oxford University Press., New Delhi, 2018
4. Yashavant P. Kanetkar, "Let Us C", BPB Publications., 2017,Sixteenth edition
5. H. M. Deitel, P. J. Deitel, "C How to Program", Pearson Education., New Delhi, 2013,Seventh edition

23N214 ENGINEERING GRAPHICS

0 0 2 1

INTRODUCTION: (8)

1. Introduction to Engineering Drawing.
2. Lettering and Line Practice
3. Dimensioning practice as per BIS conventions
4. Geometrical construction

THEORY OF ORTHOGRAPHIC PROJECTION: (10)

5. Projection of points
6. Projection of Straight lines
7. Projection of Planes
8. Projection of Solids
9. Orthographic projection of simple engineering components

ISOMETRIC PROJECTION: (6)

10. Construction of Isometric to Orthographic Projection
11. Construction of Orthographic to Isometric Projection

MODELLING OF 3D SOFTWARE: (6)

12. Modeling of simple engineering components using CAD software

Total P: 30

TEXT BOOKS:

1. Venugopal K, Prabhu Raja V "Engineering Graphics", New Age International Publishers., Chennai, 2015
2. Natarajan K. V "Engineering Drawing and Graphics", M/s Dhanalakshmi N., Chennai, 2016

REFERENCES:

1. K.C. John "Engineering Graphics for Degree", PHI Learning Pvt Ltd.., New Delhi, 2011 , 1st Edition
2. Bhatt N. D., Panchal V. M. "Elementary Engineering Drawing", Charotar Publishing House., 2014
3. Warren J Luzadder, Jon M Duff "Fundamentals of Engineering Drawing: with an Introduction to Interactive Computer Graphics for Design and Production", PHI Learning., New Delhi, 2011
4. Bureau of Indian Standards "Engineering Drawing Practices for Schools and Colleges SP 46-2003", BIS., New Delhi, 2003

23Q213 FOUNDATIONS FOR PROBLEM SOLVING

0 0 2 0

PROBLEM SOLVING:

1. Speed Mathematics (SAW, Oz, Mirror methods)
2. Speed Mathematics (High5, Minion, Butterfly methods)
3. Speed Mathematics (Inception, Goldeneye methods)
4. Thinking with Numbers
5. Problem Solving with Visual information
6. Words Puzzles
7. Resume Writing Essential

Total P: 30

REFERENCES:

1. R.S. Aggarwal, “Quantitative Aptitude for Competitive Examination”, S Chand Publishing, New Delhi, 2017.

SEMESTER – III

23N301 LINEAR ALGEBRA

3 1 0 4

LINEAR EQUATIONS: Introduction to systems of linear equations - gauss elimination - linear systems and invertible matrices. (9+3)

VECTOR SPACES: General vector spaces- real vector spaces - Euclidean n-space - subspaces - basis and dimension - row space, column space and null space - rank and nullity. (9+3)

LINEAR TRANSFORMATIONS: General linear transformation - kernel and range - matrices of linear transformations - change of basis - geometry of linear operators on R^2. (9+3)

INNER PRODUCT SPACES: Inner products- angle and orthogonality in inner product spaces, orthonormal bases, Gram Schmidt process, QR decomposition, best approximation- least squares. (9+3)

EIGENVALUES AND EIGENVECTORS: Eigen values and eigen vectors, diagonalization- orthogonal diagonalization - spectral decomposition- singular value decomposition- principal component analysis - discrete dynamical systems. (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Howard Anton and Chris Rorres, "Elementary Linear Algebra", Wiley India, New Delhi, 2018.

2. Gilbert Strang, "Linear Algebra and its Applications", Cengage, New Delhi, 2012.

REFERENCES:

1. Gareth Williams, "Linear Algebra with Applications", Narosa Publishing House, New Delhi, 2012.
2. David C Lay, "Linear Algebra and its Applications", Pearson, New Delhi, 2016.
3. Friedberg, Insel and Spence, "Linear Algebra", Pearson Education, USA, 2015.
4. Kenneth Hoffman and Ray Kunze, "Linear Algebra", Prentice Hall, New Jersey, 2015.

23N302 PROBABILITY, STOCHASTIC PROCESSES AND STATISTICS

3 1 0 4

PROBABILITY AND DISCRETE RANDOM VARIABLES: Probability, axioms, Conditional Probability, Partitions and the Law of TOTAL Probability, Baye's theorem, Independence, discrete random variables-definitions, probability mass function, families of discrete random variables - binomial, Poisson and geometric random variables, cumulative distribution functions, expectations. (9+3)

CONTINUOUS RANDOM VARIABLES: Continuous Sample Space, cumulative distribution functions, probability density function, families of continuous random variables - uniform, exponential, Erlang and Gaussian random variables, expectations. (9+3)

MULTIPLE RANDOM VARIABLES: Joint cumulative distribution function – joint probability mass function – marginal probability mass function - joint probability density function - marginal probability density function - independent random variables- expected values– Covariance, Correlation and Independence. (9+3)

STOCHASTIC PROCESSES: Definitions and Examples - Random Variables from Random Processes - Independent, Identically Distributed Random Sequences - The Poisson Process-Properties of the Poisson Process - Markov Process - Discrete-Time Markov Chains - Higher Transition Probabilities: Chapman–Kolmogorov Equations - Long-Run Behavior of Markov Chains. (9+3)

STATISTICAL INFERENCE: Foundations for inference - Variability in estimates - Confidence intervals - Hypothesis testing - Central Limit Theorem - Inference for numerical data - t-distribution - ANOVA and the F Test - Inference forcategorical data - Inference for a single proportion - Difference of two proportions - Testing for goodness of fit using chi-square - Testing for independence in two-way tables. (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Roy D Yates and David J Goodman, "Probability and Stochastic Processes", Third Edition, Wiley India, New Delhi, 2021.
2. David M Diez, Christopher D Barr, Mine Cetinkaya - Rundel, “Open Intro Statistics”, 3rd Edition, Creative Commons license, 2017

REFERENCES:

1. Saeed Ghahramani, "Fundamentals of Probability with Stochastic Processes", CRC Press, Taylor & Francis Group, USA, 2018.
2. Douglas C Montgomery and George C Runger, "Applied Statistics and Probability for Engineers", Wiley India, New Delhi, 2018.
3. Ronald E. Walpole, Raymond H Myers, Sharon L Myers and Keying Ye, "Probability and Statistics for Engineers and Scientists", Pearson, New Delhi, 2016.
4. David Spiegelhalter, "The Art of Statistics: How to Learn from Data”, Pelican Books, 2020.
5. Michael J. Evans, Jeffrey S. Rosenthal,” Probability and Statistics: The Science of Uncertainty”, second Edition, WH Freeman, 2010.

23N303 DATA STRUCTURES

3 0 0 3

INTRODUCTION: Need for Data Structures - Types of Data Structures - Abstract Data Type –Algorithm Complexity Analysis - Best case and worst-case complexities - Asymptotic notations (6)

ARRAYS AND LISTS: Array Representation and Operations - Matrix representation using Multi-dimensional arrays - Linked List Representation - Operations on a Singly Linked List - Types of Linked List - Polynomial Addition - Sparse Matrices (11)

STACKS AND QUEUES: Stack ADT - Representation and Operations - Expression Handling - Role of Stack in implementing recursive algorithms - Queue ADT - Representation and Operations - Types of Queues - Circular Queue - Deque - Priority Queue (8)

TREES: Terminologies - Binary Tree - Traversal - Expression Trees - Binary Heap - Priority Queue implementation using Binary Heap - Binary Search Tree - AVL Tree - m-way search trees - B Tree - B+ Tree - Applications - Trie Structure (11)

HASHING AND GRAPHS: - Hash Table - Hash Functions - Resolving Collisions - Rehashing. GRAPHS**:** Graph Terminologies - Types of Graphs - Representation - Breadth First Search - Depth First Search - Topological Sort. (9)

TOTAL L: 45

TEXT BOOKS:

1. Mark Allen Weiss, "Data Structures and Algorithm Analysis in C", 2nd Edition, Pearson Education, 2016.

2. Venkatesan R, Lovelyn Rose S, "Data Structures", 2nd Edition, Wiley India Pvt Ltd, 2019.

REFERENCES:

1. Jean Paul Tremblay, Sorenson, "An Introduction to Data Structures with Applications", McGraw Hill Publishing Company, New Delhi, 2017.
2. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "Fundamentals of Computer Algorithms", 2nd Edition, Universities Press, 2019.
3. Salaria R S, "Data Structures and Algorithms using C", 5th Edition, Khanna Book Publishing, New Delhi, 2017.
4. Amol M. Jagtap, Ajit S. Mali, "Data Structures Using C - A Practical Approach for Beginners", Chapman and Hall / CRC; 1st edition , 2021
5. Aaron M Tanenbaum, Moshe J Augenstein and Yedidyah Langsam, "Data structures using C and C++", Prentice Hall, 2016.

23N304 SOFTWARE ENGINEERING

3 0 0 3

PRELIMINARIES: Definition of Software – Software characteristics – Types of Software – Evolution of Software – Software Development Lifecycle – Waterfall Model – Incremental Model – Prototyping – Spiral Model – Unified Process – Agile Development Approach – Software Myths. (8)

REQUIREMENTS ENGINEERING: Requirements elicitation – Functional and Non-functional Requirements – Prioritization – Use cases – Use case diagram — Data flow diagrams – UML – Object Model – Class diagram – State diagram – Sequence diagram – Activity diagram – CRC cards – Software Requirements Specification document (SRS) (12)

SOFTWARE DESIGN: Architectural Design: Views and Viewpoints – Styles and Patterns – Layered Architecture. Detailed Design: Functional Decomposition – Object Oriented Design – User Interface Design – Good Design Attributes – Design Specifications – Coding Standards (8)

SOFTWARE TESTING: Testing fundamentals: Black Box and White Box Testing – Test Cases – Equivalence Partitioning – Boundary Value Analysis – Basis Path Testing – Cyclomatic Complexity. Testing Strategies: Unit Testing – Integration Testing – System Testing – Acceptance Testing. Special Testing: Regression Testing – Smoke Testing – Stress Testing - User Interface Testing – Test Automation – Test Documenting and Reporting – Testing Object Oriented System (8)

SOFTWARE ENGINEERING MANAGEMENT: Software Quality: Views of Quality – Quality Attributes – Cost of Quality – Quality Control vs. Quality Assurance – Formal Technical Reviews – Guidelines for Reviews – Quality Metrics. Software Configuration Management: Software Baselines – Version Control – Change Control – Software Configuration Audit. Software Maintenance: Maintenance Phase – Activities – Reengineering – Metrics (9)

TOTAL L: 45

TEXT BOOKS:

1. Roger S Pressman and Bruce Maxim, "Software Engineering - A Practitioner's Approach", McGraw Hill International Edition, Singapore, 2020.
2. Pankaj Jalote, "A Concise Introduction to Software Engineering", Springer, New Delhi, 2011

REFERENCES:

1. Ian Sommerville, "Software Engineering", Pearson Addison Wesley, Boston, 2017
2. Shari Lawrence Pfleeger, "Software Engineering: Theory and Practices", Pearson Education, New Delhi, 2009
3. Orlando Karam, Frank Tsui, "Essentials of Software Engineering", SBS Publishers and Distributors (P) Ltd, 2007 New Delhi.

23O305 ENGINEERING ECONOMICS

3 1 0 4

Fundamentals of Economic Analysis: Introduction – Scope of Engineering Economics – Circular Flow in Economy – Demand and Supply– Types of Efficiency. (7+2)

Cost and Inventory Control: Cost Concepts – Types of Cost – Short-run and Long-run - Cost-Output Relationship - Make or Buy Decision – Criteria – Approaches - Inventory Models – Purchase Model – Manufacturing Model - With or Without Shortages- Depreciation Methods – Problems in Straight Line and Diminishing Balance Method. (9+4)

Pricing, Revenue and Value Engineering: Pricing Practices and Strategies -Revenue Concepts – CVP Analysis- Problems in Break-Even Analysis – Value Analysis - Value Engineering. (9+2)

Project Management: Capital Budgeting – Decisions – Steps Involved in Capital Budgeting – Methods of Project Appraisal – Pay-back Period – Net Present Value and Internal Rate of Return - Project Management - Techniques – PERT – CPM Models – Case Analysis. (10+5)

Economic Growth and Development: Concepts of Macro Economics – National Income – Inflation – Control Measures - Monetary Policy – Fiscal Policy – Technological Innovation in Banking and Economic Development - Sustainable Development Goals – Circular Economy. (10+2)

Total L: 45 + T: 15 =60

TEXT BOOKS

1. Panneerselvam. R., "Engineering Economics", PHI Learning Private Limited, 2012.

2. Metha P.L, “Managerial Economics – Analysis, Problems and Cases”, Sultan Chand & Sons, 2016.

REFERENCES

1. Zahid A. Khan., Arshad N. Siddiquee, Brajesh Kumar, Mustufa H. Abidi, “Principles of Engineering Economics with Applications”, Cambridge University Press, 2018.
2. Varshney, R.L and K.L. Maheshwari, “Managerial Economics”, Sultan Chand & Sons, 2014.
3. McEachern and Indira., “Macro ECON”, Cengage India Private Limited, 2017.
4. Shalini Goyal Bhalla, “Circular Economy (Re) Emerging Movement”, Invincible Publishers, 2020.

23N310 OBJECT ORIENTED PROGRAMMING LABORATORY

0 0 4 2

1. Study of JDK,JRE,JVM, IDE and REPL
2. Classes and Methods
3. Inheritance
4. Interfaces and packages
5. Exception Handling
6. I/O and Files
7. Collection classes
8. Generics and Lambda Expressions

TOTAL P: 60

REFERENCES:

1. Schildt H, "Java: The Complete Reference", Eleventh Edition, McGraw-Hill Education, 2018.
2. Deitel P and Deitel H, "Java: How to Program", Eleventh Edition, Prentice Hall, 2018.
3. Hortsmann CS and Cornell G, "Core Java: Volume 1 - Fundamentals", Eleventh Edition, Prentice Hall, 2018.
4. James Gosling, Bill Joy, Guy Steele, Gilad Bracha, Alex Buckley and Daniel Smith, "The Java Language Specification – Java SE", Thirteenth Edition, Oracle America Inc., USA, 2019.
5. Daniel Liang L, "Introduction to Java Programming", Tenth Edition, Pearson Education, New Delhi, 2015.
6. Matt Weisfeld, "The Object Oriented Thought Process", Fifth Edition, Addison-Wesley Professional, US, 2019.

23N311 DATA STRUCTURES LABORATORY

0 0 4 2

1. Solving Problems using arrays
2. Searching and Sorting algorithms
3. Implementation of Linked List
4. Applications of Linked List
5. Implementation Stack and queue
6. Applications of Stack
7. Operations on Binary Search Trees
8. Applications of Binary Search Tree, AVL tree
9. Graphs - Depth First Search and Breadth First Search
10. Hashing and Collision Resolution

TOTAL P: 60

REFERENCES:

1. Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran, “Fundamentals of Computer Algorithms”, Second Edition, Universities Press, 2011.
2. Jean Paul Tremblay and Sorenson, “An Introduction to Data Structures with Applications”, McGraw Hill Publishing Company, New Delhi, 2012.
3. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest and Clifford Stein, “Introduction to Algorithms”, MIT Press, England, 2009.
4. Salaria R S, "Data Structures and Algorithms using C", 5th Edition, Khanna Book Publishing, New Delhi, 2017.
5. Amol M. Jagtap, Ajit S. Mali, ”Data Structures Using C - A Practical Approach for Beginners”, Chapman and Hall/CRC; 1st edition , 2021

23Q313 BUILDING COMMUNICATIONS SKILLS

0 0 2 1

1. Speed Mathematics (SAW, Oz, Mirror methods)
2. Speed Mathematics (High5, Minion, Butterfly methods)
3. Speed Mathematics (Inception, Goldeneye methods)
4. Thinking with Numbers
5. Problem Solving with Visual information
6. Words Puzzles
7. Resume Writing Essential

TOTAL P: 30

REFERENCES:

1. R.S. Aggarwal, “Quantitative Aptitude for Competitive Examination”, S Chand Publishing, New Delhi, 2017.

23K312 ENVIRONMENTAL SCIENCE

2 0 0 0

INTRODUCTION TO ENVIRONMENT: Environment - Definition, scope and importance. Types and composition of atmosphere - particles, ions and radicals. Ozone layer - significance, formation and depletion. Ecosystems- Structure and functions, components, energy flow, food chains, food web, Biodiversity - levels, values and threats - India as a mega - diversity nation -hotspots of biodiversity -endangered and endemic species of India - conservation of

biodiversity. (6)

ENERGY RESOURCES: Introduction - National and International status- exploitation - sustainable strategies - Fossil fuels- classification, composition, physio - chemical characteristics and energy content of coal, petroleum and natural gas; solar energy - introduction, harnessing strategies. Wind energy - availability, wind power plants, wind energy conversion systems, site characteristics, and types of wind turbines. Supporting renewable energy resources – tidal – geothermal- hydroelectric. (6)

ENVIRONMENTAL POLLUTION: Definition - Sources, causes, impacts and control measures of (a) Air pollution (b) Water pollution (c) Soil pollution (d) Marine pollution (e) Noise pollution (f) Thermal pollution (g) Nuclear hazards (h) RF hazards - Role of an individual in prevention of pollution. Disaster Management: Floods, earthquake, cyclone and landslides – Case studies, consequences and rescue measures. (6)

WASTE MANAGEMENT: Waste water - Characteristics of domestic and industrial waste water - COD and BOD –Various stages of treatment - primary, secondary, tertiary treatment – Biological and advanced oxidation processes. Solid waste management– Characteristics of municipal solid waste (MSW), biomedical, automobile and e-wastes and their management –landfills, incineration, pyrolysis, gasification and composting. (6)

SOCIAL ISSUES AND THE ENVIRONMENT: Environmentally Sustainable work practices –Rain water harvesting–Role of non-governmental organizations. Human ethics and rights –impact on environment and human health - role of information technology on environment and human kind. Green IT policies, Process of EIA - ISO14000. Legislation- Environment protection act - Air (Prevention and Control of Pollution) act - Water (Prevention and control of Pollution) act –Wild life protection act –Forest conservation act. (6)

TOTAL L: 30

TEXT BOOKS

1. Gilbert M. Masters, "Introduction to Environmental Engineering and Science", Pearson Education, New Delhi, 2004.
2. De A K , "Environmental Chemistry", New Age International P Ltd, New Delhi, 2006.

REFERENCES

1. Benny Joseph, "Environmental Science and Engineering", Tata McGraw-Hill, New Delhi, 2006.
2. Koteswara Rao MVR, "Energy Resources: Conventional & Non-Conventional", BSP Publications, New Delhi, 2006.
3. Deswal S and Deswal A,"A Basic Course in Environmental Studies", Dhanpat RaiandCo, NewDelhi, 2004.

23TC01 தமிழர் மரபு / HERITAGE OF TAMILS

1 0 0 1

அலகு – I மொழி மற்றும் இலக்கியம்: இந்திய மொழிக்குடும்பங்கள் – திராவிட மொழிகள் – தமிழ் ஒரு செம்மொழி – தமிழ் செவ்விலக்கியங்கள் – சங்க இலக்கியத்தின் சமயச் சார்பற்ற தன்மை – சங்க இலக்கியத்தில் பகிர்தல் அறம் – திருக்குறளில் மேலாண்மைக்கருத்துக்கள் – தமிழ்க்காப்பியங்கள், தமிழகத்தில் சமண பௌத்த சமயங்களின் தாக்கம் – பக்தி இலக்கியம், ஆழ்வார்கள் மற்றும் நாயன்மார்கள் – சிற்றிலக்கியங்கள் – தமிழில் நவீன இலக்கியத்தின் வளர்ச்சி – தமிழ் இலக்கிய வளர்ச்சியில் பாரதியார் மற்றும் பாரதிதாசன் ஆகியோரின் பங்களிப்பு. (3)

அலகு – II மரபு – பாறை ஓவியங்கள் முதல் நவீன ஓவியங்கள் வரை – சிற்பக்கலை: நடுகல் முதல் நவீன சிற்பங்கள் வரை – ஐம்பொன் சிலைகள் – பழங்குடியினர் மற்றும் அவர்கள் தயாரிக்கும் கைவினைப் பொருட்கள், பொம்மைகள் – தேர் செய்யும் கலை – சுடுமண் சிற்பங்கள் – நாட்டுப்புறத் தெய்வங்கள் – குமரிமுனையில் திருவள்ளுவர் சிலை – இசைக்கருவிகள் – மிருதங்கம், பறை, வீணை, யாழ், நாதஸ்வரம் – தமிழர்களின் சமூக பொருளாதார வாழ்வியல் கோவில்களின் பங்கு. (3)

அலகு – III நாட்டுப்புறக்கலைகள் மற்றும் வீர விளையாட்டுகள்: தெருக்கூத்து, கரகாட்டம், வில்லுப்பாட்டு, கணியான் கூத்து, ஒயிலாட்டம், தோல்பாவைக்கூத்து, சிலம்பாட்டம், வளரி, புலியாட்டம், தமிழர்களின் விளையாட்டுகள். (3)

அலகு - IV தமிழர்களின் திணைக் கோட்பாடுகள்: தமிழகத்தின தாவரங்களும், விலங்குகளும் – தொல்காப்பியம் மற்றும் சங்க இலக்கியத்தில் அகம் மற்றும் புறக்கோட்பாடுகள் – தமிழர்கள் போற்றிய அறக்கோட்பாடு – சங்க காலத்தில் தமிழகத்தில் எழுத்தறிவும், கல்வியும் – சங்ககால நகரங்களும் துறைமுகங்களும் – சங்ககாலத்தில் ஏற்றுமதி மற்றும் இறக்குமதி – கடல் கடந்த நாடுகளில் சோழர்களின் வெற்றி. (3)

அலகு - V இந்திய தேசிய இயக்கம் மற்றும் இந்திய பண்பாட்டிற்குத் தமிழர்களின் பங்களிப்பு: இந்திய விடுதலைப் போரில் தமிழர்களின் பங்கு – இந்தியாவின் பிறப்பகுதிகளில் தமிழ்ப்பண்பாட்டின் தாக்கம் – சுயமரியாதை இயக்கம் – இந்திய மருத்துவத்தில், சித்தமருத்துவத்தின்பங்கு – கல்வெட்டுகள், கையெழுத்துப்படிகள் – தமிழ்ப்புத்தகங்களின் அச்சுவரலாறு. (3)

TOTAL L: 15

TEXT- CUM - REFERENCE BOOKS

1. தமிழக வரலாறு – மக்களும் பண்பாடும் – கே.கே.பிள்ளை (வெளியீடு – தமிழ்நாடு பாடநூல் மற்றும்

கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ் – முனைவர் இல.சுந்தரம். (விகடன் பிரசுரம்)
3. கீழடி – வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருனை – ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils – The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.)
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D.Thirunavukkarasu) (Published by: International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies).
9. Keeladi – 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by : The Author)
11. Porunai Civilization (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigal (R.Balakrishnan) (Published by: RMRL) – Reference Book.

SEMESTER – IV

23N401 FOUNDATIONS OF OPTIMIZATION TECHNIQUES

3 1 0 4

INTRODUCTION: Components of Optimization Problem – Terminology - Classes of Mathematical Programs - Linear Programming Models - Real-world problems – Linear Programming Formulations - Changing Form - Linearization of Piecewise Linear Functions. (9+3)

ITERATIVE ALGORITHMS: Iterative Search and Constructive Algorithms - Improving Directions and Optimality - Computational Complexity and Correctness - Convexity - Convex Sets - Convex and Concave Functions. (9+3)

GEOMETRY AND ALGEBRA OFLP & DUALITY THEORY: Geometry and Algebra of LPs - Extreme Points and Basic Feasible Solutions - Optimality of Extreme Points - Linear Programs in Canonical Form - Optimality Conditions - Duality Theory - Dual of a Linear Program - Duality Theorems - Complementary Slackness - Lagrangian Duality - Farkas' Lemma and Optimality. (9+3)

SIMPLEX METHOD & APPLICATIONS OF DUALITY: Simplex Method From a Known Feasible Solution - Degeneracy and Correctness - Finding an Initial Feasible Solution - Computational Strategies and Speed - Dual Simplex Method - Network Simplex Method - Primal-Dual Interior Point Method (9+3)

CONVEX PROGRAMMING: KKT Optimality Conditions - Lagrangian Duality - Convex Optimization Models - Separable Programs - Unconstrained Optimization - Quadratic Programming - Primal-dual Interior Point Method (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:
1. Michael H Veatch, "Linear and Convex Optimization", John Wiley and Sons, Inc, 2021.
2. Richard W Cottle, Mukund N Thapa, "Linear and Nonlinear Optimization", Springer Science+Business Media, LLC 2017.

REFERENCES:
1. Hamdy A Taha, "Operations Research – An Introduction", Pearson Education Limited, 2017.
2. Stephen Boyd, Lieven Vandenberghe, "Convex Optimization", Cambridge University Press, 2004.
3. David G Luenberger, Yinyu Ye, "Linear and Nonlinear Programming", Springer Nature Switzerland, 2021.
4. Edwin K P Chong, Stanislaw H Zak, "Introduction to Optimization", Wiley India, 2017.
5. Charu C Aggarwal, "Linear Algebra and Optimization for Machine Learning", Springer Nature Switzerland, 2020

23N402 DATABASE MANAGEMENT SYSTEMS

3 0 0 3

INTRODUCTION TO DATABASE: Purpose of Database System - Data Models - Database System Architecture – DBMS ComponentsConceptual design withEntity-Relationship model - Enhanced-ER Model -Traps (9)

RELATIONAL MODEL: E-R to relational model - Codd's rules- Integrity constraints- Relational Algebra- SQL Basics. Normalization: Armstrong rules - Functional Dependencies –First Normal Form to Fifth Normal Form - Denormalization (9)

DATA STORAGE AND QUERYING: RAID - File Organization - Organization of Records in Files – Indexing and Hashing – Ordered Indices - B+ tree Index Files - B tree Index Files - Static Hashing - Dynamic Hashing - Query Processing stages – Query Optimization using Heuristics and Cost Estimation (9)

TRANSACTION MANAGEMENT: Transaction Concepts - ACID Properties - Schedules - Serializability - Concurrency Control - Need for Concurrency - Locking Protocols - Two Phase Locking - Deadlock - Transaction Recovery-Savepoints-Isolation Levels –SQL Facilities for Concurrency and Recovery-Backup and Recovery System. (9)

ADVANCED TOPICS: Parallel Database - Multidimensional Database – Datawarehouse-Multimedia Database - Web Database - Mobile Database- NoSQL database. (9)

TOTAL L: 45

TEXT BOOKS:

1. Abraham Silberchatz, Henry F Korth and Sudarshan S, “Database System Concepts”, Seventh Edition,Tata McGraw-Hill, New Delhi, 2021.
2. Ramez Elmasri and Shamkant B Navathe, “Fundamentals of Database Systems”, Seventh Edition, Addison Wesley, USA, 2010.

REFERENCES:

1. Raghu Ramakrishnan and Johannes Gehrke, “Database Management Systems”, Fourth edition, Tata McGraw-Hill, New Delhi, 2008.
2. Gupta G K, “Database Management System”, Tata McGraw-Hill, New Delhi, 2011.
3. Jeffrey A. Hoffer, Ramesh Venkataraman, Heikki Topi , "Modern Database Management", 12th Edition, Pearson, New Delhi, 2016.
4. Atul Kahate, "Database Managment Systems", 1st Edition, Pearson, New Delhi, 2011.

23N403 DESIGN AND ANALYSIS OF ALGORITHMS

3 1 0 4

DIVIDE AND CONQUER: Introduction to Algorithm Design techniques - Divide and Conquer Methodology - Solving recurrence relations - Finding Maximum and Minimum Element - Quick sort - Merge sort - Convex Hull. (9+3)

GREEDY METHOD: Greedy Strategy - Knapsack Problem - Minimum Spanning Trees - Single Source Shortest Path Method - Huffman Trees (9+3)

DYNAMIC PROGRAMMING: Principle of Optimality - Knapsack Problem - All Pairs Shortest Path - Optimal Binary Search Tree - Multistage Graphs (9+3)

BACKTRACKING: State Space Tree - Knapsack Problem - The Eight Queens Problem - Sum of subsets - Graph Coloring (9+3)

BRANCH AND BOUND: Bounding Functions - 0/1 Knapsack Problem - Traveling Sales Person Problem - Assignment Problem (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Anany Levitin, "Introduction to the Design and Analysis of Algorithms", 3rd Edition, Prentice Hall of India, New Delhi, 2017.
2. Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran, "Fundamentals of Computer Algorithms", Galgotia Publications, New Delhi, 2010.

REFERENCES:

1. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein, "Introduction to Algorithms", 3rd Edition, MIT Press, England, 2009.
2. Donald E. Knuth, —The Art of Computer Programming‖, Volumes 1& 3 Pearson Education, 2 009.
3. Jeffrey J McConnell, "Analysis of Algorithms", Jones and Bartlett Publishers, 2008.
4. Parag Himanshu Dave, Himanshu Bhalchandra Dave, "Design and Analysis of Algorithms", Pearson Education, 2008.

23N404 OPERATING SYSTEMS

3 1 0 4

INTRODUCTION: Operating System Concepts - System Calls - Services - User Operating System Interface - Design and Implementation- shell scripting basics - Introduction to Virtual Machines. (6+3)

PROCESS MANAGEMENT: Process Model - Creation – Termination - Hierarchies - States - Implementation - Scheduling Criteria - Scheduling Algorithms - Multithreading Models - Thread Libraries - Threading Issues - Thread and Multiprocessor Scheduling Algorithms - Interprocess Communication. (10+3)

PROCESS SYNCHRONIZATION AND DEADLOCKS: Race Conditions - Critical Section - Mutual Exclusion - Peterson’s Solution – **Synchronization:** Hardware - Semaphores - Mutex - Monitor - Message Passing, Dining Philosophers Problem - Readers Writers Problem. **Deadlocks:** Conditions - Detection - Recovery - Prevention - Avoidance. (11+3)

MEMORY MANAGEMENT: Main Memory: Swapping - Contiguous Memory Allocation - Paging - Structure of Page Table - Segmentation - Examples. **Virtual Memory:** Demand Paging - Copy on Write - Page Replacement - Allocation of Frames - Thrashing - Memory Mapped Files - Allocating Kernel Memory (10+3)

STORAGE MANAGEMENT: Files: Naming - Structure - Types - Access - Attributes - Operations - Implementation. Directories: Operations - Path Names - Hierarchical Directory System - Implementation - Allocation Methods - Free Space Management - Efficiency and Performance - Recovery. Mass Storage: Disk Structure - Disk Scheduling Algorithms - Swap Space Management (8+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Silberschatz A, Galvin P, Gagne G, "Operating Systems Concepts", John Wiley and Sons, Singapore, 2018.
2. William Stallings, "Operating Systems: Internals and Design Principles", Pearson Education, New Delhi, 2018.

REFERENCES:

1. Andrew S. Tanenbaum, "Modern Operating System", Fourth Edition, PHI Learning, New Delhi, 2018.

2. Dhamdhere, "Operating Systems: A Concept Based approach", Third Edition, Tata McGraw Hill, New Delhi, 2015.
3. Harvey M Deitel, Paul J Deitel, David R Choffnes, "Operating Systems", Third Edition, Pearson Education, New Delhi, 2013.
4. Remzi Arpaci-Dusseau , Andrea Arpaci-Dusseau, “Operating Systems: Three Easy Pieces”, First edition,Arpaci-Dusseau Books,2015
5. Achyut S Godbole, AtulKahate, “Operating Systems”, McGraw Hill Education, 2016.

23N405 MACHINE LEARNING-1

3 0 0 3

INTRODUCTION TO MACHINE LEARNING AND CONCEPT LEARNING: Machine Learning applications - Concept Learning Task - FIND-S - Version Spaces and Candidate Elimination - Inductive Bias. (8)

SUPERVISED LEARNING AND BAYESIAN DECISION THEORY: Learning a Class from Examples - Vapnik-Chervonenkis Dimension - Noise - Learning Multiple Classes - Regression - Model Selection and Generalization - Dimensions of a Supervised Machine Learning Algorithm – Bayesian Classification - Losses and Risks - Discriminant Functions. (9)

PARAMETRIC METHODS AND MULTIVARIATE METHODS: Maximum Likelihood Estimation - The Bayes’ Estimator - Parametric Classification – Parametric Regression - Bias/Variance Dilemma - Model Selection Procedures - Multivariate Data - Parameter Estimation - Multivariate Normal Distribution - Multivariate Classification (9)

LINEAR DISCRIMINATION AND ASSESSING CLASSIFIER PERFORMANCE: Generalizing the Linear Model - Geometry of the Linear Discriminant - Gradient Descent - Logistic Discrimination - Strategy, Principles and Guidelines of Experimentation - Cross-Validation and Resampling Methods - Measuring Classifier Performance - Interval Estimation – Binomial test - t test - K-Fold Cross-Validated Paired t Test (10)

DIMENSIONALITY REDUCTION AND CLUSTERING: Subset Selection - Feature Embedding - Factor Analysis - MDS - LDA - k Means Clustering - Expectation Maximization Algorithm - Spectral Clustering - Hierarchical Clustering - Cluster number Selection (9)

TOTAL L: 45

TEXT BOOKS:

1. EthemAlpaydin, "Introduction to Machine Learning", 4thEdition, PHI Learning, 2020.
2. Tom Mitchell, "Machine Learning", McGraw Hill, 2017.

REFERENCES:

1. Christopher M Bishop, "Pattern Recognition and Machine Learning Learning", Springer, 2011.
2. Yaser S. Abu-Mostafa, "Learning from Data", AML, 2017.
3. Richard O. Duda, Peter E. Hard, David G. Stork, “Pattern Recognition”, 2nd Edition, Wiley, 2020
4. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, “Mathematics for Machine Learning”, Cambridge University Press, 2020.

23N410 DATABASE MANAGEMENT SYSTEMS LABORATORY

0 0 4 2

1. SQL Practice: DDL, DML, DCL, TCL commands.
2. Implement Views and stored procedures.
3. Implement functions, Cursors, Triggers and Embedded SQL
4. Study of performance monitoring and tuning Tools
5. Study of NoSQL databases
6. Mini project- Application development

TOTAL P: 60

REFERENCES:

1. Abraham Silberchatz, Henry F Korth and Sudarshan S, “Database System Concepts”, 7th Edition, Tata McGraw-Hill, New Delhi, 2021.
2. RamezElmasri and Shamkant B Navathe, “Fundamentals of Database Systems”, 7th Edition, Addison Wesley, USA, 2016.
3. Ivan Bayross, “SQL, PL/SQL the Programming Language of Oracle”, 4th Edition,BPB Publications New Delhi, 2020.
4. Rosenzweig,”Oracle PL/SQL by Example 5/e, For Dummies”, 5th Edition, Pearson Education India, 2015.
5. Raghu Ramakrishnan and Johannes Gehrke, “Database Management Systems”, 4th edition, Tata McGraw-Hill, New Delhi, 2015.

23N411 MACHINE LEARNING LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Data Preparation for Machine Learning
2. Candidate elimination Algorithm
3. Implement Linear regression for data fitting
4. Implement Naïve Bayes for Classification
5. Implement Kmeans algorithm
6. Analyse how bias-variance trade-off relates to underfitting and overfitting using Polynomial regression
7. Compare Dimensionality Reduction techniques for a real world dataset
8. 8.Compare Spectral and Hierarchical Clustering Methods for a real world dataset
9. Design and analysis of machine learning experiment

TOTAL P: 60

REFERENCES:

1. EthemAlpaydin, "Introduction to Machine Learning", 4thEdition, PHI Learning, 2020.
2. Willi Richert, Luis Pedro Coelho, Building Machine Learning Systems with Python, Packt Publishing, 2018.
3. Aurelien Geron, Hands-On Machine Learning with Scikit-Learn, Keras and Tensor Flow: Concepts, Tools and Techniques to Build Intelligent Systems, O'Reilly, 2019.
4. Tom Mitchell, "Machine Learning", McGraw Hill, 2017
5. Richard O. Duda, Peter E. Hard, David G. Stork, "Pattern Recognition", 2nd Edition, Wiley, 2020

23Q413 PROBLEM SOLVING

0 0 2 1

1. Algorithmic Thinking, Branching & Repetition Problems
2. Logical Reasoning - Data Arrangements & Relations
3. Solving problems based on Coding & decoding, Series, Analogy, Odd man out and Visual reasoning
4. Problems based on Ages, Logical Connectives, Syllogisms, Data Interpretation & Data Sufficiency
5. Solving problems on Clocks Calendars, Direction Sense & Cubes
6. Problems based on Number system, Percentages, Simple & Compound Interest
7. Resume Update

TOTAL P: 30

REFERENCES:

1. R.S. Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017.

23O414 INDIAN CONSTITUTION

2 0 0 0

INTRODUCTION: Evolution of Indian Constitution; significance of constitution; Composition; Preamble and its Philosophy. (4)

RIGHTS, DUTIES AND DIRECTIVE PRINCIPLES: Fundamental Rights- Writs and Duties, Directive Principles of State Policy (5)

UNION GOVERNMENT: Union Government, President and Vice President, Houses of the Parliament and their functions; Types of Bills, Stages of passing of Bill into an Act, Veto Power, Constitution Amendment Procedure, Various Amendments made and their significance for India. (6)

STATE GOVERNMENT AND FEDERALISM: Composition of State Legislature; Powers, Functions and Position of Governor, Function of Chief Ministers, Council of Ministers; The Indian Federal System, Administrative Relationship between Union and States (8)

JUDICIARY: Supreme Court, High Court; District Court and Lower Courts - Functions and Powers – Judges – Qualifications and Powers - Judicial Review. (7)

Total L: 30

TEXT BOOKS:

1. Subash C. Kashyap, "Our Political System", National Book Trust, 2011.
2. Praveenkumar Mellalli, E., "Constitution of India, Professional Ethics and Human Rights", Sage Publications India Pvt. Ltd., 2015.

REFERENCES:

1. Briji Kishore Sharma, "Introduction to the Constitution of India", Prentice Hall of India, 2010.
2. Basu D.D., "Introduction to the Constitution of India", Prentice Hall of India, 2016.
3. Jain. M. C., "The Constitution of India", Law House, New Delhi, 2001.
4. Shukla. V. N., "Constitution of India", Eastern Book Company Ltd., New Delhi, 2011.

23TC02 தமிழரும் தொழில் நுட்பமும் / TAMILS AND TECHNOLOGY

1 0 0 1

அலகு – I நெசவு மற்றும் பானைத் தொழில்நுட்பம்: சங்க காலத்தில் நெசவுத் தொழில் – பானைத் தொழில்நுட்பம் – கருப்பு சிவப்பு பாண்டங்கள் – பாண்டங்களில் கீறல் குறியீடுகள். (3)

அலகு – II வடிவமைப்பு மற்றும் கட்டிடத் தொழில்நுட்பம்: சங்க காலத்தில் வடிவமைப்பு மற்றும் கட்டுமானங்கள், சங்க காலத்தில் வீட்டுப் பொருட்களில் வடிவமைப்பு – சங்க காலத்தில் கட்டுமான பொருட்களும் நடுகல்லும் – சிலப்பதிகாரத்தில் மேடை அமைப்பு பற்றிய விவரங்கள் – மாமல்லபுரச் சிற்பங்களும், கோவில்களும் – சோழர் காலத்துப் பெருங்கோயில்கள் மற்றும் பிற வழிபாட்டுத் தலங்கள் – நாயக்கர் காலக் கோயில்கள் – மாதிரி கட்டமைப்புகள் பற்றி அறிதல், மதுரை மீனாட்சி அம்மன் ஆலயம் மற்றும் திருமலை நாயக்கர் மஹால் – செட்டிநாட்டு வீடுகள் – பிரிட்டிஷ் காலத்தில் சென்னையில் இந்தோ-சாரோசெனிக் கட்டிடக் கலை. (3)

அலகு – III உற்பத்தித் தொழில் நுட்பம்: கப்பல் கட்டும் கலை – உலோகவியல் – இரும்புத் தொழிற்சாலை – இரும்பை உருக்குதல், எஃகு – வரலாற்றுச் சான்றுகளாக செம்பு மற்றும் தங்க நாணங்கள் – நாணயங்கள் அச்சடித்தல்- மணி உருவாக்கும் தொழிற்சாலைகள் – கல்மணிகள், கண்ணாடி மணிகள் – சுடுமண் மணிகள் – சங்கு மணிகள் – எலும்புத் துண்டுகள் – தொல்லியல் சான்றுகள் – சிலப்பதிகாரத்தில் மணிகளின் வகைகள். (3)

அலகு – IV வேளாண்மை மற்றும் நீர்ப்பாசனத் தொழில் நுட்பம்: அணை, ஏரி, குளங்கள், மதகு – சோழர்காலக் குமுழித் தூம்பின் முக்கியத்துவம் – கால்நடை பராமரிப்பு – கால்நடைகளுக்காக வடிவமைக்கப்பட்ட கிணறுகள் – வேளாண்மை மற்றும் வேளாண்மைச் சார்ந்த செயல்பாடுகள் – கடல்சார் அறிவு – மீன்வளம் – முத்து மற்றும் முத்துக்குளித்தல் – பெருங்கடல் குறித்த பண்டைய அறிவு – அறிவுசார் சமூகம். (3)

அலகு V – அறிவியல் தமிழ் மற்றும் கணித்தமிழ்: அறிவியல் தமிழின் வளர்ச்சி – கணித்தமிழ் வளர்ச்சி – தமிழ் நூல்களை மின்பதிப்பு செய்தல் – தமிழ் மென்பொருட்கள் உருவாக்கம் – தமிழ் இணையக் கல்விக்கழகம் – தமிழ் மின் நூலகம் – இணையத்தில் தமிழ் அகராதிகள் – சொற்குவைத் திட்டம். (3)

TOTAL L: 15

TEXT-CUM-REFERENCE BOOKS

1. தமிழக வரலாறு – மக்களும் பண்பாடும் – கே.கே.பிள்ளை (வெளியீடு: தமிழ்நாடு பாடநூல் மற்றும் கல்வியியல் பணிகள் கழகம்).
2. கணினித் தமிழ் – முனைவர் இல.சுந்தரம் (விகடன் பிரசுரம்).
3. கீழடி – வைகை நதிக்கரையில் சங்ககால நகர நாகரிகம் (தொல்லியல் துறை வெளியீடு)
4. பொருநை – ஆற்றங்கரை நாகரிகம். (தொல்லியல் துறை வெளியீடு)
5. Social Life of Tamils (Dr.K.K.Pillay) A joint publication of TNTB & ESC and RMRL – (in print)
6. Social Life of the Tamils – The Classical Period (Dr.S.Singaravelu) (Published by: International Institute of Tamil Studies.)
7. Historical Heritage of the Tamils (Dr.S.V.Subatamanian, Dr.K.D.Thirunavukkarasu) (Published by : International Institute of Tamil Studies).
8. The Contributions of the Tamils to Indian Culture (Dr.M.Valarmathi) (Published by: International Institute of Tamil Studies).
9. Keeladi – 'Sangam City Civilization on the banks of river Vaigai' (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
10. Studies in the History of India with Special Reference to Tamil Nadu (Dr.K.K.Pillay) (Published by : The Author)
11. Porunai Civilization (Jointly Published by : Department of Archaeology & Tamil Nadu Text Book and Educational Services Corporation, Tamil Nadu)
12. Journey of Civilization Indus to Vaigal (R.Balakrishnan) (Published by: RMRL) – Reference Book.

SEMESTER – V

23N501 MACHINE LEARNING-2

3 1 0 4

Non Parametric Methods and Decision Trees: Nonparametric Density Estimation - Generalization to Multivariate Data - Nonparametric Classification - Condensed Nearest Neighbor - Distance-Based Classification - Outlier Detection - Nonparametric Regression: Smoothing Models - Univariate Trees - Pruning - Rule Extraction from Trees - Learning Rules from Data - Multivariate Trees (9+3)

Kernel Machines and Local Models: Optimal Separating Hyperplane - Soft Margin Hyperplane - v-SVM - Kernel Trick - Vectorial Kernels - Defining Kernels - Adaptive Resonance Theory - Self-Organizing Maps - Radial Basis Functions - Learning Vector Quantization (9+3)

Graphical Models and HMM: Canonical Cases for Conditional Independence – Generative Models - d-Separation - Belief Propagation - Markov Random Fields - Discrete Markov Processes - Hidden Markov Models - Three Basic Problems of HMMs - Evaluation Problem- Finding the State Sequence - Learning Model Parameters (9+3)

Modelling Reinforcement Learning Problem: Introduction – Elements of Reinforcement Learning - Tic-Tac-Toe - A k-armed Bandit Problem - Action-value Methods - Incremental Implementation - Finite Markov Decision Processes: Policies and Value Functions - Optimal Policies and Optimal Value Functions (9+3)

Solving Reinforcement Learning Problem: Dynamic Programming: Policy Evaluation - Policy Improvement- Policy Iteration - Value Iteration - Monte Carlo Methods: Prediction, Estimation of Action Values, Control - Off-policy Prediction - Incremental Implementation -Off-Policy Monte Carlo Control - Temporal Difference Learning: - TD Prediction -Optimality of TD(0) - Sarsa - Q-Learning - Planning and Learning – Dyna. (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. EthemAlpaydin , "Introduction to Machine Learning", 4thEdition, PHI Learning, 2020.
2. Richard S. Sutton and Andrew G. Barto , "Reinforcement Learning: An Introduction", The MIT Press, 2014.

REFERENCES:

1. Christopher M Bishop, "Pattern Recognition and Machine Learning Learning", Springer, 2011.
2. Yaser S. Abu-Mostafa , "Learning from Data", AML, 2017.
3. Richard O. Duda, Peter E. Hard, David G. Stork, "Pattern Recognition", 2nd Edition, Wiley, 2020
4. Marc Peter Deisenroth, A. Aldo Faisal, Cheng Soon Ong, "Mathematics for Machine Learning", Cambridge University Press, 2020.

23N502 ARTIFICIAL INTELLIGENCE

3 1 0 4

INTRODUCTION: Artificial Intelligence - The state of art - Intelligent Agents - Rationality - Nature of Environments – Structure of

Agents - Examples. (9+3)

PROBLEM SOLVING AGENTS: Searching for solutions: Uninformed search - BFS, DFS, Uniform cost search, Iterative deepening search - Informed Search - Greedy Best First search, A* search, AO* search - Adversarial search – Games - Optimal decisions in Games, alpha - beta pruning (9+3)

KNOWLEDGE AND REASONING: Representations and mappings – Approaches to knowledge representation – Property inheritance algorithm - First Order Predicate logic- Instance and ISA relationships – Computable functions and predicates - Unification-Resolution – Question Answering. (9+3)

ACTING UNDER UNCERTAINTY: Quantifying uncertainty – Efficient representation of conditional distributions – Probability and Bayes Theorem - Bayesian Networks - Exact and approximate inferences - Making simple decisions – Utility theory - Decision networks – Value of information – Decisions with multiple agents (9+3)

PLANNING: Classical Planning- Algorithms for Planning as state space search - Planning Graphs – Classic planning as Boolean Satisfiability - Analysis of planning approaches –Time, schedules and resources – Hierarchical planning (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Stuart J Russell and Peter Norvig, "Artificial Intelligence – A Modern Approach", 3rd Edition, Prentice Hall of India, Pearson Education, New Delhi, 2021.
2. Elaine Rich, Kevin Knight and Shivashankar B Nair, "Artificial Intelligence", 3rd Edition, Tata McGraw Hill Publishing Company, New Delhi, 2019.

REFERENCES:

1. George F Luger, "Artificial Intelligence: Structures and Strategies for Complex Problem Solving", 6th Edition, Pearson Education, New Delhi, 2021.
2. Deepak Khemani, “A First Course in Artificial Intelligence”, McGraw Hill Education, New Delhi, 2017.
3. John Paul Mueller, “Artificial Intelligence For Dummies”, Wiley, 2018
4. Lavika Goel, “Artificial Intelligence: Concepts and Applications”, Wiley,2021
5. Pradeepta Mishra, “Practical Explainable AI Using Python: Artificial Intelligence Model Explanations Using Python-based Libraries, Extensions, and Frameworks”, Apress,2021

23N503 EMBEDDED SYSTEMS

2 2 0 4

INTRODUCTION: Fundamental Components of Embedded Systems - Architecture of Embedded Systems - Embedded Design Life Cycle - Development Environment (5+5)

MEMORY AND INTERRUPTS: Types of Memory - Memory Access Procedure - Memory Management techniques - Memory Testing - Common Memory problems - Interrupts - Interrupt Service Routines (6+6)

COMMUNICATION INTERFACES AND PROTOCOLS: Interfacing Buses - Serial Interfaces - RS232/UART - UART Programming - RS422/RS485 - I2C Interface - SPI Interface - I2C/SPI Programming - USB (7+7)

REAL TIME OPERATING SYSTEMS: Real-Time Concepts - Task Management - Task Scheduling - Classification of Scheduling Algorithms - Clock Driven Scheduling - Event Driven Scheduling - Resource Sharing - Priority Inheritance Protocol - Priority Ceiling Protocol - Commercial RTOS (7+7)

HOST and TARGET TESTING: Validation - Host and Target Testing - Debugging tool **- CASE STUDY** - Development of Automatic Vending Machine - Protocol converter - Development of a navigation (5+5)

TOTAL L: 30 +T: 30 = 60

TEXT BOOKS:

1. Arnold S Berger, “Embedded Systems Design - An Introduction to Processes, Tools andTechniques”, Elsevier, New Delhi, 2011
2. Prasad K V K K, “Embedded/Real-Time Systems: Concepts, Design and Programming - The Ultimate Reference”, Himal Impressions, New Delhi, 2003

REFERENCES:

1. Raj Kamal, "Embedded Systems - Architecture, Programming and Design", 3rd Edition, Tata McGraw Hill, 2017
2. Rajib Mall, “Real-Time Systems: Theory and Practice”, First Edition, Pearson, 2009
3. Sriram V Iyer and Pankaj Gupta, “Embedded Real-time Systems Programming”, TataMcGraw-Hill Publishing Company Limited, New Delhi, 2006.

23N504 COMPUTER NETWORKS

3 0 0 3

INTRODUCTION AND MEDIUM: Building a Network - Network Edge and Core - Layering and Protocols - TCP/IP Protocol suite - OSI Reference Model - Network Topologies - Internet Architecture–Physical Layer: Signal Characteristics – Transmission media – Signal Encoding Techniques – Performance Metrics. (10)

LINK LAYER SERVICES: Link Layer Services - Framing - Flow Control - Error Control - Media Access Control - Ethernet - Wireless LAN – Introduction about Bluetooth, Zigbee. (8)

SWITCHING AND ROUTING: Switching: Circuit Switching - Packet Switching - IPV4 - Global Address - Datagram Forwarding - Subnetting - CIDR - ARP- ICMP - Routing Algorithms: Distance Vector Routing and Link State Routing - IPV6 Addressing – IPV6 Protocol. (9)

CONNECTION-ORIENTED AND CONNECTIONLESS SERVICES: Overview of Transport Layer - UDP - TCP – Reliable Byte Stream - Connection Management - Flow Control - Congestion Control - SCTP. (9)

APPLICATION LAYER SERVICES: Needs/Principles of Application Layer Protocols – Role of proxy, Web and HTTP - FTP - Electronic Mail (SMTP - POP3 - IMAP - MIME) - DHCP - DNS - DASH – QUIC (9)

TOTAL L: 45

TEXTBOOKS:

1. Larry L Peterson and Bruce S Davie," Computer Networks: A systems approach", Morgan Kaufmann Publishers, USA, 6th Edition 2021.
2. James F Kurose, Keith W Ross," Computer Networking - A Top-Down Approach Featuring the Internet", Pearson Education, New Delhi, Sixth Edition, 2012.

REFERENCES:

1. Behrouz A. Forouzan ," Data Communications and Networking with TCP/IP Protocol Suite", 6th Edition, McGraw Hill, 2021.
2. Andrew S Tanenbaum and David J Wetherall, "Computer Networks", Prentice Hall of India/ Pearson Education, New Delhi,6th Edition, 2021
3. Prakash C Gupta," Data Communication and Computer Networks", Prentice Hall of India, New Delhi, 2014.
4. Prakash C Gupta," Data Communication and Computer Networks", 2nd Edition, Prentice Hall of India, New Delhi, 2013
5. Ajit Pal, "Data Communication And Computer Networks", 1st Edition, PHI Learning,

23N510 COMPUTER NETWORKS LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

1. Study of Network Components,Basic Network Commands and Network Configuration Commands 2. Chat Program using TCP Sockets using C language
2. Sliding Window Protocol using TCP Sockets using C language
3. DNS using UDP Sockets using C language
4. Study of Wireshark Tool
5. Capturing of packet header at each layer using Wireshark
6. Tracing of TCP and UDP Connection using Wireshark
7. Study of any Simulator Tool
8. Performance comparison of TCP and UDP protocols using Simulation tool
9. Set up a typical network in a lab
10. Set up a typical network in a lab

TOTAL P: 60

REFERENCES:

1. Behrouz A. Forouzan ," Data Communications and Networking with TCP/IP Protocol Suite", 6th Edition, McGraw Hill, 2021.
2. Andrew S Tanenbaum and David J Wetherall, "Computer Networks", Prentice Hall of India/ Pearson Education, New Delhi,6th Edition, 2021
3. Prakash C Gupta," Data Communication and Computer Networks", Prentice Hall of India, New Delhi, 2014.
4. Prakash C Gupta," Data Communication and Computer Networks", 2nd Edition, Prentice Hall of India, New Delhi, 2013
5. Ajit Pal, "Data Communication And Computer Networks", 1st Edition, PHI Learning,

23N511 APPLICATION DEVELOPMENT LABORATORY

0 0 4 2

LIST OF EXPERIMENTS:

<Problem Formulation and Design>:

1. Problem Identification
2. Requirements Analysis and Design

<Implementation>:

1. Development of required packages
2. Implementation of the application using coding standards and the packages developed.

<Testing and Documentation>:

1. Application Testing
2. Documentation of the development process

NOTE: The students can develop Web based Application or Mobile Application of their choice. The Language for development can be chosen based on their application requirement.

Total P: 60

REFERENCES:

1. Love, Chris. Progressive Web Application Development by Example: Develop Fast, Reliable, and Engaging User Experiences for the Web. United Kingdom, Packt Publishing, 2018.
2. Robbins, Jennifer. Learning Web Design: A Beginner's Guide to HTML, CSS, JavaScript, and Web Graphics. United States, O'Reilly Media, 2018.

3. User Interface Design: Bridging the Gap from User Requirements to Design. United States, CRC Press, 2018
4. Carlson Ph D, John R, and Carlson, John. Cross-Platform Mobile Application Development: A Beginner's Guide Using the Corona SDK. N.p., Amazon Digital Services LLC - KDP Print US, 2019.
5. Mobile Apps Engineering: Design, Development, Security, and Testing. United Kingdom, CRC Press, 2018.

23Q513 APTITUDE SKILLS

0 0 2 1

1. Algorithmic Thinking, Branching & Repetition Problems
2. Logical Reasoning - Data Arrangements & Relations
3. Solving problems based on Coding & decoding, Series, Analogy, Odd man out and Visual reasoning
4. Problems based on Ages, Logical Connectives, Syllogisms, Data Interpretation & Data Sufficiency
5. Solving problems on Clocks Calendars, Direction Sense & Cubes
6. Problems based on Number system, Percentages, Simple & Compound Interest
7. Resume Update

TOTAL P: 30

REFERENCES:

1. R.S. Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017.

SEMESTER – VI

23N601 DATA PRIVACY AND SECURITY

3 1 0 4

INTRODUCTION: Security goals –threats and attacks. – Services and mechanisms – A model for Network Security crypto Systems: Introduction – symmetric key cryptography-substitution cipher – transposition cipher -stream ciphers and block ciphers – Data Encryption Standard - Advanced Encryption Standard (AES) (10+3)

PUBLIC KEY CRYPTOSYSTEM: Basic Concepts of Number Theory - Principles of Public key cryptosystem – RSA cryptosystem - attacks on RSA – Diffie – Hellman key agreement–Elliptic Curve cryptosystem (8+3)

MESSAGE INTEGRITY, AUTHENTICATION: Message digest – Cryptographic hash function - Message authentication code – Message authentication Requirements- Message Authentication Functions - Digital signatures- Digital Signature standard.
DATA BASE SECURITY: Security Requirements – database administration security – SQL injection and exploitation and defense methods - database roles and permissions – Object level security - Sensitive data (9+3)

DATA PRIVACY: Foundations of privacy: Privacy and Contextual Integrity, Privacy regulations, Goals of Privacy engineering, taxonomy of privacy, Attacks on private data - DATA PRIVACY MODELS AND DISCLOSURE RISK MEASURES:Formal model of Data privacy, K-anonymity: Definition of k-anonymity, Practical Implications, Mechanism design, Evaluating the risk of disclosure- Limitations of K-anonymity. Disclosure Risk Measures: Attribute disclosure, identity disclosure, Synthetic datasets and network trace analysis (8+3)

MASKING METHODS & DIFFERENTIAL PRIVACY: Perturbative methods - non-Perturbative methods - Synthetic Data Generators. DIFFERENTIAL PRIVACY**:**Definition of Differential Privacy (DP), Privacy and databases, Noiseless differential privacy- Promise of DP, Formalizing DP, Lower bounds, Mechanism Design, Machine learning and Differential Privacy- Differential privacy for large data- Differentially private social network analysis – Web privacy: online tracking and advertisement Applications of Differential privacy (10+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. William Stallings, " Cryptography and Network Security: Principles and Practice”, 8th Edition, Prentice Hall of India, Pearson Education, New Delhi, 2020.
2. Vicenc Torra,” Data Privacy: Foundations, New Developments and Big Data Challenge”, Springer, 2017.

REFERENCES:

1. Behrouz A.Forouzan and DebdeepMukhopadhyay, “Cryptography and Network Security”, Tata McGraw Hill, 2011.
2. AtulKahate,” Cryptography and Network Security”, Tata McGraw Hill, 2013.
3. TianqingZhu, GangLi, WanleiZhou, Philip S. Yu, "Differential Privacy and Applications", Springer, 2017
4. Benjamin C.M. Fung, Ke Wang, Ada Wai-Chee Fu and Philip S. Yu, “Introduction to Privacy-Preserving Data Publishing: Concepts and Techniques”, Chapman & Hall/CRC, 2010.
5. Charu C. Aggarwal, “Privacy-Preserving Data Mining: Models and Algorithms”, Springer, 2008.

23N602 PARALLEL AND DISTRIBUTED SYSTEMS

3 1 0 4

INTRODUCTION TO PARALLEL COMPUTING: Flynn’s Taxonomy-Types of Parallel Computers-Characteristics of Parallel Systems-Parallel Computing Models: Shared Memory Models-Interconnection Network Models-Dataflow Model-Parallel Algorithms: Classes of Problems Solvable through Parallelization-Types of Parallelization-Assigning Computational Tasks to Processors. (9+3)

PARALLEL COMPUTING ARCHITECTURE: Introduction-Shared Memory Architecture: Shared Memory Paradigm3-Cache Coherency-Memory Consistency-Message-Passing Architecture: Message-Passing Paradigm-Routing- Switching-Stream Processing Architecture: Dataflow Paradigm-Parallel Accelerators- Stream Processors (9+3)

PARALLEL PROGRAMMING WITH OpenMP and MPI: OpenMP: Overview of Features-Additional Feature Details: OpenMP

Directives-Synchronization-Runtime Library Routines-MPI: Introduction to MPI-Basic Point-to-point Communication Routines-Basic MPI Collective Communication Routines-Environment Management Routines-Point-to-point Communication Routines-Collective Communication Routines (9+3)

DISTRIBUTED COMPUTING: Definition, motivation, communication protocols – RPC-RMI, Coordination – clocks- mutual exclusion – leader election, Replication management – Consistency protocols. (9+3)

GPU COMPUTING: CPU vs GPU, Architecture of CUDA capable GPU – GPU computing – Data parallelism – CUDA program structure – Vector addition – CUDA thread Organisation – Matrix-matrix multiplication – Blocks – OpenCL data parallel model, device architecture (9+3)

TOTAL L: 45 +T: 15 = 60

TEXT BOOKS:

1. Vivek Kale, Parallel Computing Architectures and APIs: IoT, big data stream processing", CRC Press, 2020.
2. Andrew S.Tanenbaum and Maarten van Steen, "Distributed Systems, Principles and Paradigm" Prentice Hall,2016

REFERENCES:

1. AnanthGrama, Anshul Gupta, George Karypis and VipinKumar,"Introduction to Parallel Computingll", Second Edition, Pearson Education, 2009
2. David Kirk, Wen Mei W Hwu, "Programming Massively Parallel Processors", Third Edition: A Hands-on Approach", Morgan Kaufmann Publishers Inc, 3rd edition, 2016
3. Roman Trobec BoštjanSlivnik , PatricioBulić, Boru"tRobič , "Introduction to Parallel Computing", Springer,2018
4. Shane Cook,"CUDA Programming: A Developer's Guide to Parallel Computing with GPU's (Applications of GPU Computing),MK Publishers,2013
5. Wan Fokkink, " Distributed Algorithms: An Intuitive Approach", PHI Learning Pvt. Ltd,2nd Edition, 2018

23N603 DEEP LEARNING

3 0 0 3

INTRODUCTION: Need for Deap Learning - tensor operations - Activation Functions – Loss Functions - Deep Feed forward Networks – Regularization for Deep Learning (DL) – Optimization for Training Deep Models – Applications of DL (9)

CONVOLUTIONAL NETWORKS: Convolution operation – Motivation – Pooling – Convolution variants – Down sampling, stride and padding –Local, convolution, tiled and full connections– CNN training – Structured outputs – Data types – Efficient convolution algorithms – Random or unsupervised features – Neuro scientific basis of CNN – Applications (9)

SEQUENCE MODELING: Recurrent Neural Networks (RNN) – Unfolding computational graphs – RNN design patterns - Backpropagation through time – Bidirectional RNN – Encoder Decoder Sequence-to-Sequence Architectures – Deep recurrent networks – Recursive neural networks - Challenge of long-term dependencies – LSTM and GRU –Attention and the Transformer (9)

AUTOENCODERS, REPRESENTATION LEARNING AND STRUCTURED MODELS: Autoencoders (AE) – AE variants - Applications of AE - Representation Learning – Greedy pre-training – Transfer learning and domain adaptation - Structured Probabilistic Models for Deep Learning - Using Graphs to Describe Model Structure- Sampling from Graphical Models - Restricted Boltzmann Machine (9)

DEEP GENERATIVE MODELS: Sampling and Monte Carlo Methods – Deep Boltzman Machines —Directed Generative Nets – VariationalAutoencoders - Generative Adversarial Networks– Normalizing flow models – Diffusion models (9)

Total L: 45

TEXT BOOKS:

1. Ian Goodfellow, YoshuaBengio, Aaron Courville, "Deep Learning", The MIT Press, 2017.
2. Magnus Ekman, Learning Deep Learning, Addison-Wesley Professional, 2021

REFERENCES:

1. Charu C. Aggarwal, "Neural Networks and Deep Learning", Springer 2018.
2. Adam Gibson, Josh Patterson "Deep Learning: A Practitioner's Approach ", OReilly, 2017.
3. Umberto Michelucci "Applied Deep Learning. A Case-based Approach to Understanding Deep Neural Networks" Apress, 2018.
4. Nicholas Locascio and Nikhil Buduma "Fundamentals of Deep Learning: Designing Next-Generation Machine Intelligence Algorithms", OReilly, 2017
5. Giancarlo Zaccone, Md. Rezaul Karim, Ahmed Menshawy "Deep Learning with TensorFlow: Explore neural networks with Python", Packt Publisher, 2017.

23N604 NATURAL LANGUAGE PROCESSING

3 0 0 3

WORDS: Introduction - Mathematical Foundations – Linguistic Essentials - Regular Expressions, Text Normalization, Edit Distance - Finite State Transducers - Language Modeling with N-grams - Naive Bayes Classification and Sentiment - Neural Nets and Neural Language Models - Hidden Markov Models - Part-of-Speech Tagging. (10)

SYNTAX: Formal Grammars of English - Syntactic Parsing – Ambiguity – Cocke Kasami Younger (CKY) algorithm – PCFG – Inside and Outside probabilities – Dependency Grammar – Transition based parsing – MST based Dependency parsing. (8)

SEMANTICS: Vector Semantics - Semantics with Dense Vectors - Word Senses: WSD and WordNet - Lexicons for Sentiment

and Affect Extraction - Representation of Sentence Meaning - Computational Semantics - Information Extraction – Named Entity Recognition - Semantic Role Labeling and Argument Structure - Coreference Resolution and Entity Linking (9)

PRAGMATICS AND APPLICATIONS: Discourse Coherence – Neural Sequence modeling: RNN, LSTM – Text Summarization - Topic Modelling - Question Answering. (9)

NATURAL LANGUAGE GENERATION: Architecture of NLG Systems- Generation Tasks and Representations- Application of NLG. Machine Translation: Language similarities and differences – The transfer metaphor – Direct translation – Statistical translation - Translation involving Indian Languages – Generative Pre-Training – GPT-n (9)

TOTAL L: 45

TEXTBOOKS:

1. Daniel Jurafsky and James H. Martin, “Speech and Language Processing”, Prentice-Hall, Inc., 2017
2. Christopher D. Manning, Hinrich Schütze, “Foundations of Statistical Natural Language Processing”, The MIT Press, 2018.

REFERENCES:

1. Yoav Goldberg, “Neural Network Methods for Natural Language Processing”, Synthesis Lectures on Human Language Technologies, April 2017.
2. Steven Bird, Ewan Klein, and Edward Loper, “Natural Language Processing with Python - Analyzing Text with the Natural Language Toolkit”, O,Reilly. 2019
3. Li Deng and Yang Liu, "Deep Learning in Natural Language Processing", Springer, Germany. 2018.
4. Dipanjan Sarkar, “Text Analytics with Python - A Practitioner's Guide to Natural Language Processing, Apress, 2019
5. JalajThanaki, “Python Natural Language Processing - Advanced machine learning and deep learning techniques for natural language processing, Packt Publisher, 2017.

23N610 DEEP LEARNING LABORATORY

0 0 4 2

Deep learning Packages Basics: Tensorflow, Keras

Implement the following from scratch

1. Train Simple Perceptron with Gradient Descent for regression
2. Train MLP with Backpropagation for classification
3. CNN with maxpooling for MNIST Dataset
4. Bahdanau and Loung Attention
5. Transformer

Implement the following using existing packages.

1. Face Recognition using SSD
2. Time Series Prediction using LSTM
3. Image generation using GAN
4. Generating fake videos using autoencoders
5. Recommender systems using RBM

TOTAL P: 60

REFERENCES:

1. Ian Goodfellow, YoshuaBengio and Aaron Courville, “Deep Learning”, MIT Press, 2017.
2. Josh Patterson, Adam Gibson "Deep Learning: A Practitioner's Approach", O'Reilly Media, 2017
3. Giancarlo Zaccone, Md. Rezaul Karim, Ahmed Menshawy, Deep Learning with TensorFlow: Explore neuralnetworks with Python, Packt Publisher, 2017.
4. Antonio Gulli, Sujit Pal, Deep Learning with Keras, Packt Publishers, 2017.
5. Francois Chollet, Deep Learning with Python", Manning Publications, 2017.

23N611 INNOVATION PRACTICES

0 0 2 1

This course involves preparing students to think innovatively and present possible solutions to identified industry/academic problem or issue

TOTAL P: 60

23Q613 ENHANCING ARITHMETIC PROBLEM SOLVING

0 0 2 1

ENHANCING ARITHMETIC PROBLEM SOLVING:

1. Logarithms, Progressions, Geometry and Quadratic equations
2. Attention to detail&Image Interpretation
3. Voices and Speech
4. Algebraic Expressions
5. Surds, Indices and Simplification
6. Set Theory & Functions
7. Cryptarithmetic & Trigonometry
8. Advance Quantitative
9. Advance Verbal
10. Advance Logical
11. Preparing for Business roles, Case studies, &Guesstimation techniques
12. Handling Stress
13. Handling Peer pressure
14. Resume progress check

TOTAL P: 30

REFERENCES:

1. R.S. Aggarwal, "Quantitative Aptitude for Competitive Examination", S Chand Publishing, New Delhi, 2017.
2. P.C.Wren and H.Martin, "High school English grammar & composition ", S Chand Publishing, New Delhi, 2017.
3. Norman Lewis, "Word Power Made Easy", Goyal Publisher, New Delhi, 2011.
4. Arun Sharma, “How to Prepare for Quantitative Aptitude for the CAT”, 8th edition, McGraw Hill Education, 2018.

SEMESTER – VII

23N701 BIG DATA AND ADVANCED DATABASE SYSTEMS

3 0 0 3

BIG DATA OVERVIEW: Big data- Needs and challenges - Architecture of big data systems, NoSQL data models, CAP theorem, Components of Hadoop Ecosystem, Map-Reduce programming – Columnar Database - HBase, big data software lifecycle. (9)

DOCUMENT DATABASE: Introduction to MongoDB - Installation of MongoDB – Create Database, collection and documents – CRUD Operations - Query and Criteria. (9)

GRAPH DATABASE: - Overview of Neo4j - Neo4j Architecture - - Exploring Neo4j using Cypher Query Language: Create Nodes and Relationships - Importing Data with Neo4j. (9)

MINING DATA STREAMS: Concepts - Stream Data Model and SPARK Architecture – RDD – Transformations– Bloom Filters - parallel programming - Data Frames and Spark SQL. (9)

DATABASE INTEGRATION: Importance – Types and Complexity - ETL – data lakes- Schema Extraction Approaches– Data Integration Framework - Semantic Integration - RDF Schema –Traditional Ontology Languages: OWL (9)

Total L: 45

TEXT BOOKS:

1. G. Sudha Sadasivam,R. Thirumahal, "Big Data Analytics",Oxford University Press,2020.
2. April Reeve, “Managing Data in Motion”, Elsevier 2013.

REFERENCES:

1. EMC Educational Series, "Data Science and Big Data Analytics", Wiley, New Delhi, 2015.
2. David Stevenson, "Big Data Demystified", Pearson Education, 2018.
3. Mohammed Guller, Big Data Analytics with Spark, Apress,2015
4. Jeffrey A. Hoffer, Ramesh Venkataraman, Heikki Topi, "Modern Database Management", 12th Edition, Pearson, New Delhi, 2016.
5. Ian Robinson, Jim Webber, Emil Eifrem, “Graph Databases: New Opportunities for Connected Data”, Second Edition, O’Reilly, January 2016.

23N710 BIG DATA AND ADVANCED DATABASE SYSTEMS LABORATORY

0 0 4 2

1. Map Reduce Program 1
2. Map Reduce Program 2
3. CRUD Operations in MongoDB
4. Mongo DB – case study.
5. CRUD Operations in Neo4j
6. Neo4j – Case study
7. Spark streaming using R - Case Studies
8. Local and global schema generation through XML.

TOTAL P: 60

REFERENCES

1. Tom White “Hadoop: The Definitive Guide” Third Edit on, O’reily Media, 2012. • Seema Acharya, Subhasini Chellappan, "Big Data Analytics" Wiley 2015.
2. Tom Plunkett, Mark Hornick, “Using R to Unlock the Value of Big Data: Big Data Analytics with Oracle R Enterprise and Oracle R Connector for Hadoop”, McGraw-Hill/Osborne Media (2013), Oracle press
3. Ian Robinson, Jim Webber, Emil Eifrem, “Graph Databases: New Opportunities for Connected Data”, Second Edition, O’Reilly, January 2016
4. GuyHarrison, “Next Generation database: NoSQL and big data”, APress, 2015
5. Shannon Bradshaw, Eoin Brazil, Kristina Chodorow,” MongoDB: The Definitive Guide: Powerful and Scalable Data Storage”, O’Reilly, 3rd Edition, December 2019.

23N720 PROJECT WORK-1

0 0 4 2

The project I involves the following:

- Identification of Real-World Problem
- System Requirement Analysis and Specification
- Developing a Model and Solution for the Identified Problem
- Consolidated Report Preparation and Presentation

TOTAL P: 60

SEMESTER – VIII

23N820 PROJECT WORK - 2

0 0 8 4

The Project work II involves

- Preparing a project - brief proposal including
- Problem Identification
- A statement of system / process specifications proposed to be developed
- List of possible solutions including alternatives and constraints
- Cost benefit analysis
- Time Line of activities
- Presentation highlighting the
- Design based on functional requirements
- Implementation
- Testing and Validation
- Results and future work
- Consolidated report based on standards

TOTAL P: 120

LANGUAGE ELECTIVES

23G001 COMMUNICATION SKILLS FOR ENGINEERS

0 0 4 2

COMMUNICATION CONCEPTS: Process of Communication – Inter and Intrapersonal Communication – Essentials for effectiveness (9)

ORAL COMMUNICATION: Oral presentations with visual aids and Group discussions. (16)

FOCUS ON SOFT SKILLS: Etiquette – Work Place etiquette – Telephone etiquette- Body Language – Critical Reasoning and Conflict Management based on Case Studies – Group Communication- Meetings -Interview Techniques (14)

TECHNICAL WRITING: Technical Writing Principles - Style and Mechanics - Technical Definitions – Physical, Functional and Process Descriptions -– Technical Report Writing – Preparing Instructions – Interpretation of Technical Data (14)

BUSINESS CORRESPONDENCE: Writing Emails, Preparing Resumes. (7)

Total: 60

TEXT BOOK:

1. Course materials prepared by the Faculty, Department of English.

REFERENCES:

1. Jeff Butterfield, “Soft Skills for Everyone”, Cengage Learning, New Delhi, 2020.
2. Sabina Pillai and Agna Fernandez, “Soft skills and Employability Skills”, Cambridge University Press, New Delhi, 2019.
3. Prashant Sharma, “Soft Skills Personality Development for Life Success”, BPB Publications, New Delhi, 2021.
4. Shoba K N and Praveen Sam D, “Technical English”, Cambridge University Press, New York, 2020

23G002 BASIC GERMAN

0 0 4 2

Guten Tag! - Learning: To greet, learn numbers till 20, practice telephone numbers & e mail address, learn alphabet, speak about countries & languages ; **Vocabulary:** related to the topic; **Grammar:** W – Questions, Verbs & Personal pronouns I. (10)

Freunde, Kollegen und ich - Learning: To speak about hobbies, jobs, learn numbers from 20; **Vocabulary:** related to the topic; **Grammar:** Articles, Verbs & Personal pronouns II, sein &haben verbs, ja/nein Frage, singular/plural. (10)

In der Stadt – **Learning**: To know places, buildings, question, know transport systems, understand international words; **Vocabulary:** related to the topic; **Grammar:** Definite & indefinite articles, Negotiation, Imperative with Sie. (12)

GutenAppetit! – Learning: To speak about food, shop, converse; **Vocabulary:** related to the topic; **Grammar:** Sentence position, Accusative, Accusative with verbs. (13)

Tag für Tag and ZeitmitFreunden – Learning: To learn time related expressions, speak about family, ask excuse, fix appointments on phone, birthdays, understand & write invitations, converse in the restaurant ; **Vocabulary:** related to the topic; **Grammar: Preposition –** am, im, um, von…bis, Possessive articles, Modalverbs. (15)

Total: 60

TEXT BOOK:

1. Dengler, Stefanie et al.*,, Netzwerk A1.1*, Klett-Langenscheidt Gmbh, München, 2013.

REFERENCES:

1. Dengler, Stefanie et al.*, Netzwerk A1*,Klett-Langenscheidt Gmbh, München, 2013.

2. Sandra Evans,Angela Pude, Franz Specht-Menschen A1–Hueber Verlag ,2012.
3. Hermann Funk, Christina Kuhn, Silke Demme, Studio d A1 , Goyal Publishers & Distributors Pvt. Ltd ,2009.
4. Rosa-Maria Dallapiazza, Eduard von Jan, Til Schönherr, *Tangram Aktuell 1 (Deutsch als Fremdsprache)* , Max Hueber Verlag ,2004.

23G003 BASIC FRENCH

0 0 4 2

Unité 1: inviter et répondre à une invitation, Pronoms sujets, L'article définis, l'article indéfinis, Conjugation : présent, adjectifs possessifs, interrogation, décrire les personnes. La vie de quatre parisiens de profession différentes (15)

Unité 2: Exprimer l'ordre et l'obligation demander et commander, l'adjectif possessifs, l'articles partitif, l'article démonstratif, négation ne....pas, l'article contracté, verbe pronominaux, prépositions. (15)

Unité 3: Raconter et reporter-donner son avis, Futur simple, pronom complètement d'objet direct, passé composé, plusieurs région de France, imparfait, pronom y/en, imparfait (15)

Unité 4: Demander l'autorisation-passé récent, futur proche,La vie administrativé et régionale, Pluriel des noms, moyens de transport (15)

Total : 60

TEXT BOOK :
1. Christine Andant étal, À propos (livre de l'élève), LANGER, NEW DELHI, 2012.

REFERENCES:
1. Dondo Modern French Course ---Mathurin Dondo
2. Modern French Grammar---Margaret Lang and Isabelle Perez.

23G004 BASIC JAPANESE

0 0 4 2

Course contents :

Orientation Session, Geographic & Socio, economic perspective to Japan, Japanese people and culture and Basic greetings and responses (3)

Basic script, Method of writing hiragana and katakana, and Combination sounds and simple words (3)

Topic marker “wa”, Desu / dewaarimasen cupolas, Interrogative particle “ka”, Grammar particles “mo”, “no”, ‘’ Introducing some one: “Kochirawa ~“ and Self introductions: Hajimemashite” (3)

Demonstratives “Kore”, “Sore”, “Are”, Demonstrative “Kono”, “Sono”, “Ano” , Possessive noun particle “no” and Japanese apartments: Greeting your neighbour (3)

Place marakers “Koko”, “Soko”, “Asoko”, Direction markers “Kochira”, “Sochira”, “Achira” and Japanese department stores: Asking for and buying something (3)

Asking for and telling the time, Paticle “ni (at)” for time, kara (from) ~ made (until), Particle “to (and)”, Time periods: Days of the week, months, time of day, Verbs (Present / future and past tense) and Telephone enquiry: Asking for a phone no. And business hours (3)

Destination particle “e”, Particles “de (mode of transportation)” and “to (with) and Japanese train station: Asking for Fare and track no. / types of trains (3)

Direct object particle “o”, Particle “de (place of action)” , Verbs (“~masenka”, “~mashou”) and “Ohanami” Cherry blossom viewing (3)

Particle “de (by means of)” , Particle “ni (to)”, ,Aaemasu (give) and Moraimasu (receive) and Visiting a Japanese house (3)

Adjectives (“i” and “na” type), Adjectives (Positive and negative useage), Particle “ga (however, but), “Dore which?)” and Leaving a room, thanking someone for hospitality (3)

Likes and dislikes, Potential verbs (wakarimasu and dekimasu), “Kara (~ because)”, Adverbs and Asking someone out over the phone (3)

Verbs denoting presence: “Imasu” and “arimasu”, Particle “ni (in)”, “Dare (who?)” , Adverbs (“Chikakuni ~“), Particle “dare mo (negative ~ no one)” , Dare ka (anyone), dare ga (who) , Nanika (anything) , naniga (what) - ~ya (and) ~ nado (etc.) and Asking for directions (3)

Counters and Counting suffixes (3)

Introduction to Adjectives (na and ii type), Different usages of adjectives, Comparison, Likes and dislikes and Going to a trip (3)

Need and desire (gahoshii), Wanting to … (Tabetidesu), Going for a certain purpose (mi –niikimasu) and Choosing from a menu (3)

Verb groups, I, II and III and Exercises to group verbs (3)

Please do (tekudasai), Present continuous tenses (teimasu), Shall I? (~mashouka) and Describing a natural phenomenon (It is raining) (3)

To grant permission (~temo ii desu), Asking for permission (~ temo ii desuka) and Should not do (~ tewaikemasen) (3)

Describing a continuing state and Describing a habitual action (2)

Roleplays in Japanese (2)

A demonstration on usage of chopsticks and Japanese tea party (2)

Total: 60

TEXT BOOK

1. Minna no nohongo – Romaji ban (first 10 lessons of this book)

REFERENCE

1. Minna no Nihongo I Honsatsu Roma – ji ban (Main Textbook Romanized Version). International publisher – 3A Corporation,
2. Tokyo, Indian distributor – Goyal Publishers & Distributors, New Delhi

PROFESSIONAL ELECTIVES (REGULAR, PE I TO PE VI)

23N001 SOFTWARE TESTING AND QUALITY ASSURANCE

3 0 0 3

TESTING FUNDAMENTALS: Objectives and Principles - Fundamental Test Process - Software Testing Life Cycle (STLC) - V Model Cost of defect repair - Risk of inadequate testing - Developing Risk Matrix. (9)

TESTING TECHNIQUES: Dynamic Testing: Black Box testing - White box testing - Grey box testing. Functional Testing: GUI testing - Compatibility Testing, Security testing, Portability testing, Configuration Testing, Recovery testing. Performance testing: Load testing, Stress testing, Soak testing, Spike testing, Scalability testing, Volume testing. (9)

TEST MANAGEMENT: People and organizational issues in testing Organization structures for testing teams testing services Test Planning Test Plan Components Test Plan Attachments Locating Test Items test management test process Reporting Test Results Introducing the test specialist Skills needed by a test specialist Building a Testing Group- The Structure of Testing Group. (9)

TEST AUTOMATION: Software test automation skills needed for automation scope of automation design and architecture for automation requirements for a test tool challenges in automation Test metrics and measurements project, progress and productivity metrics Case Study on Test Tools: Junit, Selenium (9)

SOFTWARE QUALITY ASSURANCE: Views of Quality- Quality Control vs Quality Assurance- Cost of Quality- Quality Assurance Group Software Quality Challenges - SQA Activities- Formal Technical Reviews- SQA plan. Models and standards: ISO Standards- Role of CMM- TMM. (9)

Total L: 45

TEXT BOOKS:

1. Roger Pressman S,"Software Engineering: A Practitioner", Tata McGraw Hill, New Delhi, 2010**.**
2. Srinivasan Desikan and Gopalaswamy Ramesh, "Software Testing Principals and Practices", Pearson Education, New Delhi, 2017

REFERENCES:

1. Naresh Chauhan, Naresh Chauhan, Software Testing: Principles and Practices, OUP India, 2010.
2. Milind Limaye, "Software Quality Assurance", Tata McGraw Hill, New Delhi, 2011.
3. Ron Patton, "Software Testing", Pearson Education, Second Edition, 2021.
4. Ilene Burnstein, "Practical Software Testing: A Process – Oriented Approach", Springer India, 2022.

23N002 SOFTWARE PROJECT MANAGEMENT

3 0 0 3

INTRODUCTION TO PROCESS MANAGEMENT: The Management Spectrum - The People - The Product TheProcess - The Project- Process Improvement - CMM and its variants. (9)

PROJECT PLANNING, SCHEDULING AND TRACKING: The Project Planning Process - Software Scope and Feasibility - Basic concepts in Project scheduling - Defining a Task Set for the Software Project - Defining a Task Network - Scheduling - Earned Value Analysis. (9)

RISK MANAGEMENT AND CONTRACT MANAGEMENT: Reactive Vs Proactive Risk Strategies - Software Risks - Risk Identification- Risk Projection - Risk Mitigation - Monitoring - Management - RMMM Plan - Introduction - Types of Contracts - Stages in Contract Management - Typical terms of a Contract - Contract Management - Acceptance. (9)

SOFTWARE MAINTENANCE: Maintenance Processes - Problem Reporting - Problem Resolution - Software Quality Assurance activities for Maintenance - People issues in maintenance and support Software maintenance from customer perspective - Global

Maintenance teams. (9)

PEOPLE MANAGEMENT: Understanding Organizational Behavior - Selecting the Right Person for the Job - Motivation - Working in Groups - Becoming a Team - Decision Making - Leadership - Organizational Structures. (9)

Total L: 45

TEXT BOOKS:
1. Mike Cotterell, Bob Hughes, "Software Project Management", Tata McGraw-Hill, 2010.
2. Gopalaswamy Ramesh , "Managing Global Software Projects", Tata McGraw-Hill, 2017.

REFERENCES:
1. Robert K Wysocki, Robert Beck Jr, David B Crane, "Effective Project Management, Traditional, Agile, Extreme", John Wiley & Sons Inc, 2011.
2. Walker Royce, "Software Project Management: A Unified Framework", PEARSON Education, 2021.
3. Pressman R S, "Software Engineering – A Practitioner's Approach," 8th Edition, Tata McGraw-Hill Book Company, 2014.
4. Ian Sommerville , "Software Engineering", Pearson Addison Wesley, Boston, 2017.

23N003 RECOMMENDER SYSTEMS

3 0 0 3

INTRODUCTION: Recommender system functions- Understanding user ratings - Applications of recommendation systems- Issues with recommender systems Recommender Systems as a Multi-Disciplinary Field. (9)

RECOMMENDATION TECHNIQUES: Matrix Factorization - Collaborative Filtering: User-based nearest neighbor recommendation, Item- based nearest neighbor recommendation, Model Based Techniques- Content-based recommendation - Knowledge based recommendation: Knowledge representation and reasoning Overspecialization. (10)

HYBRID RECOMMENDATION SYSTEM: Opportunities for hybridization- Monolithic hybridization design- Parallelized hybridization design- Pipelined hybridization design-Feature weighted linear stacking. (10)

EVALUATING RECOMMENDER SYSTEMS: General Properties of evaluation evaluation design Evaluation on historical dataset Alternate evaluation designs. (8)

SECURITY ISSUES IN RECOMMENDER SYSTEMS: Attack dimensions Attack types Evaluation of effectiveness and countermeasures Privacy aspects Case study: Personalized recommendations on the mobile internet. (8)

Total L: 45

TEXT BOOKS:
1. Jannach D., Zanker M. and FelFering A., Recommender Systems: An Introduction , Cambridge University Press, 2011.
2. Michael D. Ekstrand, John T. Riedl, and Joseph A. Konstan Publishers Inc, 2011.

REFERENCES:
1. Ricci F., Rokach L., Shapira D., Kantor B.P., Recommender Systems Handbook, Springer, 2011.
2. Manouselis N., Drachsler H., Verbert K., Duval E., Recommender Systems For Learning, Springer, 2013.
3. Aggarwal Charu C,"Recommender Systems The TexBook", Springer,2016.
4. Bobadilla. J, Ortega. F, Hemando.A, and Gutierrez.A, "Recommander Systems Survey, Knowledge – based Systems", 2013.

23N004 AGILE SOFTWARE DEVELOPMENT

3 0 0 3

AGILE METHODOLOGY: Theories for Agile Management – Agile Software Development – Traditional Model vs. Agile Model - Classification of Agile Methods – Agile Manifesto and Principles – Agile Project Management – Agile Team Interactions – Ethics in Agile Teams - Agility in Design, Testing – Agile Documentations – Agile Drivers, Capabilities and Values (9)

AGILE PROCESSES: Lean Production - SCRUM, Crystal, Feature Driven Development- Adaptive Software Development - Extreme Programming: Method Overview – Lifecycle – Work Products, Roles and Practices. (9)

AGILITY AND KNOWLEDGE MANAGEMENT: Agile Information Systems – Agile Decision Making - Earl_S Schools of KM – Institutional Knowledge Evolution Cycle – Development, Acquisition, Refinement, Distribution, Deployment , Leveraging – KM in Software Engineering – Managing Software Knowledge – Challenges of Migrating to Agile Methodologies – Agile Knowledge Sharing – Role of Story-Cards – Story-Card Maturity Model (SMM). (9)

AGILITY AND REQUIREMENTS ENGINEERING: Impact of Agile Processes in RE–Current Agile Practices – Variance – Overview of RE Using Agile – Managing Unstable Requirements – Requirements Elicitation – Agile Requirements Abstraction Model – Requirements Management in Agile Environment, Agile Requirements Prioritization – Agile Requirements Modeling and Generation – Concurrency in Agile Requirements Generation. (9)

AGILITY AND QUALITY ASSURANCE: Agile Product Development – Agile Metrics – Feature Driven Development (FDD) – Financial and Production Metrics in FDD – Agile Approach to Quality Assurance - Test Driven Development – Agile Approach in Global Software Development. (9)

TEXT BOOKS:
1. David J. Anderson and Eli Schragenheim, — Agile Management for Software Engineering: Applying the Theory of Constraints for Business Results‖, Prentice Hall, 2003.
2. Hazza and Dubinsky, —Agile Software Engineering, Series: Undergraduate Topics in Computer Science‖, Springer, 2009.

REFERENCES:

1. Craig Larman, — Agile and Iterative Development: A Manager_s Guidell, Addison-Wesley, 2004.
2. Kevin C. Desouza, — Agile Information Systems: Conceptualization, Construction, and Managementll, Butterworth - Heinemann, 2007.

23N005 OPEN SOURCETECHNOLOGY

3 0 0 3

OPEN-SOURCE PRINCIPLES AND METHODOLOGY: Need and Principles of OSS - Open-Source Standards - OSS success - Free vs. Proprietary Software vs. Open-Source Software - Proprietary vs. Open-Source Licensing Model - History: BSD, The Free Software Foundation and the GNU Project Challenges in Open Source - Open Source Initiatives, Open Standards Principles, Methodologies, Philosophy, Software freedom, Open-Source Software Development. (9)

LICENSING: Creating Licenses, Important FOSS Licenses (Apache, BSD, PL, LGPL), copyrights and copy lefts, Patent - Zero marginal cost, Income-generation Opportunities, Internationalization. (9)

OPEN-SOURCE PROJECTS: Starting and maintaining own Open-Source Project, Open-Source Hardware, Open-Source Design, Open-source Teaching, Opens source media. Collaboration: Community and Communication, Contributing to Open Source Projects GitHub: interacting with the community on GitHub, Communication and etiquette, testing open-source code, Wikipedia: contributing to Wikipedia or contributing to any prominent open-source project of choice. (9)

OPEN-SOURCE ETHICS, SOCIAL IMPACT AND ECOSYSTEM: Open source vs. closed source, Ethics of Open-source, Social and Financial impacts of open-source technology, Shared software, Shared source, Open Source as a Business Strategy - Open-Source Operating Systems: GNU/Linux, Android, Free BSD, Open Solaris. Open-Source Hardware. (9)

DEVELOPMENT TOOLS AND CASE STUDIES: IDEs, Debuggers, Programming languages, LAMP, Open-Source Database technologies - Contributing to Open Source as a Beginner, Different Roles In a Typical Open Source Project, Choosing Open Source Projects - Overview of Python open-source projects - Apache Web server, BSD, GNU/Linux - Android, Mozilla (Firefox), Wikipedia, Drupal, WordPress, Git, GCC, GDB, GitHub, Open Office, LibreOffice , Understanding the developmental models, licensing, mode of funding, commercial/non-commercial use. (9)

Total L: 45

TEXT BOOKS:

1. VM Brasseur, Forge Your Future with Open Source: Build Your Skills. Build Your Network. Build the Future of Technology, Pragmatic Bookshelf; 1st edition, 2018.
2. Regina Obe,Leo heu, PostgreSQL: Up and Running: A Practical Guide to the Advanced Open Source Database" O'REILLY publications, 2017.

REFERENCES:

1. Prem Kumar Ponuthorai, Jon Loeliger, "Version Control with Git", O'Reilly Media, Inc. 2022.
2. Kailash Vadera &Bhavyesh Gandhi "Open-Source Technology", University Science Press, Laxmi Publications, 2016.
3. Nihad A. Hassan, Rami Hijazi, "open Source Intelligence Methods and Tools: A Practical Guide to Online Intelligence", APRESS publications, 2018.
4. M. N. Rao, "Fundamentals of Open Source Software", Eastern Economy Edition, PHI Learning, 2014.

23N006 HUMAN COMPUTER INTERACTION

3 0 0 3

INTRODUCTION: The Human: I/O channels Memory Reasoning and problem solving; The Computer: Devices Memory processing and networks; Interaction: Model frameworks Ergonomics styles elements interactivity- Paradigms. - Case Studies (9)

DESIGN AND PROCESS: Interactive Design: Basics process scenarios navigation screen design Iteration and prototyping. HCI in software process: Software life cycle usability engineering Prototyping in practice design rationale. Design rules: principles, standards, guidelines, rules. Evaluation Techniques Universal Design (9)

MODELS AND THEORIES: HCI Models: Cognitive models- Socio-Organizational issues and stakeholder requirements Communication and collaboration models-Hypertext, Multimedia and WWW. (9)

MOBILE HCI: Mobile Ecosystem: Platforms, Application frameworks - Types of Mobile Applications: Widgets, Applications, Games- Mobile Information Architecture, Mobile 2.0, Mobile Design: Elements of Mobile Design, Tools. - Case Studies (9)

WEB INTERFACE DESIGN:Designing Web Interfaces Drag & Drop, Direct Selection, Contextual Tools, Overlays, Inlays and Virtual Pages, Process Flow - Case Studies (9)

Total L: 45

TEXT BOOKS:

1. BenSchneiderman, "Designing the user interface : strategies for effective human-6th edition, Pearson; 2016.
2. Scott MacKenzie, "Human- Computer Interaction:An Empirical Research Perspective" ,1st Edition , Elsevier, 2012

REFERENCES:

1. Bill Scott and Theresa Neil, "Designing Web Interfaces", First Edition, O'Reilly, 2009.
2. Alan Dix, Janet Finlay, Gregory Abowd, Russell Beale, "Human Computer Interaction", 3rd Edition, Pearson Education, 2004.
3. Helen Sharp, Yvonne Rogers, "Interaction Design: Beyond Human-Computer Interaction", 5th Edition, Wiley,2019.
4. Battacharya A, "Human Computer Interaction", Mc Graw Hill India.

23N007 INFORMATION RETRIEVAL

3 0 0 3

INTRODUCTION: Boolean retrieval - IR problem - Inverted index - Processing Boolean queries - Extended Boolean model and ranked retrieval - Document delineation - Determining vocabulary of terms - Skip pointers Search structures for dictionaries - Wildcard queries - Spelling and phonetic correction (9)

INDEX CONSTRUCTION: Blocked sort-based indexing - Single-pass in-memory indexing - Distributed indexing -Dynamic indexing-Statistical properties of terms in IR -Dictionary compression -Postings file compression (9)

VECTOR SPACE MODEL AND EVALUATION: Term frequency and weighting - Vector space model - Queries as vectors - Computing vector scores - IR system evaluation - Standard text collections - Evaluation of unranked and ranked retrieval sets (9)

PROBABILISTIC AND LANGUAGE MODELS: Probability ranking principle - Binary independence model - Appraisal of probabilistic models - Language models - Query likelihood models - Merits and demerits of language models (9)

WEB SEARCH: Web characteristics - Search user experience - Index size and estimation - Near-duplicates and shingling - Web crawler features and architecture - URL frontier - Link analysis - Web as a graph - PageRank algorithm - Hubs and authorities (9)

Total L: 45

TEXT BOOKS:

1. Ricardo Baeza-Yates, Berthier Ribeiro-Neto, "Modern Information Retrieval: The Concepts and Technology behind Search", Addison Wesley, USA, 2012.
2. David A.Grossman and Ophir Frieder," Information Retrival: Algorithms and Heuristics", Springer 2012.

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1. Bruce Croft W, Metzler D, StrohmanT, "Search Engines: Information Retrieval in Practice", Addison Wesley, USA,2013.
2. Gerald K, "Information Retrieval Architecture and Algorithms", Springer, Heidelberg, 2013.
3. Hang Li, "Learning to Rank for Information Retrieval and Natural Language Processing", 2nd Edition, Morgan & Claypool Publishers, USA, 2014.
4. Stefan Büttcher, Charles L. A. Clarke, Gordon V. Cormack, "Information Retrieval: Implementing and Evaluating Search Engines", MIT Press, Cambridge, USA, 2016.

23N008 FULL STACK DEVELOPMENT

3 0 0 3

JAVASCRIPT AND BASICS OF MERN STACK: JavaScript Fundamentals - Objects - Generators, advanced iteration - Modules - DOM tree - Node properties - browser events - Event delegation - UI Events -Forms, controls - Document and resource loading - Mutation observer - Event loop: microtasks and macrotasks - MERN Components - React - Node.js - Express - MongoDB - Need for MERN - Server-Less Hello World - Server Setup - nvm - Node.js - npm. (9)

REACT: React Introduction - React ES6 - React Render HTML - React JSX - Components -React Classes - Composing Components - Passing Data - Dynamic Composition - React state - setting State - Async State Initialization - Event Handling Communicating from Child to Parent - Stateless Components - Designing components- React Forms - React CSS - React SaaS. (9)

NODE.JS AND EXPRESS: Node.js basics - Local and Export Modules - Node Package Manager - Node.js web server - Node.js File system - Node Inspector - Node.js EventEmitter - Frameworks for Node.js - Express.js Web App - Serving static Resource - Node.js Data Access - Express REST APIs - REST - Resource Based - HTTP Methods as Actions - JSON- Express - Routing - Handler Function - Middleware - The List API - Automatic Server Restart - Testing - The Create API - Using the List API - Using the Create API- Error Handling - Template Engine. (9)

MONGODB: MongoDB - MongoDB Basics - Documents - Collections - Query Language - Installation - The mongo Shell - Schema Initialization - MongoDB Node.js Driver - Reading from MongoDB - Writing to MongoDB - CRUD operations - projections - Indexing - Aggregaton - Replication - Sharding - Creating backup – Deployment. (9)

ADVANCED FEATURES: Modularization and Webpack - Routing with React Router - Forms - More Filters in the List API - UI Components - Update API - Delete API - React-Bootstrap - Bootstrap Installation - Navigation - Table and Panel - Forms - Alerts - Modals -Server Rendering - Basic Server Rendering - Handling State - MongoDB Aggregate - Pagination - Higher Order Components - Session Handling. (9)

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TEXT BOOKS:

1. Nader Dabit, Full Stack Serverless: Modern Application Development with React, AWS, and GraphQL, 2020, O′Reilly, First Edition.
2. Shama Hoque, Full-Stack React Projects: Learn Mern Stack Development, Packt Publishing Limited, 2020, First Edition.

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1. Frank Zammetti, Modern Full-Stack Development: Using Type Script, React, Node.js, Apress, First Edition.
2. Chris Northwood, The Full Stack Developer: Your essential guide to the everyday skills, Apress, 2018, First Edition.
3. JuhaHinkula, Hands-On Full Stack Development with Spring Boot 2 and React, Packt Publishing, 2019.
4. Jack Chan, Ray Chung, & Jack Huang, Python API Development Fundamentals, Packt Publishing, 2018.

23N009 DEVOPS

3 0 0 3

INTRODUCTION: Dev vs Ops Practices - Adopting DevOps - Identifying the Target State - Assessing the Current State - Choosing the Transformation Plays - Adopting the Transformation Plays - Business Case for a DevOps Transformation- DevOps plays for optimizing delivery pipeline. DevOps process Best practices - Maturity Life Cycle - Maturity Map - Progression Framework - Readiness Model - Maturity Checklists - Agile Framework. Source code management (Git): Version controlling with SVN and Git, Branching Workflows in SVN & GitHub Flow (9)

CONTINUOUS INTEGRATION AND DELIVERY: Best Practices for CI/CD - Jenkins overview -Jenkins Master, Node, Agent and Git (SCM) Integration with Jenkins - Integrating GitHub with Jenkins - Maven (Build) Tool Integration with Jenkins - Building Jobs with Jenkins - Source Code Review Gerrit - Repository Management - Testing with Jenkins - Setting up Unit Testing - Automated Test Suite - Continuous Delivery- Build Pipeline - Jenkins Features - Security in Jenkins Case study : Building CI/CD pipeline for Android, iOS, Angular, .Net, Python and NodeJs platforms (9)

CONTINUOUS DEPLOYMENT: Chef : Chef Landscape Components - Chef Server Features of Chef Server - Extended Features of Chef - Chef Automate Workflow - Compliance - Ansible: Prominent Features - Benefits of Ansible - Ansible Terminology, Key Concepts, Workflow, and Usage - Testing Strategies with Ansible - Monitoring - Splunk - Nagios Monitoring Tool for Infrastructure Artifact management: Nexus, JFrog containerization using Kubernetes (OPENSHIFT): Kubernetes Namespace & Resources. AWS & AZURE CLOUD: Introduction to AWS & Azure Clouds, Pipeline of AWS & Azure Clouds (9)

DEVSECOPS: DevOps to DevSecOps - Types of attacks - Adversaries and their weaponsDevSecOps - Security implied in DevOps - Points of contention between DevOps and security teams - A layered approach to effective DevSecOps - Three layers overview - Security Education - Secure By Design - Security Automation - DevSecOps Lifecycle DevSecOps Methodologies DevSecOps Tools Container Platforms: Docker Podman Kubernetes Docker Security principles and best practices Docker capabilities content trust registry - Host Security images security Kubernetes Security (9)

MLOPS: Overview People - Key Features - MLOps Lifecycle - Developing Models - Preparing for Production - Monitoring and Feedback Loop Model Governance - MLOps in Practice (9)

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TEXT BOOKS:

1. Sanjeev Sharma The DevOps Adoption Playbook: A Guide to Adopting DevOps in a Multi-Speed IT Enterprise, John Wiley & Sons, Inc., IBM Press, 2017.
2. Gene Kim, John willis, Patrick Debois, Jez Humb, “the DevOps Handbook: How to create world- class Agility, Reliability and Security in Technology Organization”, O’Reilly Publications, 2016.

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1. Noah Gift, Alfredo Deza, Practical MLOps, O'Reilly Media, Inc.2021.
2. Mark Treveil, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, Lynn Heidmann Introducing MLOps , O'Reilly Media, Inc, 2020
3. SricharanVadapalli, DevOps - Continuous Delivery, Integration, and Deployment with DevOps ,Packt Publishing, 2018
4. Ankita Patil, Mitesh Soni, Hands-on Pipeline as Code with Jenkins CI/CD Implementation for Mobile, Web, and Hybrid Applications Using Declarative Pipeline in Jenkins , BPB publications, 2021.

23N010 ADVANCED DATA STRUCTURES

3 0 0 3

AMORTIZED ANALYSIS AND SETS : Amortization - Methods - Applications. DISJOINT SETS: Equivalence Relations Dynamic Equivalence Problem Basic Data Structure Smart Union Algorithms Path Compression. (9)

HEAP STRUCTURES: Min - Max Heaps - Deaps - Leftist Heaps - Binomial Heaps - Fibonacci Heaps. (8)

SEARCH TREES: Red-Black Tree - AA Tree - Interval Tree - Splay Trees. (7)

MULTIDIMENSIONAL STRUCTURES: K - D Trees - Point Quad Trees - MX- Quad Trees - R - Trees - TV Trees. (9)

GRAPH ALGORITHMS: Network Flow Problems Biconnectivity Euler circuits, RANDOMIZED ALGORITHMS ; Random number generators Skip lists primality testing (12)

Total L: 45

TEXT BOOKS:

1. Mark Allen Weiss, "Data structures and Algorithm Analysis in C++", 6th edition, Pearson Education, New Delhi, 2013.
2. Ellis Horowitz, Sartaj Sahni and Dinesh Mehta, "Fundamentals of Data Structures in C++", University Press, New Delhi, 2013.

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1. Peter Brass, "Advanced Data Structures", Cambridge University Press, USA, 2014.
2. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest and Clifford Stein, "Introduction to Algorithms", MIT Press, Massachusetts 2009.
3. Subrahmanian V S , "Principles of Multimedia Database Systems", Morgan Kaufman, USA, 2001.
4. Peter Sanders, Kurt Mehlhorn, Martin Dietzfelbinger, Roman Dementiev, "sequential and parallel algorithms and data structures " , Springer, 2019

23N011 ETHICAL HACKING

3 0 0 3

ETHICAL HACKING OVERVIEW & VULNERABILITIES: Understanding the basics of hacking-Types of hackers- Importance of security-Concept of ethical hacking and essential Terminologies- Threat- Attack- Vulnerabilities- Target of Evaluation-Exploit. Phases involved in hacking. (9)

FOOT PRINTING, RECONNAISSANCE AND SCANNING NETWORKS: Footprinting Concepts - Footprinting through Search Engines, Web Services, Social Networking Sites, Website, Email - Competitive Intelligence - Footprinting through Social Engineering -Footprinting Tools - Network Scanning Concepts - Port-Scanning Tools - Scanning Techniques -Scanning Beyond IDS and Firewall. (9)

SYSTEM HACKING: Aspect of remote password guessing- Role of eavesdropping -Various methods of password cracking- Keystroke Loggers- Understanding Sniffers Comprehending Active and Passive Sniffing- ARP Spoofing and Redirection- DNS and IP Sniffing- HTTPS Sniffing. (9)

HACKING WEB SERVICES & SESSION HIJACKING: Web application vulnerabilities- Application coding errors- SQL injection into Back-end Databases- Cross-site scripting- cross-Site request forging- Authentication bypass- Web services and related flaws- Protective http headers Understanding Session Hijacking- Phases involved in Session Hijacking-Types of Session Hijacking- Session Hijacking Tools. (9)

HACKING WIRELESS NETWORKS: Hacking Wireless Networks - Wireless Concepts - Wireless Encryption - Wireless Threats - Wireless Hacking Methodology - Wireless Hacking Tools - Bluetooth Hacking - Countermeasures- Wireless Security Tools - Wi-Fi Pen Testing. (9)

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TEXT BOOKS:

1. Rafay Baloch, “Ethical Hacking and Penetration Testing Guide”, CRC Press, 2014.
2. Ethical Hacking -A Comprehensive Beginner's Guide to Learn and Master Ethical Hacking, Hein smith, hilarymorrison · 2018.

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1. Kevin Beaver, “Ethical Hacking for Dummies”, Sixth Edition, Wiley, 2018.
2. Michael T. Simpson, "Hands-on Ethical Hacking & Network Defense", Course Technology, 2010.
3. Patrick Engebretson, “The Basics of Hacking and Penetration Testing” Ethical Hacking and Penetration Testing Made Easy, Syngress Media, Second Revised Edition, 2013.
4. Ramachandran V, “Wireless Penetration Testing Beginner’s Guide“, 3rd edition, Packt Publishing, 2011.

23N012 COMPILER DESIGN

3 0 0 3

INTRODUCTION TO COMPILERS: Translators - Compilation and Interpretation - The Phases of Compiler Errors Encountered in Different Phases - The Grouping of Phases - Compiler Construction Tools. (7)

LEXICAL ANALYSIS: Need and Role of Lexical Analyzer - Input Buffering - Lexical Errors - Expressing Tokens by Regular Expression - Finite Automata: NFA- DFA - Converting NFA to DFA - Minimization of DFA- Converting Regular Expression to DFA. LEX Tool: Structure of LEX Program–Predefined Variables–Library routines–Design of Lexical Analyzer for a Sample Language. (9)

SYNTAX ANALYSIS: Need and Role of the Parser - Context Free Grammars - Top Down Parsing: Recursive Descent Parser- Predictive Parser. Bottom Up Parsers: Shift Reduce Parser - LR Parser - LR (0) Item - Construction Of SLR Parsing Table - CLR Parser - LALR Parser. Error Handling and Recovery in Syntax Analyzer YACC Tool: Structure of YACC Program Communication between LEX and YACC - Design of a Syntax Analyzer for a Sample Language. (10)

INTERMEDIATE CODE GENERATION: Benefits- Intermediate Languages - Generation of Three Address Code - Declarations - Assignment Statements - Arrays - Boolean Expressions - Backpatching - Flow of Control Statements Procedure calls. (9)

RUN-TIME ENVIRONMENT, CODE OPTIMIZATION AND GENERATION: Source Language Issues - Storage Organization- Storage Allocation - Symbol Tables. Principal Sources of Optimization - Optimization of Basic Blocks - Global OptimizationGlobal Data Flow Analysis - Issues in Design of A Code Generator - A Simple Code Generator Algorithm. (10)

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TEXT BOOKS:

1. Alfred V Aho, Monica Lam, Ravi Sethi and Jeffrey D Ullman, “Compilers- Principles, Techniques and Tools”, Essex Pearson, Harlow, 2014.
2. V Raghavan, “Principles of Compilar Design”, tata Mcgraw Hill Publishing Co Ltd, 2016.

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1. Dick Grone, Henn E Bal,Ceriel J H Jacobs and Koen G Langendoen, “ Modern Compiler Design”, John Wiley & Sons, USA, 2000.
2. Dhamdhere D M,” Compiler Construction Principals & practice”, Macmillan India Limited, NewDelhi, 1997.
3. Jean Paul Tremblay and Paul G Sorenson, “the Theory & Practice of Compiler Writing”, McGraw Hill Publishing Company, New Delhi, 1985.
4. Sudha SadasivamG,”Compiler Design”, Scitech Publications (India) Private Limited, Chennai, 2010.

23N013 BLOCKCHAIN TECHNOLOGY

3 0 0 3

INTRODUCTION: Distributed System, P2P system, Distributed Hash Table, Distributed Ledger Technology- Private, public and

permissioned ledgers - Cryptographic primitives- public key cryptography- Digital Signature Algorithm -Hashing- Blockchain evolution- Structure of Blockchain Life of Blockchain application - consensus Byzantine General problem and Fault Tolerance (11)

BLOCKCHAIN 1.0 - BITCOIN AND CRYPTOCURRENCY :Block Hash - structure of Block syntax , structures, and validation - Transaction life cycle- Transaction types Hash computation and Merkle Hash Tree -Bit coin and importance- Creation of coins Bitcoin P2P Network-, Bitcoin protocols - Mining strategy and rewards PoW and PoS Difficulty, Hash rate Wallets- Double spending forking- Token, Coinbase. (12)

BLOCKCHAIN 2.0 - ETHEREUM : Distributed applications (Dapps), Smart contracts, Ethereum Virtual Machines, Ethereum high level design, Ethereum addresses, Ethereum accounts, Transactions, Currency, Gas, Tokens, Decentralized Autonomous Organizations(DAOs), Bitcoin vs Ethereum Trie- Solidity programming writing smart contracts remix IDE TestNet- sample exercises - issues in solidity programming (12)

BLOCKCHAIN 3.0 HYPERLEDGER: Fabric- Architecture, Identities and Policies, Membership and Access Control, Channels, Transaction Validation, Writing smart contract using Hyperledger Fabric. (5)

APPLICATIONS: Know Your Customer (KYC), Food Security, Mortgage over Block chain, Block chain enabled Trade, Cross border payments (5)

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TEXT BOOKS:

1. Bina Ramamurthy, Blockchain in Action , Manning Publications, 1st Edition, 2020.
2. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder“ Bitcoin and CryptoCurrency Technologies” PrinstoneUniversity Press, 2016.

REFERENCES:

1. Josh Thompson, “BlockChain: The Blockchain for Begginings, Guide to BlockChain Technology and BlockChain Programming” Create Space Independent Publishing platform, 2017.
2. Vikram Dhilon, “David Metcalf and Max Hooper, “BlockChain Enabled Applications”, press, 2017.
3. Antony Lewis,” The Basics of Bitcoins and BlockChain: An introduction to Cryptocurrencies and the technology that Powers Them (Cryptography,Derivatives Investments, Futures Trading, Digital Assets, NFT)” Mango Media, 2018.
4. Ashwani Kumar, “Hyperleger Fabric In Depth by Ashwani Kumar”, !st Edition, BPB Publications, 2020.

23N014 CLOUD COMPUTING

3 0 0 3

INTRODUCTION TO CLOUD COMPUTING: The Vision of Cloud Computing - Defining a Cloud - A Cloud Computing Reference Model - Characteristics and Benefits - Challenges Ahead - Historical Developments - Building Cloud Computing Environments - Study of Cloud computing platforms and Technologies Case Study - Google Cloud, AWS. (9)

VIRTUALIZATION: Introduction - Hypervisors - Main Categories of Virtualization - Application Server - Application Network - Storage Service - Benefits of Virtualization - Cost of Virtualization Case study: Creation of Virtual machines in Google Cloud, AWS. (9)

CLOUD COMPUTING ARCHITECTURE AND CLOUD SERVICES: Introduction to Cloud Computing Architecture - Cloud Reference Model -Microservices - Architecture - RESTful Web Services- Storage as a Service - Database as a Service - Information as a Service - Process as a Service - Application as a Service - Platform as a Service - Integration as a Service - Security as a Service - Management as a Service - Testing as a Service - Infrastructure as a Service - Case Study - Google Cloud, AWS. (10)

CLOUD APPLICATIONS: Scientific Applications - Gene Expression Data Analysis for Cancer Diagnosis - Business and Consumer Applications - Social Networking - Media Applications. (8)

CLOUD COMPUTING INFRASTRUCTURE: Introduction to Dockers Deploying, maintaining and scaling containerized applications using cloud infrastructure - Cloud storage Identity Access Management Cloud monitoring - Case Study: Google Cloud, AWS. (9)

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TEXT BOOKS:

1. Rajkumar Buyya, Christian Vecchiola, ThamaraiSelvi, “Mastering Cloud Computing”, Tata McGraw Hill Education Private Limited, New Delhi, 2017.
2. Kai Hwang, “Distributed and Cloud Computing: From Parallel Processing to the Internet of Things”, 1st edition, Morgan Kaufmann, 2013

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1. Shailendra Singh, “Cloud Computing” 1st edition, Oxford University Press, 2018.
2. Sandeep Bhowmik, “Cloud Computing” 1st edition, Cambridge University Press, July 2017.
3. Kris Jamsa, “Cloud Computing” 2nd edition, Jones &BArtiett Learning Publisher, 2022.
4. Priyanka Vergadia, “Visualizing Google Cloud: Illustrated References for Cloud Engineers & Architects: 101 Illustrated References for Cloud Engineers & Architects” 1st edition , John Wiley & sons Inc, May 2022.

23N015 DATA EXPLORATION AND VISUALIZATION

3 0 0 3

EXPLORATORY DATA ANALYSIS :EDA fundamentals – Understanding data science – Significance of EDA – Making sense of data – Comparing EDA with classical and Bayesian analysis – Software tools for EDA – Visual Aids for EDA- Data transformation techniques-merging database, reshaping and pivoting, Transformation techniques – Grouping Datasets – data aggregation – Pivot tables and cross-tabulations. (9)

VISUALIZING: Simple line plots – Simple scatter plots – visualizing errors – density and contour plots – Histograms – legends – colors – subplots – text and annotation – customization – three dimensional plotting – Geographic Data with Basemap – Visualization with Seaborn. (9)

UNIVARIATE ANALYSIS: Introduction to Single variable: Distributions and Variables – Numerical Summaries of Level and Spread – Scaling and Standardizing – Inequality – Smoothing Time Series. (9)

BIVARIATE ANALYSIS: Relationships between Two Variables – Percentage Tables – Analyzing Contingency Tables – Handling Several Batches – Scatterplots and Resistant Lines – Transformations. (9)

MULTIVARIATE AND TIME SERIES ANALYSIS: Introducing a Third Variable – Causal Explanations – Three-Variable Contingency Tables and Beyond – Longitudinal Data – Fundamentals of TSA – Characteristics of time series data – Data Cleaning – Time-based indexing – Visualizing – Grouping – Resampling. (9)

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TEXT BOOKS:

1. Suresh Kumar Mukhiya, Usman Ahmed, “Hands-On Exploratory Data Analysis with Python”, Packt Publishing, 2020.
2. Catherine Marsh, Jane Elliott, “Exploring Data: An Introduction to Data Analysis for Social Scientists”, Wiley Publications, 2nd Edition, 2008.

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1. Jake Vander Plas, “Python Data Science Handbook: Essential Tools for Working with Data”, Oreilly, 1st Edition, 2016.
2. Eric Pimpler, Data Visualization and Exploration with R, GeoSpatial Training service, 2017.
3. Claus O. Wilke, “Fundamentals of Data Visualization”, O’reilly publications, 2019.
4. Matthew O. Ward, Georges Grinstein, Daniel Keim, “Interactive Data Visualization: Foundations, Techniques, and Applications”, 2nd Edition, CRC press, 2015.

23N016 INTERNET OF THINGS

3 0 0 3

THE FUNDAMENTALS OF AN IOT ECOSYSTEM: Introduction: Definitions and Functional Requirements - IoT Architecture - Basics of Sensors and Actuators - Analog Sensors - Digital Sensors - Actuators. IoT Connectivity Protocols**:** IEEE 802.15.4 Zigbee Architecture 6LOWPAN LoRA-COAP MQTT (9)

IOT ARCHITECTURE FOR ENTERPRISES: Enterprise IoT Technology Stack-Middle ware layer Micro service Vs Monolithic Architecture. Moving Intelligence To The Edge**:** The integration of IoT and Cloud- Limitations of centralized IoT data processing The Need for Edge Analytics - Edge Analytics Architecture - Capabilities needed at Edge Devices - Running Data Analytics at Edge Devices (9)

ELECTRONIC PROTOTYPING: Prototypes and Production - Open Source versus Closed Source - Prototyping Embedded Devices- Prototyping IoT Projects with Arduino - Prototyping IOT Projects with Raspberry PI (9)

BUSINESS MODELS FOR IOT PROJECTS: Design Thinking Principles- Basic building blocks of a business model – The Osterwalder’s Business Model Canvas- Lean business model - Business model for IoT based startups- Pitching an idea – Funding opportunities for startups (9)

DESIGN AND IMPLEMENTATION OF ENTERPRISE IOT SOLUTIONS: Enterprise IoT development Platforms - Case studies: Connected Vehicles - Connected Healthcare - Smart Grid - Industry 4.0 (9)

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TEXT BOOKS:

1. Internet of Things, Samuel Greengard, MIT Press, 2021
2. Internet of Things : Enabling Technologies, Platforms, and Use Cases, Pethuru Raj, Anupama C. Raman, CRC Press, Taylor & Francis Group, 2017.

REFERENCES:

1. Cognitive Internet of Things : Enabling Technologies, Platforms, and Use Cases, Pethuru Raj, Anupama C. Raman, CRC Press, Taylor & Francis Group, 2022.
2. Enterprise IoT: Strategies and Best Practices for Connected Products and Services, Dirk Slama (Author), Frank Puhlmann (Author), Jim Morrish (Author), O,Reilley , 2015
3. Wei Chang, Jie Wu ,Fog/Edge Computing For Security, Privacy, and Applications, Springer 2021.
4. L.S. Jayashree, G,Selvakumar, “Getting Started with Enterprise IoT: Design Approches and Software Architecture Models”, Springer, 2020.

23N017 COMPUTER VISION

3 0 0 3

IMAGE PROCESSING TECHNIQUES: Introduction to vision challenges - Classical filtering operations: mode, median, rank order filters - Gaussian filters Thresholding techniques: Region-Growing methods - Adaptive thresholding - Edge detection techniques:

Template matching - Canny, Laplacian operators Corner and Interest point detection: Harris Interest point operator - Local Invariant Feature Detectors and Descriptors. (9)

SHAPES AND REGIONS: Binary shape analysis object labeling and counting size filtering distance functions skeletons and thinning deformable shape analysis boundary tracking procedures active contours shape models and shape recognition centroidal profiles handling occlusionboundary length measures boundary descriptors. (9)

LINE, CIRCLE AND ELLIPSE DETECTION: Line detection Hough Transform (HT) for line detection foot-of-normal method line localization line fitting RANSAC for straight line detection HT based circular object detection accurate center location speed problem ellipse detection Case study: Human Iris location - hole detection. (9)

3D VISION: Methods for 3D vision projection schemes shape from shading photometric stereo shape from texture shape fromfocus active range finding surface representations point-based representation volumetric representations 3D object recognition (9)

3D MOTION:3D reconstruction - introduction to motion - triangulation - bundle adjustment translational alignment parametric motion -spline-based motion -optical flow -layered motion. Application: Photo album - Face detection-Face recognition-OpenCV. (9)

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TEXT BOOKS:
1. E.R. Davies, “Computer & Machine Vision” 4th Edition, Academic Press, 2012.
2. R. Szeliski , "Computer Vision: Algorithms and Applications", 2nd Edition, Springer, 2022

REFERENCES:
1. David A Forsyth, Jean Ponce., ”Computer Vision : A Modern Approach”, 2nd Edition, Pearson, 2015.
2. Simon J.D. Prince,”ComputerVision:Models,Learning, and Inference”,Cambridge University Press, 2012.
3. Martin Goerner, Ryan Gillard, ValliappaLakshmanan,”Practical Machine Learning for Computer Vision: End- to- End Machine Learning for Images”, O'Reilly Media, 2021.
4. Jan Erik Solem , "Programming Computer Vision with Python: Tools and algorithms for analyzing images", O'Reilly Media, 2012.

23N018 EVOLUTIONARY COMPUTING ALGORITHMS

3 0 0 3

HEURISTIC AND META HEURISTIC APPROACHES: Challenges in Solving Complex Problems - Evolutionary Algorithms: Principles, Historical Development, Features, Classification and Components, Advantages, Applications. Heuristic Search: Hill Climbing: Principles, Local and Global maxima, Ridges, Plateau - Steepest Ascent - Simulated Annealing: Annealing schedule, Parameter Selection (9)

GENETIC ALGORITHM: Biological Background - Simple Genetic Algorithm (SGA) - Representation types - Recombination Types - Mutation Types - GA Algorithm - Schema Theorem - Variations of GA: Adaptive GA, Real Coded GA (7)

DIFFERENTIAL EVOLUTION: Principles, Mutation, Crossover, Selection. SWARM INTELLIGENCE: Particle Swarm Optimization: Swarms, Operating principles, PSO Algorithm, Neighborhood Topologies - Variations of PSO: Binary, weighted (10)

ANT COLONY OPTIMIZATION: Ant Foraging Behavior, Theoretical Considerations, ACO Algorithm, Variations of ACO: Elitist Ant System (EAS), Min Max Ant System (MMAS) and Rank Based Ant Colony System (RANKAS) (7)

MULTI-OBJECTIVE OPTIMIZATION AND MEMETIC ALGORITHMS: Multi-Objective Principles - Classical Methods - Challenges - Evolutionary algorithms for Multi-Objective Optimization - Multimodal Function Optimization - Non-Dominated Sorting Genetic Algorithm (NSGA): Non-Elitist, Elitist - Controlled Elitism in NSGA . (12)

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1. Eiben A E and Smith J E, “ Introduction to Evolutionary Computing”, 2nd edition, Springer, Heidelberg,2015.
2. Deb K, “Multi –Objective Optimization Using Evolutionary algorithms”, Wiley- Blackwell, USA, 2011.

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1. Rich E and Knight K,”Artificial Intelligence”, Tata McGraw Hill Education Private Limitted, India, 2011
2. Xin-She Yang, “nature Inspired Computation and Swarm Intelligence: Algorithms,Theroy of Applications “, Elsevier Science, United Kingdom, 2020.
3. Aboul Ella Hassanien and Eid/emery, Swarm Intelligence: Principles, Advances and Applications, CRC Press,New York, 2016.
4. Kenneth A.De Jong, “Evolutionary Computation: A Unified Approach”, Bradford Books, Reprint Edition, 2016.

23N019 NEURO FUZZY SYSTEM

3 0 0 3

FUNDAMENTALS OF ANN AND SIMPLE MODELS : Soft computing vs. hard computing–Artificial Neural networks :Introduction - Evolution of neural networks - characteristics- Gradient descent technique – ANN architectures and Learning algorithms, Activation functions, Bias, threshold and other parameters, McCulloch Pitts Model, Simulation of Logic functions, Perceptron Network, Hebbian Network,Linear Separability Problem, ADALINE and MADELINE networks (9)

SUPERVISED AND UNSUPERVISED LEARNING: Supervised learning network: perceptron networks - adaptive linear neuron, multiple adaptive linear neuron, BPN, RBF, TDNN associative memory network: auto-associative memory network, hetero-associative memory network, BAM, hopfield networks, unsupervised learning networks: Kohonen self-organizing feature maps. (10)

GENETIC ALGORITHM: Introduction —Basic concepts, Genetic modeling– Encoding- Genetic operators – Genetic programming – Multilevel optimization – Real life case studies- Advances in GA. (7)

FUZZY LOGIC : Introduction to Fuzzy logic - Fuzzy sets and membership functions - Operations on Fuzzy sets - Fuzzy relations, rules, propositions, implications and inferences - Defuzzification techniques - Fuzzy logic controller design -Some applications of Fuzzy logic. (9)

HYBRID SOFT COMPUTING TECHNIQUES & APPLICATIONS: NEURO-FUZZY MODELING : Adaptive Neuro-Fuzzy inference systems – Fuzzy Clustering Algorithms - Neuro-Fuzzy Control , Genetic neuro hybrid systems – Genetic fuzzy hybrid and fuzzy genetic hybrid systems — Applications: A fusion approach of multispectral images with SAR, optimization problems using genetic algorithm approach. Inference and Decision Support Systems: Fuzzy Cognitive Maps (10)

Total L: 45

TEXT BOOKS:

1. S.Rajasekaran, G.A.Vijayalakshmi Pai , "Neural Networks, Fuzzy Logic and Evolutionary Algorithm: Synthesis & Applications", Prentice-Hall of India Pvt. Ltd, 2017.
2. Lavika goel, “Artificial Intelligence: Concepts and Applications”,Wiley, 2021

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1. Charu C. Aggarwal, “Neural Networks and Deep Learning: A Textbook”, Springer Nature, 2023.
2. S.N.Sivanandam, S.N.Deepa , "Principles of Soft Computing", 3rd Edition, Wiley India Pvt Ltd, 2018.
3. George Timothy J. Ross, Fuzzy Logic with engineering applications , Fourth Edition, John Wiley & Sons, 2021
4. Eyal Wirsansky , “ Hands-On Genetic Algorithms with Python: Applying genetic algorithms to solve real-world deep learning and artificial intelligence problems”, Packt publishing, 2020
5. Simon Haykin, ”Neural Networks and Learning Machines”, Third Edition, Pearson, 2019.
6. David E. Goldberg , "Genetic Algorithm in Search Optimization and Machine Learning", Pearson Education India, 2013.

23N020 GENERATIVE AI

3 0 0 3

GAI MODELS AND PROMPT ENGINEERING: Introduction to generative AI - Main drawbacks of RNNs and CNNs – Transformer Architecture - Prompt Engineering: Definition - Elements of a good prompt design - Prompt parameters - Prompt Types - Langchain: Models – Prompt - Output Parsers – Chain. Case Study: Chatbot Integration for Customer Support. (9)

RAG: Stages in RAG – Chunking – Vector databases – Vector Search Strategies – Embeddings – Knowledge Graphs – RAG stages and evaluation with Langchain and LlamaIndex. Case Study : Q&A System for a Digital Library. (8)

AGENTS AND AGI: Types of agents (reactive, model-based, learning, social) – Agent Architecture - Communication and coordination between agents - Agents, Agent Executors, Tools, Toolkits, LangGraph in LangChain- Introduction to AGI – Types and Applications of AGI – Autogen. Case study: Text to sql. (10)

FINE TUNING: Introduction and Need - Fine-tuning applications - Pre-training and fine-tuning processes - Techniques for fine-tuning models - RLHF - Evaluate fine-tuned model performance – InstructGPT – Finetune model with SFT. Case Study: Fine-Tuning a LLM for Sentiment Analysis of Product Reviews. (9)

ADVANCED GAI: LLMOPs – Security threats and Security controls – Privacy – Multimodal GAI. (9)

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TEXT BOOKS:

1. Paul Singh, Anurag Karuparti, “Generative AI for Cloud Solutions”, Packt Publishing (O’Reilly), 2024.
2. Ben Auffarth, “Generative AI with LangChain: Build large language model (LLM) apps with Python, ChatGPT and other LLMs”, Packt Publishing , 2023.

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1. ZonunfeliRalte and IndrajitKar, “Learn Python Generative AI: Journey from autoencoders to transformers to large language models”, BPB Publications, 2024,
2. Denis Rothman, “Transformers for Natural Language Processing and Computer “, Packt Publishing, 2024.
3. Patrick Parra Pennefather, “Creative Prototyping with Generative AI: Augmenting Creative Workflows with Generative AI (Design Thinking)”, APress, 2023.
4. Joseph Babcock and Raghav Bali , “Generative AI with Python and TensorFlow 2: Create images, text, and music with VAEs, GANs, LSTMs, Transformer models”, Packt publishing, 2021

PROFESSIONAL ELECTIVE VERTICALS FOR HONORS

CYBER-PHYSICAL SYSTEMS

23N031 VIRTUALIZATION

3 0 0 3

INTRODUCTION - Virtual machines - Virtualization - need for virtualization - Before Virtualization VS After Virtualization - Benefits of Virtualization - Cost of Virtualization - Virtualization Limitations - Case study – Creation of Virtual machines using Google Cloud Platform, AWS. (9)

HYPERVISORS - Introduction to Hypervisors – Bare Metal Hypervisors: Architecture – Hosted Hypervisors: architecture – Embedded hypervisors: Architecture - Virtual Machine Creation - Virtual Machine Management - Virtual Machine Migration – Case Study - Xen-KVM. (9)

OS VIRTUALIZATION - Main Categories of Virtualization: Full virtualization - Para virtualization - Server virtualization - OS level virtualization - emulation–binary translation – Case Study - VMware ESXi, Xen. (9)

TYPES OF VIRTUALIZATION - Load Balancing - Application Virtualization – Storage Virtualization – Network Virtualization - Desktop Virtualization- Application Server Virtualization - Case Study – Application Streaming. (9)

SYSTEM LEVEL VIRTUALIZATION - Memory Virtualization – Ballooning - Sharing Memory across Virtual Machines - Memory Allocation Policies -CPU Virtualization - Device Virtualization - Containers – Case Study - VMware ESXi running with multiple virtual machines on a physical server. (9)

Total L: 45

TEXT BOOKS:

1. Priyanka Vergadia, “Visualizing Google Cloud: Illustrated References for Cloud Engineers & Architects: 101 Illustrated References for Cloud Engineers and Architects”,1st edition, John Wiley & Sons Inc, May 2022.
2. Lee Chao, “Virtualization and Private Cloud with VMware Cloud Suite”, Auerbach Publications, 1st edition, 2019.

REFERENCE BOOKS:

1. Dac-Nhuong Le, Raghvendra Kumar, Gia Nhu Nguyen, Jyotir Moy Chatterjee, “Cloud Computing and Virtualization”, Scrivener Publishing, Wiley, 2018.
2. Nadar Benmessaoud, Cj Williams, Uma Mahesh Mudigonda, Mitch Tulloch, “Microsoft System Center: Network Visualization and Cloud Computing”, Microsoft Press US, 2014.
3. Kai Hwang, Geoffrey C. Fox and Jack J. Dongarra, “Distributed and cloud computing from Parallel Processing to the Internet of Things”, Morgan Kaufmann, Elsevier – 2012.
4. Matthew Portney, Virtualization Essentials, John Wiley & Sons, 2012.
5. Dave Shackleford, Virtualization security, protecting virtualized environment, John Wiley, 2012.

23N032 EDGE COMPUTING

3 0 0 3

INTRODUCTION OF EDGE AND FOG COMPUTING: Internet of Things (IoT) and New computing paradigms, Fog computing: A platform for Internet of Things and analytics, Emergence of edge computing, Legal aspects of operating IoT applications in the fog. (9)

EDGE ARCHITECTURE: MULTI-TIER CLOUD COMPUTING FRAMEWORK;Data services with clouds at home; Leveraging mobile devices to provide cloud service at the edge; Fast, scalable and secure onloading of edge functions- The cloud-edge interplay in off –loading services (9)

NETWORKING FOR EDGE &FOG: Integrating IoT + Fog + Cloud Infrastructures: System modeling and research Challenges, Management and Orchestration of network slices in 5G, Fog, Edge, and Clouds. (9)

SYSTEM DESIGN: Optimization problems in fog and edge computing, Middleware for fog and edge Computing: Design issues, A Lightweight container middleware for edge cloud architectures –Edge analytics tools - Open source edge computing simulation platforms: - Case studies (9)

DATA PROCESSING: Data management in fog computing, Using machine learning for protecting the security and privacy of Internet of Things (IoT) systems, Edge Computing realization for Big data analytics, designing Edge Analytics solutions for different business applicatons. Edge Computing models - Case Studies: Cloud gaming , Manufacturing , Smart city applications, Smart surveillance video stream processing at - smart transportation applications. (9)

Total L: 45

TEXT BOOKS:

1. R. Buyya, S.N. Srirama, “Fog and Edge computing: Principles and Paradigms”, Wiley Blackwell, 2019.
2. Mohiuddin Ahmed, Paul Haskell-Dowland , “Secure Edge Computing, Applcations Techniques and Cheallenges “, O’Reilley, 2021

REFERENCES:

1. Wei Chang, Jie Wu, Fog/Edge Computing For Security, Privacy, and Applications, Springer 2021.
2. Cloud and Distributed Computing: Algorithms and Systems”;, Rajiv Misra, Yashwant Patel,Wiley 2020
3. The Internet of Things Enabling Technologies, Platforms, and Use Cases, By Pethuru Raj, Anupama C. Raman · 2017
4. L S Jayashree, G Selvakumar, “Getting Started with Enterprise IoT”, 1st edition, Springer, 2020.

23N033 NETWORK DESIGN

3 0 0 3

NETWORK DESIGN: Overview of Analysis, Architecture, and Design Processes - A Systems Methodology -System and Service Description - Performance Characteristics. Requirement Analysis: User, Application, Device, Network Requirements.(9)

NETWORK ARCHITECTURE: Component Architectures - Reference Architecture - Architectural Models - Addressing and Routing Architecture - Network Management Architecture - Performance Architecture - Security and Privacy Architecture. (9)

LOGICAL NETWORK DESIGN I: Hierarchical Network Design - Redundant Network Design Topologies -Modular Network Design - Designing the Enterprise Edge Topology - Secure Network Design Topologies Designing models for addressing and Numbering. (9)

LOGICAL NETWORK DESIGN II: Selecting Switching and Routing protocols Developing Network Security Strategies: Network Security Design Security Mechanisms - Modularizing Security Design Developing Network management Strategies: Network Management Design - Selecting Network Management Tools and Protocols. (9)

PHYSICAL NETWORK DESIGN: Selecting Technologies and Devices for Campus Networks: LAN Cabling Plant Design - Selecting Internetworking Devices for a Campus Network Design - Selecting Technologies and Devices for Enterprise Networks: Remote- Access Technologies - Selecting Remote-Access Devices for an Enterprise Network Design. (9)

Total L: 45

TEXT BOOKS:

1. P. Oppenheimer, "Top-Down Network Design," Cisco Press, 2010
2. Michel Thomatis,2nd Edition, Lulu.com Publisher, 2019

REFERENCES:

1. Marwan Al-shawi, Andre Laurent, " Designing for Cisco Network Service Architectures, 4th edition, ," Cisco Press, 2017
2. J. McCabe, "Practical Computer Network -- Analysis and Design," Morgan Kaufmann Publishers, Inc. 1998
3. T. Quinn-Andry and K. Haller, "Designing Campus Networks," Cisco Press, 1998
4. Teresa C Piliouras Network Design: Management and Technical Perspectives," CRC Publisher, 1988

23N034 SOFTWARE DEFINED NETWORKS

3 0 0 3

INTRODUCTION: History and Evolution of Software Defined Networking (SDN) - IETF Forces - Active Networking. - Control and Data Plane Separation: Concepts, Advantages and Disadvantages - OpenFlow protocol - South Bound Architecture - North Bound Architecture (9)

NETWORK FUNCTION VIRTUALIZATION: Concepts - Applications - Existing Network Virtualization Frameworks - Mininet based examples. - Control Plane - Overview - Existing SDN Controllers including Floodlight and OpenDaylight projects - Customization of Control Plane - Switching Implementation using SDN Concepts (10)

DATA PLANE: Software-based and Hardware-based Approaches - Programmable Network Hardware - Programming SDNs: Northbound Application Programming Interface - Current Languages and Tools - Composition of SDNs (8)

SOFTWARE DEFINED NETWORKS FOR THE INTERNET-OF-THINGS: Challenges - Understanding the nature of IoT traffic flows in different use cases- - A software defined end-to-end IoT Infrastructure - Resource provisioning in the IoT Multinetwork environments - Addressing scalability and security issues- Adding SDN automation and verification in IoT infrastructure. (9)

USE CASES OF SDNS: Data Centers - Internet Exchange Points, - Backbone Networks - Home automation Systems - Industrial automation Systems and Smart grids. (9)

Total L: 45

TEXT BOOKS:

1. Thomas D. Nadeau, Ken Gray, "SDN: Software Defined Networks, An Authoritative Review of Network Programmability Technologies", O'Reilly Media, 2013.
2. Paul Goransson and Chuck Black, "Software Defined Networks: A Comprehensive Approach", June, Morgan Kaufmann, 2014.

REFERENCES:

1. Vivek Tiwari, "SDN and Open Flow for Beginners", ASIN, 2013.
2. Fei HU, "Network Innovation through Open Flow and SDN: Principles and Design", CRC Press, 2014.
3. Sriram Subramanian, Sreenivas Voruganti , "Software-Defined Networking (SDN) with Open Stack", Pact Publishing, India, 2016.
4. Ken Gray Thomas Nadeau , "Network Function Virtualization", Elsevier, 2016.

23N035 SEMANTIC WEB TECHNOLOGY

3 0 0 3

SEMANTIC WEB VISION AND STRUCTURED WEB DOCUMENTS: Introduction to Semantic web - Evolution of web-Semantic Web Technologies - Recommended Layered Architectures. Structured web documents- The XML Language: Structuring - Namespaces - Addressing and Querying XML Documents - Processing. (9)

DESCRIBING WEB RESOURCES: Introduction - RDF: Basic Ideas - `-Based Syntax. RDF Schema: Basic Ideas - RDF and RDF Schema IN RDF Schema - An Axiomatic Semantics for RDF and RDF Schema Querying in SPARQL (9)

ONTOLOGY ENGINEERING AND OWL: Introduction - Constructing Ontologies Manually - Reusing Existing Ontologies - Using Semiautomatic Methods - On-to-Knowledge Semantic Web Architecture OWL Language Ontology Examples- OWL In OWL - Future Extensions. (9)

LOGIC AND INFERENCE: Rules - Monotonic Rules: Syntax - Semantics - Representing Family Relationships. Non monotonic Rules: Syntax - Brokered Trade as an Example - Monotonic and Non monotonic Rule Markup Case Study. (9)

TOOLS AND APPLICATIONS: Development tools for semantic web- Jena Framework- Semantic Wikis-Semantic web service, Horizontal Information Products at Elsevier - Data Integration at Audi - Skill Finding at Swiss Life.(9)

Total L: 45

TEXT BOOKS:

1. Grigoris Antoniou, Frank vanHarmelen, "Semantic Web Primer", MIT press, USA, 2008.
2. Michael C Daconta, Leo J Obrst, Kevin T Smit, "The Semantic Web: A Guide to the Future of XML, Web Services, and Knowledge Management", Wiley, USA, 2003.

REFERENCES:

1. Pascal Hitzler, Markus Krotzsch, Sebastian Rudolph, "Foundations of Semantic Web Technologies", CRC Press, 2009.
2. John Hebeler, Matthew Fisher, Ryan Blace, Andrew Perez-Lopez, "Semantic Web Programming", 1st Edition, Wiley, 2009.
3. Liyang Yu, "A Developer's Guide to the Semantic Web", First, Springer, 2011.
4. Ducharme B, "Learning SPARQL", 1st Edition, O'Reilly Media, 2011.

23N036 SPEECH PROCESSING

3 0 0 3

BASIC CONCEPTS: Speech Fundamentals: Articulatory Phonetics – Production and Classification of Speech Sounds; Acoustic Phonetics – acoustics of speech production; Review of Digital Signal Processing concepts; Short-Time Fourier Transform, Filter-Bank and LPC Methods. (9)

SPEECH ANALYSIS: Features, Feature Extraction and Pattern Comparison Techniques: Speech distortion measures – mathematical and perceptual – Log Spectral Distance, Cepstral Distances, Weighted Cepstral Distances and Filtering, Likelihood Distortions, Spectral Distortion using a Warped Frequency Scale, LPC, PLP and MFCC Coefficients, Time Alignment and Normalization – Dynamic Time Warping, Multiple Time – Alignment Paths. (9)

SPEECH MODELING: Hidden Markov Models: Markov Processes, HMMs – Evaluation, Optimal State Sequence – Viterbi Search, Baum-Welch Parameter Re-estimation, Implementation issues. (9)

SPEECH RECOGNITION: Large Vocabulary Continuous Speech Recognition: Architecture of a large vocabulary continuous speech recognition system – acoustics and language models – n-grams, context dependent sub-word units; Applications and present status. (9)

SPEECH SYNTHESIS: Text-to-Speech Synthesis: Concatenative and waveform synthesis methods, sub-word units for TTS, intelligibility and naturalness – role of prosody, Applications and present status. (9)

TEXT BOOKS:

1. Lawrence Rabiner and Biing-Hwang Juang, "Fundamentals of Speech Recognition", Pearson Education, 2009.
2. DanielJurafsky and James H Martin, "Speech and Language Processing – An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition", Third Edition, Pearson Education, 2023

REFERENCES:

1. Thomas F Quatieri, "Discrete –Time Speech Signal Processing", Pearson Education - India, 2015.
2. Rabiner L R , Schaffer R W , "Digital Processing of Speech Signals", Pearson Education - India, 2015.
3. Owens FJ, "Signal Processing of Speech", Macmillan, 2015.
4. John R Deller Jr, John H L Hansen, John G Proakis , "Discrete Time Processing of Speech Signal", IEEE press, 2015.

23N037 COGNITIVE ANALYTICS

3 0 0 3

PHILOSOPHY, PSYCHOLOGY AND NEUROSCIENCE: Philosophy: Mental-physical Relation – From Materialism to Mental Science – Detour before the naturalistic turn – The Philosophy of Science – The Mind in Cognitive Science – Logic and the Sciences of the Mind – Psychology: Place of Psychology within Cognitive Science – Science of Information Processing – Neurosciences: Cognitive Neuroscience –Perception – Decision – Learning and Memory – Language Understanding and Processing.

COMPUTATIONAL INTELLIGENCE: Machines and Cognition – Artificial Intelligence – Architectures of Cognition – Knowledge Based Systems – Logical Representation and Reasoning – Logical Decision Making – Decision making under Uncertainty – Learning – Language – Vision – Robotics.

PROBABILISTIC PROGRAMMING LANGUAGE : WebPPL Language – Syntax – Using JavaScript Libraries – Manipulating probability types and distributions – Finding Inference – Exploring random computation – Coroutines: Functions that receive continuations –Enumeration – Other basic computation.

IMPLEMENTING THE INFERENCE MODELS OF COGNITION: Generative Models – Conditioning – Causal and statistical dependence – Conditional dependence – Data Analysis – Algorithms for Inference.

IMPLEMENTING THE LEARNING MODELS OF COGNITION: Learning as Conditional Inference – Learning with a Language of Thought – Hierarchical Models – Occam's Razor– Learning (Deep) Continuous Functions – Mixture Models.

Total L: 45

TEXT BOOKS:

1. Hurwitz, Kaufman, and Bowles, "Cognitive Computing and Big Data Analytics", Wiley, Indianapolis, IN, 2005.
2. Masood, Adnan, Hashmi, Adnan,"Cognitive Computing Recipes-Artificial Intelligence Solutions Using Microsoft Cognitive Servicesand TensorFlow", 2015.

REFERENCES:

1. Peter Fingar, "Cognitive Computing: A Brief Guide for Game Changers", PHI Publication, 2015
2. Gerardus Blokdyk ,"Cognitive Computing Complete Self-Assessment Guide", 2018
3. Rob High, Tanmay Bakshi, "Cognitive Computing with IBM Watson: Build smart applications using Artificial Intelligence as a Service", IBM Book Series, 2019.
4. Jeff Johnson, "Designing with the Mind in Mind – Simple Guide to Understanding", 2nd edition, Elsevier., 2014.

23N038 AUGMENTED REALITY

3 0 0 3

INTRODUCTION TO VIRTUAL REALITY: Basic features of VR systems – The three I's of virtual reality, Commercial VRtechnology and the five classic components of a VR system - VR Input hardware: Tracking systems, motion capture systems and gesture interfaces- VR Output hardware: visual displays- sound displays & haptic feedback - Content creation considerations for VR - VR health and safety issues - cyber sickness - side effects of exposures to VR environment. (9)

VR DEVELOPMENT PROCESS: Geometric Modeling Virtual Object Shape Object – Visual Appearance -Kinematics Modeling Transformation Matrices Object Position Transformation Invariants Object Hierarchies Viewing The 3D World Physical Modeling–Collision Detection – Surface Deformation Force Computation – Force Smoothing And Mapping-Behavior Modeling Model Management. (9)

AUGMENTED REALITY: System structure of Augmented Reality - Key technologies in AR - Marker-less tracking for Augmented Reality - Enhancing interactivity in AR environment - General solution for calculating geometric & illumination consistency in the augmented environment - Evaluating AR systems. (9)

AUGMENTED AND MIXED REALITY: Taxonomy, technology and features of augmented reality - difference between AR and VR - Challenges with AR - AR systems and functionality - Augmented Reality methods - Visualization techniques for Augmented Reality (9)

METAVERSE: Introduction to Metaverse and immersive experience - Metaverse value chain with 7 layers - AR, VR and MR in Metaverse – Blockchain adoption in Metaverse - Need of decentralization in Metaverse - Understanding Tokens, NFTs - NFT Token Standards - Use Cases: Gaming, Virtual Learning - Social Interaction - Personalized Avatars - Digital Identity in Metaverse. (9)

Total L: 45

TEXT BOOKS:

1. C.Burdea& Philippe Coiffet," Virtual Reality Technology", 2nd edition Gregory, John Wiley & Sons, Inc., 2013.
2. Alan B Craig," Understanding Augmented Reality", Concepts and Applications, Morgan Kaufmann, 1st edition, 2013.

REFERENCES:

1. Dieter Schmalstieg& Tobias Hollerer,"AugmentedRiality: Principles & Practices, Pearson Education (USA). Addison-Wesley Educational Publishers Inc, New Jersey, United States, 2016
2. Steve Aukstakalinis, "Practical Augmented Reality: A Guide to the technologies, Applications and Human Factors for AR and VR", Addision-Wesley Professional, 1st Edition, 2016.
3. Robert Scoble& Shel Israel, "The Fourth Transformation: how augumented Reality &aritificial Intelligence Will Change Everything", Patrick Brewster Press, 1st edition, 2016.
4. William R. Sherman, Alan B. Craig, : Understanding Virtual Reality: Interface, Application and Design", Morgan Kaufmann, 2008

23N039 ROBOTICS

3 0 0 3

INTRODUCTION : Introduction to robotics- History, growth- Robot applications- Manufacturing industry, defense, rehabilitation and medical - Laws of Robotics - Degrees of freedom of planar and spatial manipulator - Robot classifications - work envelope - Internal Grippers and External Grippers - Selection and Design Considerations -resolution, accuracy and repeatability of robot (9)

ROBOT KINEMATICS: Representation of objects in 3-D space-position and orientation - Frame transformations - Composite rotation matrix - Representation of position in cylindrical, spherical coordinate system - Representation of orientation using roll,

pitch and yaw angles - Representation of orientation using Euler angles - Denavit-Hartenberg notations- Link and Joint parameters- Rules for coordinate assignments - Forward and inverse kinematics - velocity analysis (9)

TRAJECTORY AND MOTION PLANNING: Linear trajectory function - Polynomial trajectory function - Gross and fine motion planning - Motion planning schemes- Visibility graph - Vornoi diagram - Tangent graph - Accessibility graph - Path velocity decomposition - Incremental planning - Relative velocity approach - Reactive control strategy and potential field approach (9)

ROBOT DYNAMICS: Introduction to Inverse and Forward dynamics - Determination of inertia tensor - Lagrange - Euler formation for Joint torque - Control of robotic joints (9)

ACTUATORS AND SENSORS: Actuators and types - DC motors - BLDC servo motors - Introduction to sensors, characteristics - Sensor types-Touch, Potentiometer, Encoder, Force, Range and proximity - Economic Analysis of Robots. (9)

TEXT BOOK:

1. Fu. K.S., Gonzalez. R.C., George Lee. C.S., "Robotics –Control, Sensing, Vision, and Intelligence", McGraw Hill, 2015
2. Pratihar.D.K, "Fundamentals of Robotics", Narosa Publishing House,India,2019

REFERENCE BOOKS:

1. Groover Mikell .P, "Industrial Robotics -Technology Programming and Applications", McGraw Hill, 2014
2. Deb S.R., "Robotics Technology and Flexible Automation", Tata McGraw Hill Book Co., 2013.
3. Koren Y., "Robotics for Engineers", McGraw Hill Book Co., 1992
4. Maja J Mataric, "The Robotics Primer",Universities Press, 2013.
5. John J. Craig , "Introduction to Robotics Mechanics and Control", Pearson Education India,2008

23N040 NETWORK OPTIMIZATION

3 0 0 3

INTRODUCTION : Graphs and Flows -Network Flow Models -Network Flow Algorithms – Shortest Path Problems - Label Setting Methods - Label Correcting Methods - Single Origin/Single Destination Methods - Auction Algorithms – Multiple Origin/Multiple Destination Methods. (9)

MAX-FLOW AND MIN-COST FLOW PROBLEM: Max-Flow and Min-Cut Problems - Ford-Fulkerson Algorithm - Price-Based Augmenting Path Algorithms - Transformations and Equivalences – Duality - Case study : Routing, Duality for Assignment Problem (9)

SIMPLEX METHODS FOR MIN-COST FLOW: Main Ideas in Simplex Methods - Basic Simplex Algorithm - Extension to Problems with Upper and Lower Bounds - Case Study : Adding Additional Data Structures for implementing Simplex methods in Network Optimization for computing flow and price computation. (9)

DUAL ASCENT METHODS FOR MIN-COST FLOW: Dual Ascent -The Primal-Dual (Sequential Shortest Path) Method –The Relaxation Method -Sensitivity Analysis – Case Study: Implementation Issues aligned with exiting network. (9)

AUCTION ALGORITHMS FOR MIN-COST FLOW: The Auction Algorithm for the Assignment Problem - Extensions of the Auction Algorithm -The Preflow-Push Algorithm for Max-Flow -The Relaxation Method: Case Study: Challenge's in implementation and Relation to Auction Algorithm. (9)

Total L: 45

TEXT BOOKS:

1. Dimitri Bertsekas, "Network Optimization: Continuous and Discrete Methods", 1998.
2. M. Pioro and D. Medhi, Morgan Kaufmann, "Routing, Flow and Capacity Designing in Communication and Computer Network", Elsevier, 2016.

REFERENCES:

1. Terry L. Friesz, David Bernstein, "Foundations of Network Optimization and Games", Kindle edition, Springer, 2016.
2. Panos M. Pardalos, Ding-Zhu Du, "Network Optimization Problems: Algorithms, Applications and Complexity", World Scientific publishing, 1993.
3. Panos M. Pardalos, Donald W. Hearn, William W. Hager, "Network Optimization", Springer, 1997.
4. Pablo PavónMariño, "Optimization of Computer Networks: Modeling and Algorithms: A Hands-On Approach", Wiley, 2016

23N046 NETWORK MANAGEMENT & SECURITY

3 0 0 3

INTRODUCTION: Network Management Overview, Goals, Organization, and Functions, Network and System Management, Network Management System Platform, Current Status and future of Network Management. (9)

SNMP: SNMPv1: Communication and Functional Models, SNMPv2: Major Changes in SNMPv2, SNMPv2 System architecture, SNMPv2 Structure of Management Information, The SNMPv2 Management Information Base, SNMPv2Protocol, Compatibility with SNMPv1, SNMPv3 : Architecture, Protocol, Compatibility with SNMPv1 and SNMPv2 (9)

RMON: Remote Network Monitoring: Statistics Collection, Alarms and Filters, RMON1: Protocol Directory Group, Protocol Distribution Group, Address Map Group, RMON2: Host Groups, Matrix Group, Extensions to RMON1 for RMON2 Devices. (9)

INTRUSION DETECTION AND PREVENTIONS SYSTEMS: Network Intrusion detection (ID): Analysis Approaches, Host-Based ID, Network-Based ID, Distributed ID, Intrusion Detection Exchange Format, Honeypots, Snort, Intrusion prevention systems, Trusted Computing and Multilevel Security: Bell-Lapadula Model for Computer Security, Applications of Multilevel Security. (9)

DNS SECURITY AND SECURITY PROTOCOLS: DNS security issues and defenses: DNS Resolving, Caching, DNS Vulnerabilities, Cache Poisoning, Defense using Bailiwicks Rules, DNS Poisoning Defenses, TLS/SSL, Internet Security Protocols and Standards: IPSec, VPN, SSH. (9)

Total L: 45

TEXT BOOKS:
1. Michael W Lucas, “SNMP Mastry”, Tilted Windmill Press, 2020
2. William Stallings , "Cryptography and Network Security: Principles and Practice", Prentice Hall of India, Pearson Education, 2017

REFERENCES:
1. William Stallings, “SNMP, SNMPv2, SNMPv3, and RMON 1 and 2”, Addison-Wesley, 1998
2. Mani Subrahmanian, “Network Management Principles and Practice”, Pearson Education, 2010.
3. William Stallings and Lawrie Brown, “Computer Security: Principles and Practice”, Prentice Hall Press, 2014.
4. SabyasachiPramanik, DebabrataSamanta, M. Vinay, Abhijit Guha, “Cyber Security and Network Security”, John Wiley & Sons, 2022.

SECURITY AND PRIVACY

23N041 INFORMATION SECURITY

3 0 0 3

INTRODUCTION: History - Critical Characteristics of Information - NSTISSC Security Model - Components of an Information System - Securing the Components - Balancing Security and Access - The SDLC - The Security SDL (9)

SECURITY INVESTIGATION: Need for Security - Business Needs - Threats - Attacks - Legal - Ethical and Professional Issues in Information Security (9)

RISK MANAGEMENT: Risk Identification - Risk Assessment - Risk Control Strategies - Selecting a Risk Control Strategy (9)

STANDARDS AND PRACTICES: Blueprint for Security - Information Security Policy - Standards and Practices - ISO17799/BS 7799 - NIST Models - Design of Security Architecture - Continuity Strategies (9)

PHYSICAL DESIGN: Security Technology - IDS - Scanning and Analysis Tools - Cryptographic Algorithms and Tools - Physical Security - Implementing Information Security - security and Personnel - Information Security Maintenance - Digital Forensics (9)

Total L: 45

TEXT BOOKS:
1. Michael E Whitman, Herbert J Mattord , "Principles of Information Security", 6th Edition, Cengage Learning Inc, United States,2017.
2. Micki Krause, Harold F Tipton , "Handbook of Information Security Management,Volume 1-3", CRC Press LLC, 2011.

REFERENCES:
1. Matt Bishop , "Computer Security Art and Science", Pearson/PHI, 2003.
2. William Stallings , "Cryptography and Network Security Principles and Practice", 7th Edition, Prentice Hall of India, Pearson Education, New Delhi, 2017.
3. V. K. Pachghare , "Cryptography and Information Security", 2nd Edition, PHI Learning, Private Limited, 2015.
4. Ritendra Goel, Praveen Kumar Shukla, Surya Prakash Tripathi , "Introduction to Information Security and Cyber Laws", KogentLearning Solutions Inc, 2014.

23N042 CYBER SECURITY AND DIGITAL FORENSICS

3 0 0 3

INTRODUCTION : Cyber Security – History of Internet – Impact of Internet – CIA Triad; Reason for Cyber Crime – Need for Cyber Security – History of Cyber Crime; Cybercriminals – Classification of Cybercrimes – A Global Perspective on Cyber Crimes; Cyber Laws – The Indian IT Act – Cybercrime and Punishment. (8)

ATTACKS AND COUNTERMEASURES: Open Web Application Security Project (OWASP); Malicious Attack Threats and Vulnerabilities: Scope of Cyber-Attacks – Security Breach – Types of Malicious Attacks – Malicious Software – Common Attack Vectors – Social engineering Attack – Wireless Network Attack – Web Application Attack – Attack Tools – Countermeasures. RECONNAISSANCE : Harvester – Whois – Netcraft – Host – Extracting Information from DNS – Extracting Information from E-mail Servers – Social Engineering Reconnaissance; Types of Scanning –Scanning Methodology – Ping Sweer Techniques – Nmap Command Switches – SYN – XMAS – NULL – FIN Scans. (11)

INTRUSION DETECTION &PREVENTION: Types of Intrusion Detection: Host -Based , Network -Based , Distributed or Hybrid Intrusion Detection – Honeypots – Example System Snort. Firewalls and Intrusion Prevention Systems: Need for Firewalls – Firewall Characteristics and Access Policy – Types of Firewalls– Intrusion Prevention Systems – Example Unified Threat Management Products. (8)

DIGITAL FORENSICS: Forensic Science – Digital Forensics – Digital Evidence – The Digital Forensics Process – Introduction – The Identification Phase. Digital Crime and Investigation: Digital Crime – General Conditions – Offenses – Investigation Methods for Collecting Digital Evidence – International Cooperation to Collect Digital Evidence.Digital Forensic Readiness : Introduction – Law Enforcement versus Enterprise Digital Forensic Readiness – Rationale for Digital Forensic Readiness – Frameworks, Standards and Methodologies – Enterprise Digital Forensic Readiness – Challenges in Digital Forensics (10)

iOS and ANDROID FORENSICS : Mobile Hardware and Operating Systems - iOS Fundamentals – Jailbreaking – File System – iPhone Security – iOS Forensics – Procedures and Processes – Tools . Android basics – Key Codes – Rooting Android – Boot Process – File Systems – Security – Tools – Android Forensics – Forensic Procedures – Android Debugging Bridge (ADB) – Android Only Tools – Dual Use Tools – MobilEdit – Android App Decompiling (8)

Total L: 45

TEXT BOOKS:

1. Nina Godbole, SunitBelapure, “Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives”, Wiley Publishers, 1st Edition, 2022.
2. William Stallings, Lawrie Brown, “Computer Security Principles and Practice”, 4th Edition,Pearson Education, 2021.
3. Chuck Easttom, “An In-depth Guide to Mobile Device Forensics”, !st Edition, CRC Press, 2022.

REFERENCES:

1. David Kim, Michael G. Solomon, “Fundamentals of Information Systems Security”, 5th edition, Jones & Bartlett Learning Publishers, 2023.
2. Vacca, J,” Computer Forensics, Computer Crime Scene Investigation”, 2nd Edition, Charles River Media, 2015.
3. Patrick Engebretson, “The Basics of Hacking and Penetration Testing: Ethical Hacking and Penetration Testing Made easy”, Elsevier, 2nd Edition,2013.

23N043 WEB SERVICES AND SECURITY

3 0 0 3

WEBSERVICES: Architecture - Messaging - Service Description - Service Discovery - Service Transport Security. (9)

IMPLEMENTATION OF WEB SERVICES: Messaging-Service Description-Service Discovery-Service Transport Security-REST API Architecture-WSDL-UDDI-Web Service Clients and Service Invocation-WS* Standards (9)

MICROSERVICES: Role of Web services, Web services protocol stack ,Communication model, Micro services - Architectural Benefits, Deriving Business Value, Define and Apply Goal oriented Layered Approach, Design Process, Docker and Microservices, Role of Service Discovery, Applications of Microservices. (9)

REST BASED SERVICES: Principles-Comparison with WSDL -XML Based Web Services-Design and Implementation of REST services-Resource Oriented Architecture-best practices (9)

WS SECURITY: WS overarching concern, Core concepts, Challenges, Threats and remedies, Securing the communication layer, Message level security, OAUTH2.2 ,WS security framework, WS security policy, WS trust, WSsecure conversation, Data level security (9)

Total L: 45

TEXT BOOKS:

1. Greg Lomow and Eric Newcomer, "Understanding SOA with Web Services", Pearson Education, New Delhi, 2013.
2. Sanjay Patni, "Pro RESTful APIs: Design, Build and Integrate with REST, JSON, XML and JAX-RS", Springer, USA, 2017.

REFERENCES:

1. Eric Newcomer, "Django RESTful Web Services", PACKT Publishing, Birmingham, UK, 2018.
2. Irakli Nadareishvili Media, New Delhi, 2016.
3. JobineshPurushothaman, "RESTful Java Web Services", PACKT Publishing, Birmingham, UK, 2015.
4. Mike Rosen, Boris Lublinsky, Kevin T. Smith and Marc. J. Balcer, "Applied SOA: Service Oriented Architecture and Design Strategies", 2nd Edition, Wiley, Indianapolis, USA, 2012.

23N044 PRIVACY AND SECURITY IN ONLINE SOCIAL NETWORKS

3 0 0 3

INTRODUCTION: Social Networks, Offline to Online Communities, Evolution of Online Social Networks, Analysis and Properties, Security Issues in Online Social Networks, Challenges and privacy issues in social networks, Recognizing digital friends and challenges, Encryption techniques for P2P social networks, Trust Management in Online Social Networks, Controlled Information Sharing in Online Social Networks, Identity Management in Online Social Networks, data collection from social networks, challenges, opportunities, and pitfalls, APIs. (9)

ANONYMIZATION MODELS: K-anonymity, l-diversity, t-closeness, differential privacy, Comparison, Using anonymization for privacy preservation in social networks, K anonymity and l-diversity in publishing online social networks, Link prediction techniques, Role of link prediction in compromising privacy, approaches to ensure privacy, network anonymization techniques, clustering approaches, graph modification approaches, deanonymization techniques (9)

TRUST MANAGEMENT IN ONLINE SOCIAL NETWORKS: Trust and Policies, Trust and Reputation Systems, Trust in Online Social Media, Trust Properties, Trust Components, Social Trust and Social Capital, Trust Evaluation Models; Trust, credibility,

and reputations in social systems; Online social Media and Policing, Information privacy disclosure, revelation and its effects in online social networks; Phishing in OSM, Identifying fraudulent entities in online social networks (9)

CONTROLLED INFORMATION SHARING IN ONLINE SOCIAL NETWORKS: Access Control Models, Access Control in Online Social Networks, Relationship-Based Access Control, Privacy Settings in Online Social Networks, Existing Access Control Approaches, Secure Multi-party computation and Cryptography, Privacy preserving data integration techniques (9)

IDENTITY MANAGEMENT IN ONLINE SOCIAL NETWORKS: Identity Management, Digital Identity, Identity Management Models: From Identity 1.0 to Identity 2.0 , Approaches for identity Management in Online Social Networks, Identity as Self-Presentation, Identity theft, challenges, knowledge based approaches for identity management in OSN, Machine learning for OSN privacy (9)

Total L: 45

TEXT BOOKS:

1. Chbeir, Richard, Al Bouna, Bechara, "Security and Privacy-Preserving in Social Networks", Springer, 2013.
2. Barbara Carminati, Elena Ferrari, Marco Viviani, "Security and Trust in Online Social Networks", Springer, 2014

REFERENCES:

1. Altshuler, Y., Elovici, Y., Cremers, A.B., Aharony, N., Pentland, A., " Security and Privacy in Social Networks", Springer, 2013
2. Tripathy, B.K. &Baktha, Kiran, "Security, Privacy, and Anonymization in Social Networks: Emerging Research and Opportunities" ,2018.
3. Brij B. Gupta, Somya Ranjan Sahoo,"Online Social Networks Security", 1st edition, CRC Press, 2021.
4. Elena Zheleva, Evimaria Terzi, and Lise Getoor," Privacy in Social Networks", Morgan & Claypool Publishers, 20124

23N045 MALWARE ANALYSIS

3 0 0 3

INTRODUCTION : Introduction to malware, OS securityconcepts, malware threats, evolution of malware, malware types - viruses, worms, rootkits, Trojans, bots, spyware, adware, logic bombs, Life cycle of malicious attack, malware components, malware behaviors - Downloader, Backdoors, CredentialStealers, Persistence Mechanisms, Privilege Escalation- Launchers - Process Injection, Process Replacement, Hook Injection, APC injection - malware analysis: static analysis, dynamic analysis. (9)

STATIC ANALYSIS: :X86 Architecture- Main Memory,Instructions, Opcodes and Endianness, Operands, Registers, SimpleInstructions, The Stack, Conditionals, Branching, Rep Instructions,C Main Method and Offsets. Antivirus Scanning, Fingerprint forMalware, Portable Executable File Format, The PE File Headersand Sections, The Structure of a Virtual Machine, ReverseEngineering- x86 Architecture, recognizing C code constructs inassembly, C++ analysis,Analyzing Windows programs, Anti-static analysis techniquesobfuscation, packing, metamorphism, polymorphism. (9)

DYNAMIC ANALYSIS: Live malware analysis, dead malwareanalysis, analyzing traces of malware- system-calls, api-calls, registries, network activities. Anti-dynamic analysis techniques- anti-VM, runtime-evasion techniques, , Malware Sandbox,Monitoring with Process Monitor, Packet Sniffing with Wireshark, Debugging - Kernel vs. User-Mode Debugging, OllyDbg, Breakpoints, Tracing,Exception Handling, Patching (9)

ANTI REVERSE ENGINEERING TECHNIQUES:: Anti-Disassembly- techniques, obscuring flow control, thwarting stack frame analysis, Anti-Debugging- behaviour, interfering with functionalities, Anti-Emulator, Anti-Virtual MachineTechniques – vulnerable instructions – escaping VM (9)

MALWARE DETECTION TECHNIQUES: Signature-based techniques: malware signatures, packed malware signature, metamorphic and polymorphic malware signature Non-signature based techniques: similarity-based techniques, machine-learning methods, invariant inferences - Case studies with different families of recent malwares. (9)

Total L: 45

TEXT BOOKS:

1. Michael Sikorski and Andrew Honig, "Practical Malware Analysis: The Hands-On Guide to Dissecting Malicious Software", NoStarch Press, San Francisco, 2012.
2. Christopher C. Elisan , Advanced Malware Analysis,.Cameron H. Malin, Eoghan Casey, James M. Aquilina and Curtis W. Rose, Malware, Tata McGraw Hill, 2015

REFERENCES:

1. Abhijit Mohanta and Anoop Saldanha, "Malware Analysis and Detection Engineering: A Comprehensive Approach to Detect and Analyze Modern Malware", Apress, Mangalore, 2020.
2. Michael Hale Ligh, Steven Adair, Blake Hartstein and Matthew Richard: Malware Analyst's Cookbook and DVD: Tools and Techniques for Fighting Malicious Code. Wiley
3. Dang, Gazet,Bachaalany, "Practical Reverse Engineering",Wiley,2014
4. Yin, Heng, Dawn Song. "Automatic Malware Analysis: An Emulator Based Approach", Springer Science & Business Media, 2012.
5. Malware Analysis and Detection Engineering: A Comprehensive Approach to Detect and Analyze Modern Malware, O'Reilly, 2020.
6. Dylan Barker," Malware Analysis Techniques: Tricks for the triage of adversarial software".

23N047 QUANTUM-SAFE CRYPTOGRAPHY

3 0 0 3

QUANTUM INFORMATION: Quantum computing, bits vs qubits, quantum algorithms-superposition, entanglement, quantum vs

classical algorithms, quantum parallelism, Shor's, algorithms, Hidden subgroup problem ,Grover's algorithm, Boolean and quantum Fourier analysis, HHL algorithm, combination of quantum and classical techniques, status of quantum computers (9)

CHALLENGES: Principles of cryptographic design, symmetric and assymetricsystems ,Discrete Log problem, integer factorisaton problem, Key exchange, Symmetric and public key encryption, Random Oracle Model, Elgamal encryption- threat to asymmetric cryptography (9)

POST QUANTUM CRYPTOGRAPHY TECHNIQUES-1: Code-based cryptography- error correcting codes, McElise Crypto System - Lattice-based Cryptography: Mathematical lattices, Shortest Vector Problem, Learning with Errors, NTRU Cryptosystem – possible attacks (9)

POST QUANTUM CRYPTOGRAPHY TECHNIQUES-2: Hash-based cryptography- hash functions, one time signatures and hash trees, Merkle trees and signatures, LMS systems , SPHINCH-Isogeny-based cryptography:Elliptic curves, Elliptic curve Diffie- hellman super- singular ellipticcurves, isogeny, Supersingular Isogeny-Based Diffie-Hellman (SIDH) Multivariate cryptography: multivariate polynomial –multivariate public key system – examples – possible attacks (9)

QUANTUM CRYPTOGRAPHY: Fully homomorphic encryption,no cloning theorem, Quantum key distribution, Quantum one time pad, secret key distillation, privacy amplification using hash functions, BB84 system, Quantum public key encryption, Quantum fully homomorphic encryption – security analysis of QKD, Comparison of quantum-safe approaches. (9)

Total L: 45

TEXT BOOKS:

1. Arne S, "Quantum-safe cryptography – fundamentals, current developments andrecommendations Federal Office for Information Security", 2021
2. Federico Grasselli, "Quantum Cryptography: From Key Distribution to Conference Key Agreement", Springer, 2021

REFERENCES

1. Assche, G, "Quantum Cryptography and Secret-Key Distillation", Cambridge University Press, 2006.
2. Nirbhay Kumar Chaubey, Bhavesh B. "Prajapati, Quantum Cryptography and the Future of Cyber Security (Advances in Information Security, Privacy, and Ethics) ", IGI global, 2020.
3. Daniel J. Bernstein, Johannes Buchmann and Erik Dahmen,"Post-Quantum Cryptography", Springer, 2009.
   Zhiyong Zheng, Kun Tian, Fengxia Liu, "Modern Cryptography- Post-Quantum Cryptography", Springer, 2023

OPEN ELECTIVES

23NO01 BIOINFORMATICS

3 0 0 3

INTRODUCTION: Scope, applications and limitations of Bioinformatics- Biological Databases and information retrieval – FASTA & BLAST – Protein Structure - Amino Acid, Peptide Formation, Dihedral Angles, Secondary Structures, Tertiary Structures, Determination of Protein Three-Dimensional Structure, Protein Structure Database, Visualisation. (9)

SEQUENCE ALIGNMENT: Pairwise sequence alignment- sequence similarity, homology, identity – scoring matrices – Multiple sequence alignment – exhaustive and heuristic approaches – scoring (9)

MOTIFS:Protein Motifs and Domain Prediction - Identification of Motifs and Domains in Multiple Sequence - Alignment - Motif and Domain Databases Using Regular Expressions - Motif and Domain Databases Using Statistical Models - Protein Family Databases- Motif Discovery in Unaligned Sequences (9)

PHYLOGENETICS: Molecular Evolution and Molecular Phylogenetics - Gene Phylogeny versus Species Phylogeny - Forms of Tree Representation - Procedure - Phylogenetic Tree Construction Methods - Distance-Based Methods, Character-Based Methods – Evaluation (9)

STRUCTURAL BIOINFORMATICS:Protein Secondary Structure Prediction - Protein Tertiary Structure Prediction – Methods, Homology modeling, Threading and fold recognition, AbInitio Protein Structural Prediction – Drug discovery (9)

Total L: 45

TEXT BOOKS:

1. Dev Bukhsh Singh, Rajesh Kumar Pathak, "Bioinformatics: Methods and Applications", 1st Edition, Elsevier, 2021.
2. AuthurLesk, "Introduction to Bioinformatics", Fifth edition, Oxford University Press, 2019.

REFERENCES:

1. Ion & Alexander, "Bioinformatics Algorithms", Wiley Interscience, 2009.
2. Perambur S. Neelakanta, "A textbook of bioinformatics: information-theoretic perspectives of bioengineering and biological complexes", World Scientific Publishing, 2021.
3. Andreas D. Baxevanis, Gary D. Bader, David S. Wishart, "Bioinformatics", 4th Edition, May 2020.
4. Xiong, J., "Essential Bioinformatics". Cambridge University Press, Cambridge, 2006.

23NO02 DESIGN THINKING

3 0 0 3

INTRODUCTION: Understanding Design thinking - Design Thinking as problem solving tool - Human Centric Design Process - Principles of Design Thinking - Design Thinking Process - Problems Best suited for Design Thinking - Case Study: Problem Identification. (9)

EMPATHISE WITH USERS:Empathy for design – Five Whys - Needs of user - Types of user research - Interviewing (Stories, Anecdotes) - Mind Mapping Tool - Observational Research - Ergonomics and Human Factors - Affinity analysis - Empathy map – Persona. (9)

PROTOTYPING: Ideas to presentable concepts – Concept Selection - Storyboards - Tasks in Prototyping – Prototyping techniques – High-Fidelity and Low-Fidelity Prototyping. prototyping used in the software industry – Rapid - Evolutionary - Incremental – Extreme. Case Study: Design prototypes using Prototyping tools. (9)

DESIGN AND TESTING AND EVALUATION: Designing for Tangible and Intangibles - Digital touchpoints - Product Design - Interaction Design - Service Design - Transportation Design - Communication Design - Testing Prototypes. (9)

APPLICATIONS: Design Thinking for strategic innovations - Artificial Intelligent Application - HealthCare and Science – Education- Transportation - Finance – Technology (9)

Total L: 45

TEXT BOOKS:

1. Andrew Pressman "Design Thinking A Guide to Creative Problem Solving for Everyone", Routledge Publication, 2019.
2. Christian Mueller-Roterberg, “Handbook of Design thinking”, Amazon Digital Services LLC - KDP Print US, 2018.

REFERENCES:

1. Muller-Roterberg "Design thinking for dummies" John Wiley & Sons, 2020.
2. Tim Brown, “Change by Design”, Harper Business Publisher, 2019
3. Alyssa Gallagher and Kami Thordarson, "Design Thinking in Play: An Action Guide for Educators", ASCD Book, 2020
4. HassoPlattner, Christoph Meinel and Larry Leifer, "Design Thinking: Understand – Improve – Apply", Springer, 2011

23NO03 ETHICS OF ARTIFICIAL INTELLIGENCE

3 0 0 3

INTRODUCTION TO ETHICS OF AI: Overview of Narrow AI, General AI and Responsible AI - Laws and Regulation Ethics of the Ethics of AI- Ethical Issues and relationship with artificial entities- Exploring the Ethical Considerations in Indian context. Frameworks and Modes:AI Governance by Human rights- Incompatible incentives of private sector AI - Normative Modes: Codes and Standards- Professional Norms in the Governance of AI -Legal and Regulatory Approaches for Managing AI Systems in India (9)

CONCEPTS AND ISSUES: Justice in Artificial Intelligence: Limits, Failings, and Ethics of Fairness - Accountability in computer Systems- Transparency-Responsibility and AI- Ethical Analysis and design- The future of work in the age of AI- Sentiment AIs - Autonomy –Algorithmic Governance and Law. (9)

PERSPECTIVES AND APPROACHES: Perspectives and Approaches of computer science- Social Failure modes in technology - Human centred Approach to AI Ethics: perspective from Cognitive science - Integrating ethical and economic values- Fairness through the lens of Directed Acyclic Graphs: a Statistical Modelling Perspective Designing for other worlds- Perspectives and Approaches AI in ethics: East Asia – Middle East- Policy framework for trustworthy AI. (9)

ADDRESSING ETHICAL ISSUEIN AI: Ethical theories - purpose of AI- Ethical principles of AI –Options at the policy andorganisational levels- guidance mechanisms –AI ethics stakeholders -Principles for Responsible Management of AI Systems in India (9)

NEAR FUTURE OF AI: Mass Unemployment- autonomous Weapons – Ethical Matrix –Ethics of Artificial Lover-Long term impact of super intelligence: Alignment of advanced machine learning systems- moral machines-Designing AI with Rights, Consciousness, Self-Respect, and Freedom. Applications: Transport, Defence, Healthcare, Law, Education, Robot Teaching, Social organization of work, Smart City. (9)

Total L: 45

TEXT BOOKS:

1. Markus D. Dubber, Frank Pasquale, Sunit Das, “The Oxford Handbook of Ethics of AI”, Oxford University Press, USA, 2020.
2. Bernd Carsten Stahl, “Artificial intelligence for a better future: An Ecosystem Perspective on the ethics of AI and Emerging digital Technologies”, Springer, UK, 2021.

REFERENCES:

1. S. Matthew Liao, “Ethics of Artificial Intelligence”, Oxford UniversityPress,USA, 2020.
2. Steven John Thompson, “Machine Law, Ethics, and Morality in the Age of Artificial Intelligence”, IGI Global, USA, 2021.
3. Christoph Bartneck, Christoph Lütge,Alan Wagner, Sean Welsh, “An Introduction to Ethics in Robotics and AI”, First edition,Springer, Switzerland, 2021.
4. NitiAayog, “Responsible AI”, Govt of India, 2021.

23NO04 GRAPH THEORY

3 0 0 3

INTRODUCTION: Review on Definition and Basic Terminologies of Graphs – Representations of Graphs – Walks in Graphs and Digraphs- Subgraphs-Vertex Degrees - Path and Cycles - Regular and Bipartite Graphs- Representations of graphs – adjacency and incidence lists – adjacency and incidence matrices -Graph Traversals-Applications: Four Cubes Problem- Social Networks. (9)

EULERIAN AND HAMILTONIAN GRAPHS: Exploring and Travelling – Eulerian Graphs – Konigsberg bridge problem - Hamiltonian Graphs – Applications: Dominoes – Chinese Postman Problem - Travelling salesman problem (8)

PATHS AND CONNECTIVITY: Connected Graphs and Digraphs-Menger's Theorem for Graphs-Applications: Reliable Telecommunication Networks. Network flows and applications- Flows and cuts in Networks, Maximum-flow problem, flows and connectivity– applications (10)

VERTEX-COLORING: Vertex-coloring - chromatic number of a graph, vertex coloring algorithms – sequential vertex coloring, largest degree first algorithm, applications - scheduling problem, fast register allocation for computer programming. 10)

MATCHING AND FACTORS: Matching, Perfect matching, Tutte's 1-factor theorem, weighted Bipartite matching, Hall's theorem. (8)

Total L: 45

TEXTBOOKS:

1. Jonathan L. Gross and Jay Yellen, "Graph Theory and its Applications", CRC Press, New York, 2016.
2. Douglas B West, "Graph Theory", Prentice Hall, New Delhi, 2017.

REFERENCES:

1. Bondy J.A. and Murty U.S.R., "Graph Theory", Springer, London, 2016.
2. Joan M Aldous and Robin J Wilson, "Graphs and Applications- An Introductory Approach, Springer-Verlag", New York, 2014.
3. Reinhard Diestel,"Graph Theory", Springer-Verlag, Berlin Heidelberg, 2012.
4. Willian Kocay, Donald L. Kreher, "Graphs, Algorithms, and Optimization", CRC Press, New York , 2017.

23NO05 INTELLECTUAL PROPERTY RIGHT

3 0 0 3

PATENTABLE ENGINEERING INVENTIONS: What are inventions? Commercial implications of patent protecting an invention.Introduction To Patents - Requirements For Patentability – Novelty, Inventive step, IndustrialApplication - Types of Patent Applications - Provisional And Complete Specification - Structure of Provisional And Complete Specifications - Sections And Rules of the Indian Patent Act – Non patentable Inventions (9)

TYPES OF INTELLECTUAL PROPERTY:Patents, Trademarks, Copyrights, Industrial Designs,Geographical Indication (GI), Trade Secrets, Integrated Circuit Layout Designs (SICLDR) (9)

PROCESS OF PATENTING AND PATENTABILITY SEARCHING: Patenting Process - Procedure for grant of patent - International Patenting through PCT route -What is patentability search – Patent information and databases- outcome of search – Limitations ofpatentability search – Patent search reportPractical Exercises: Prior art Searching and Reporting using open source and commercial tools –Disclosing an Invention- Comparing inventions with their closest prior arts (9)

PATENT DRAFTING: Title of the invention - Field of the invention - Background art - Objects of the invention – Patentcitations and prior art in the background of the invention- Patent of addition - Divisional application, Introduction to claims - Types of claims - Significance and scope of claims Structure of claims – Draftingand interpretation of claims (9)

SCOPE OF PATENT RIGHTS, LICENSING AND TRANSFER OF TECHNOLOGY, NEW DEVELOPMENTS IN IPR: Administration of patent system, IPR of computer software (9)

Total L: 45

TEXT BOOKS:

1. Robert P. Merges, Peter S. Menell, Mark A. Lemley, "Intellectual Property in New Technological Age", 2016.
2. "Patent IPR Licensing- Technology Commercialization – Innovation Marketing: Guide Book for Researchers, Innovators", Indian Innovators Association, 2017.

REFERENCES:

1. Kushdeep Dharani, Neeraj Pandey ,"Intellectual Property Rights", PHI, 2014
2. Halbert, "Resisting Intellectual Property", Taylor & Francis Ltd, 2007.
3. Mayall, "Industrial Design", McGraw Hill, 1992.
4. Niebel, "Product Design", McGraw Hill, 1974.

23NO06 SOCIAL AND ECONOMIC NETWORK ANALYSIS

3 0 0 3

GRAPH THEORY IN NETWORK ANALYSIS: Representing Networks - Graphs and Networks - Paths and Cycles - Components and Connected subgraphs - Neighborhood - Degree and Network Density - Eigenvectors and Eigenvalues - Degree Distributions - Cliquishness, Cohesiveness, and Clustering – Centrality. (7)

NETWORK DYNAMICS: The Small-World Phenomenon: Six Degrees of Separation - Decentralized Search- Power Laws - Rich-Get-Richer Models - The Long Tail - The Problem of Ranking - Link Analysis Using Hubs and Authorities – PageRank (8)

SOCIAL NETWORK ANALYSIS: Triadic Closure - The Strength of Weak Ties - Homophily- Affiliation - Betweenness Measures and Graph Partitioning- Communities and Blocks - Methods for Identifying Community Structures - Stochastic Block Models and

Communities - Maximum-Likelihood Estimation of Communities (10)

CASCADING BEHAVIOR IN NETWORKS:Introduction to game theory - Best Responses and Dominant Strategies - Nash Equilibrium - Multiple Equilibria: Coordination Games - Diffusion in Networks – Epidemics - The SIR Epidemic Model - The SIS Epidemic Model. (10)

ECONOMIC NETWORK ANALYSISAUCTIONS: Types of Auctions - Second-Price Auctions - Matching Markets: Bipartite Graphs and Perfect Matching - Sponsored Search Markets: Advertising Tied to Search Behavior - Advertising as a Matching Market – The VCG principle - Equilibria of the Generalized Second-Price Auction. (10)

Total L: 45

TEXT BOOKS:

1. David Easley, Jon Kleinberg, “Networks, Crowds and Markets”, Cambridge University Press, USA, 2016.
2. Matthew O. Jackson, “Social and Economic Networks”, Princeton University Press, New York, 2015.

REFERENCES:

1. Jure Leskovec, AnandRajaraman,Jeff Ullman, ”Mining of Massive Datasets”, Cambridge University Press, USA, 2014.
2. Charu C Agarwal, “Social Networks Data Analytics”, Springer, USA, 2011.
3. John Scott, Peter J. Carrington, “Sage Handbook of Social Network Analysis”, SAGE Publications, New York 2011
4. Valente, Thomas, “Social Networks and Health: Models, Methods and Applications”, Oxford University Press, New York, 2010

SELF DIRECTED LEARNING COURSES

23IC01 EXPERIENCING ENTREPRENEURSHIP

3 0 0 3

ENTREPRENEURSHIP DEVELOPMENT PROCESS: Entrepreneurial Ecosystem - Entrepreneur, Enterprise and Environment, Entrepreneurial Motivation, Entrepreneurial Competencies. ED programme - Need and objectives, EDPs in India and its phases. Stages of ED Process - Discover, evaluate, developing a plan, resources, managing the company, harvesting (10)

CREATIVITY, IDEA GENERATION: Creativity: Right environment, Components of performance, tools and techniques, motivation. Design thinking: Process, heuristics approaches, behavioural change, empathy.Idea Generation: Process, methods, sources, entrepreneurial linkages. Systematic Inventive Thinking (SIT): Principles, toolsDisruptive Innovation (8)

PRODUCT / PROCESS / TECHNOLOGY / SERVICE DEVELOPMENT: Understanding Market Needs - Market Research Fundamentals, Identifying Customer Pain Points, Target Audience Analysis. Prototyping and Design - Rapid Prototyping, User-Centered Design Principles, Iterative Design Processes. Streamlining Operations - Value Stream Mapping, Lean and Agile Principles, Continuous Improvement Strategies. Service Development - Designing Customer-Centric Services, Service Innovation, Quality Assurance in Services. (10)

BUSINESS PLAN AND FEASIBILITY ANALYSIS: Market Analysis: Preparation of Business Plan/Model. Product/Service Feasibility Analysis – Methods and Tools. Industry and Competition Analysis: GTM Strategy – Viability - FAB Analysis - USP, Gap Analysis. Environment and Technical Analysis. Financial Feasibility Analysis: Concept of Costs, Profits and Losses, Manage Cash Flow, Analyze Financial Performance and Budgeting. (8)

*** CAPSTONE PROJECT:** Students will prepare a business plan and feasibilityanalysisfor a business of their choice (9)

Total L: 45

REFERENCES:

1. Rajeev Roy, Entrepreneurship. New Delhi: 3 rd Edition, Oxford University Press, 2020
2. Pareek, UdaiNarain, and T. Venkateswara Rao. Developing motivation through experiencing: a trainers' guide for behavior simulation in motivation training. Oxford & IBH, 1991.
3. Rod Judkins, “The Art of Creative Thinking”, Hodder & Stoughton, 2015
4. Peter Drucker, “Innovation and Entrepreneurship”, Routledge, 2015
5. Osterwalder, A. and Pigneur, Y. Business model generation: a handbook for visionaries, game changers, and challengers (Vol. 1). John Wiley & Sons, 2010
6. Andy Polaine, LavransLøvlie, and Ben Reason, "Service Design: From Insight to Implementation" Rosenfeld Media, 2013.

23IC02 VENTURE DEVELOPMENT

3 0 0 3

BUSINESS MODEL CANVAS: Customer Segments, Value Propositions, Channels and customer relationships, Revenue streams and key resources, Key activities and key partnerships, Cost structure (10)

FUNDING OPPORTUNITIES: Government Funding and Incubators: Functions of an incubator. Venture capital and Angel Funding: Venture Capital Industry, Company Formation, Due diligence, Valuation and term sheets, Value addition and governance, Exits. (10)

ENTERPRISE FORMATION & LEGAL COMPLIANCES: Understanding Legal Structures / Compliance Requirements - Business Registration and Licensing, Intellectual Property Protection. Tax Compliance and Planning - Corporate Taxes and Business

Deductions, External Audits and Compliance Checks. Exit Strategies and Legal Considerations - Succession Planning and Legal Requirements, Handling Legal Challenges during Business Exits. (15)

***CAPSTONE PROJECT:** Students will demonstrate their ability to fully develop and communicate a venture concept. (10)

Total L: 45

REFERENCES:

1. Osterwalder, A. and Pigneur, Y, Business model generation: a handbook for visionaries, game changers, and challengers (Vol. 1). John Wiley & Sons, 2010
2. Rajeev Roy, Entrepreneurship, Oxford Higher Education, India, 2020
3. "A Note on the Venture Capital Industry," Harvard Business School (HBS # 9-295- 065, July 12, 2001)
4. SumitPahwa "Corporate Law Referencer 9/e",OakBridge Publishing Pvt. Ltd, 2023
5. CharantimathPoornima M, "Entrepreneurship Development and Small Business Enterprises", 3/e, Pearson Education, 2018

(* The activity will start during the 7th week of the respective courses)